

The Effect of Cognitive Skills on Fertility Timing*

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Abstract

In the NLSY79, 69% of women in the lowest cognitive ability quartile have a first birth before age 22, compared with 22% in the highest. This gradient persists within education groups and within contraceptive method, suggesting that neither education sorting nor method selection fully accounts for it. I estimate a life-cycle model in which cognitive ability shifts the effectiveness of fertility control, beyond standard opportunity-cost channels. Restricting effectiveness to be equal across ability groups causes the model to underpredict the ability–fertility gradient by a factor of five. Equalizing effectiveness to high-ability levels reduces births before age 22 by 50% and raises college attendance by 15%. A cost-reduction policy generates welfare gains of 9% of lifetime consumption for the lowest ability quartile but near-zero gains for the highest, suggesting that improving effectiveness matters more than reducing cost for disadvantaged women.

JEL codes: J13, J12, I21, J24, C61.

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1 Introduction

Early childbearing varies sharply with cognitive ability. In the National Longitudinal Survey of Youth 1979 (NLSY79), 69% of women in the lowest cognitive ability quartile have had a first birth before age 22, compared with 22% in the highest quartile. This gap reflects a pronounced difference in the *timing* of first births rather than in whether women eventually become mothers: first-birth hazard rates are sharply decreasing in cognitive ability during the teenage years but converge across ability groups by the late twenties, so that mean age at first birth differs by 5.4 years across quartiles, with higher-ability women postponing rather than forgoing motherhood.

I first document that this gap is not an artifact of education sorting or contraceptive method selection: within education groups, women with lower cognitive ability experience significantly higher birth rates even when using the same contraceptive method, and differences in method choice do not explain the pattern. I then develop and estimate a dynamic life-cycle model of schooling, work, marriage, and contraceptive effort to ask whether standard opportunity-cost channels can account for this gap. They cannot: the model without ability-dependent fertility-control effectiveness underpredicts the ability gap in early births by a factor of five. Matching the data requires a direct ability shifter in the mapping from contraceptive effort to realized births—a residual fertility-control channel, absent from the existing structural fertility literature, that operates beyond the education and wage mechanisms emphasized in standard models. Counterfactual experiments show that equalizing fertility-control effectiveness to high-ability levels reduces births before age 22 by 50% and increases college attendance by 15%, while a cost-reduction policy that lowers aggregate early births by 10% generates welfare gains of 9% of lifetime consumption for the lowest ability quartile but near-zero gains for the highest. The binding constraint for disadvantaged women is effective fertility control, not cost.

Understanding the sources of this disparity is important for several reasons. Early childbearing reduces women’s education, labor supply, and lifetime earnings ([Adda et al., 2017](#)), while delayed motherhood causally raises children’s human capital ([Miller, 2009](#)). Because early births concentrate among disadvantaged women, ability-dependent fertility timing is a channel through which inequality is transmitted across generations ([Aizer et al., 2022](#);

Seshadri and Zhou, 2022). Moreover, as of 2011, approximately 40% of U.S. pregnancies were unintended (Finer and Zolna, 2016), and early parents who are financially constrained under-invest in children’s early human capital (Caucutt and Lochner, 2020). Effective policy requires understanding whether the binding constraint is the incentive to avoid a birth or the ability to do so. These questions have become more pressing even as aggregate fertility has declined across high-income countries (Doepke et al., 2023; Kearney and Levine, 2025). Reductions in birth rates among young women—particularly those without a high school degree—are the largest contributors to the aggregate fertility decline (Kearney et al., 2022; Onofri et al., 2026), making the mechanisms that shape early childbearing central to understanding the broader demographic transition.

The paper’s findings help rationalize the success of long-acting reversible contraceptive (LARC) programs. Lindo and Packham (2017) exploit the rollout of Colorado’s Family Planning Initiative and find that expanded LARC access reduced teen birth rates by roughly 6% overall and by larger margins in higher-poverty counties. Similarly, the Contraceptive CHOICE Project found that providing no-cost LARCs with comprehensive counseling reduced teen pregnancy rates by approximately 79% relative to national rates (Secura et al., 2014). LARCs work precisely because they decouple contraceptive effectiveness from user effort. Medical evidence documents large gaps between “perfect use” and “typical use” contraceptive failure rates that vary systematically with user characteristics (Black et al., 2010; Trussell, 2011); an IUD or implant eliminates this gap by removing the need for daily compliance, correct timing, or partner negotiation. In the language of my model, LARCs raise effectiveness toward its upper bound for all women, eliminating the ability gap in the effort-to-birth-risk mapping. Because the binding constraint for low-ability women is effective fertility control rather than cost, technology-shifting interventions like LARC programs should generate the largest welfare gains for the most disadvantaged women—consistent with the evidence from LARC expansions documented above.

The empirical analysis provides direct support for the effectiveness mechanism. In the NLSY79, the ability gap in early births is not an artifact of education sorting: within education groups, women with lower cognitive ability experience significantly higher birth rates. More importantly, conditioning on the contraceptive method used does not eliminate this

gap. Among women using the same method (pill, barrier, or behavioral), those in the lowest AFQT quartile have birth rates 2.3 percentage points higher than those in the highest quartile—and this within-method gap *exceeds* the unconditional one, ruling out method selection as the driver. These patterns point to a per-use effectiveness channel that cannot be explained by method choice, education, or differential incentives to exert effort.

This paper studies live births and abstracts from abortion. Even if abortion plays a role, the available evidence suggests that the estimates presented here are a conservative lower bound on the true ability gap in contraceptive effectiveness. [Woodley of Menie et al. \(2017\)](#) find no significant relationship between the number of abortions and maternal cognitive ability in the NLSY79, and [Jones and Jerman \(2017\)](#) document that low-income women—who are disproportionately low-ability—have abortion rates six times higher than higher-income women. Differential abortion use therefore *compresses* rather than generates the live-birth gradient relative to the underlying conception gradient.

To interpret these patterns, I develop a structural life-cycle model in which cognitive ability is an early, persistent trait¹ and women jointly choose schooling, labor supply, experience accumulation, marriage, and contraceptive effort. By contraceptive effort, I mean the deliberate actions and resources women devote to preventing pregnancy—paying for and accessing methods, learning about options, negotiating use with partners, using contraception consistently, and managing the cognitive burden of planning and adherence. The model’s key innovation is a fertility-control technology in which cognitive ability shifts the *effectiveness* of contraceptive effort—the rate at which effort translates into reduced birth risk—in addition to the standard channels through which ability affects fertility via schooling and wage incentives. This mechanism is motivated by both the perfect–typical use gap in failure rates discussed above and the descriptive evidence that, within contraceptive method, cognitive ability predicts birth rates even after conditioning on education.

The model nests the standard opportunity-cost explanation as a special case. When effectiveness is restricted to not vary with ability or education, the only channel through which ability affects fertility is the incentive channel: higher wages and steeper experience profiles make early births more costly for high-ability women. The unrestricted model, in

¹Measured cognitive ability is highly stable from late adolescence through adulthood ([Almlund et al., 2011](#); [Heckman et al., 2006](#)).

which effectiveness varies with ability, substantially outperforms the restricted specification—total fit improves by 51%, with the sharpest gains on the fertility-by-ability moments (78% improvement). This shows that the standard opportunity-cost channel, even after the model has fully optimized effort by each ability type to match wages, schooling, marriage, and contraception use, cannot account for the observed ability gap in early births.

Two sets of counterfactual experiments quantify the policy-relevant implications. First, equalizing fertility-control effectiveness to high-ability levels—the model analogue of universal LARC access—reduces births before age 18 by 50% and increases college attendance by 15%; combining this with equalized schooling opportunities raises college attendance by 31% and reduces early births by 55%. Notably, equalizing schooling opportunities alone does *not* reduce early fertility—it increases it—confirming that the binding constraint is fertility control, not educational incentives. Second, a cost-reduction policy calibrated to lower aggregate early births by 10% generates welfare gains of 9% of lifetime consumption for Q1 women but near-zero gains for Q4 women. These results are consistent with the evidence from LARC evaluations ([Lindo and Packham, 2017](#)), where the largest effects also concentrate among the most disadvantaged women.

This paper makes three main contributions. First, I provide new descriptive evidence on the relationship between cognitive ability and fertility control. Using the NLSY79, I show that the ability gap in early births persists within education groups and—more importantly—within contraceptive method: women with lower cognitive ability have higher birth rates even when using the same method, and this within-method gap exceeds the unconditional one. These findings establish that neither education sorting nor method selection can account for the ability gap, pointing to a per-use effectiveness channel.

Second, I estimate a life-cycle model that jointly determines schooling, wage growth through experience, marriage, and fertility timing, where both education and cognitive ability can independently shift effective fertility control. The model introduces a novel parameter—ability-dependent fertility-control effectiveness—that governs how cognitive skill shifts the mapping from contraceptive effort to realized births. This extends the conception-risk specifications in [Choi \(2017\)](#) and [Ejrnaes and Jørgensen \(2020\)](#) by allowing heterogeneity in the productivity of effort, not merely its cost or baseline risk levels.

Third, I use the estimated model to quantify the policy-relevant implications of this decomposition. The counterfactual exercises show that the welfare gains from fertility-control interventions are concentrated among the most disadvantaged women, and that interventions targeting effectiveness—rather than cost—are the most promising policy lever for improving women’s control over fertility timing.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and presents new reduced-form evidence on within-method birth hazards by cognitive ability. Section 4 documents the life-cycle patterns that serve as estimation targets for the structural model. Section 5 presents the life-cycle model. Section 6 describes the estimation strategy and identification. Section 7 presents model fit and the specification test. Section 8 presents counterfactual experiments and policy implications. Section 9 concludes.

2 Related Literature

This paper contributes to the large literature that models fertility choices as the outcome of forward-looking household optimization. Foundational work places fertility within household decision-making and the quantity–quality trade-off (Becker, 1960; Becker and Lewis, 1973; Ben-Porath, 1976; Willis, 1973). Dynamic structural models then endogenize the timing and spacing of births in a life-cycle framework, including early discrete-choice models (Heckman and Walker, 1990; Hotz and Miller, 1988; Wolpin, 1984). Building on this tradition, a subsequent wave of life-cycle models jointly determines family formation and labor-market choices: Van der Klaauw (1996) study women’s marital status and labor supply, Francesconi (2002) estimate married women’s joint fertility–labor decisions, Sheran (2007) develop a model with endogenous schooling, marriage, and fertility, and Keane and Wolpin (2010) integrate schooling, work, marriage, fertility, and welfare participation. Related work quantifies how marriage and labor markets shape family structure and birth timing (Caucutt et al., 2002; Regalia et al., 2019).

A closely related strand emphasizes imperfect fertility control and policy-driven changes in reproductive technologies. Choi (2017) incorporates fertility risk and abortion, Ejrnæs and Jørgensen (2020) model family planning in a life-cycle framework with uninsurable income

shocks, and [Amador \(2017\)](#) analyze how abortion and contraception policy affects reproductive choices, schooling, and work. These papers formalize the idea that fertility outcomes reflect both preferences and the effectiveness of avoiding conception, but model the effort–risk mapping as common across women conditional on education and age. This paper introduces cognitive ability as a direct shifter of contraceptive effectiveness, providing a channel that explains why similarly educated women display different fertility timing profiles by cognitive skill. In the estimated model, allowing contraception costs to vary by education is not sufficient: matching the ability gradient requires an ability shifter in the conception hazard. On the interaction between fertility and careers, [Adda et al. \(2017\)](#) quantify the career costs of children; the model shows that the incentives created by career costs are not sufficient to match the ability gradient without a direct ability channel in fertility control.

Third, the paper relates to empirical work on the income–education–fertility relationship and the role of unintended childbearing. [Rosenzweig and Schultz \(1989\)](#) show that schooling increases contraceptive knowledge and effectiveness in use, and [Musick et al. \(2009\)](#) document that the education gradient in births is primarily driven by unintended childbearing. Policies and technologies that lower the cost of fertility control also shape both timing and human-capital investment: [Goldin and Katz \(2002\)](#) and [Bailey \(2006\)](#) show that pill access delayed first births and facilitated educational and career investment; [Kearney and Levine \(2009\)](#) find that Medicaid family-planning expansions reduced births via increased contraception use; and a recent randomized intervention by [Bailey et al. \(2023\)](#) shows that eliminating out-of-pocket costs at Title X clinics substantially increases uptake of highly effective methods and implies a meaningful reduction in undesired pregnancies. Finally, quasi-experimental evidence on education’s causal effect on fertility finds small or context-dependent effects ([Fort et al., 2016](#); [McCrary and Royer, 2011](#)). Relative to this reduced-form literature, I contribute new evidence showing that the ability gap in birth rates persists within education groups and within contraceptive method, ruling out both education sorting and method selection as explanations. I also provide a structural interpretation that explicitly accounts for innate cognitive skills when mapping education and contraception policies into fertility timing and educational attainment.

Fourth, the paper connects to a broader literature documenting that cognitive (and

noncognitive) skills predict a wide range of life outcomes.² In this tradition, [Heckman et al. \(2006\)](#) show that higher cognitive and noncognitive skills reduce risky behaviors, including teen pregnancy and early marriage, while [Fe et al. \(2022\)](#) links childhood cognition to adult behaviors and outcomes, including lower fertility in young adulthood. The within-education and within-method evidence documented in this paper (described above) sharpens these findings by isolating a per-use effectiveness channel that operates beyond education and method selection. I embed these patterns in a disciplined life-cycle model and rationalize them through an ability-dependent fertility-control technology that operates in addition to education and wages. To my knowledge, no existing structural model of fertility timing incorporates cognitive ability as a direct shifter of contraceptive effectiveness while simultaneously allowing for endogenous schooling, labor supply, marriage, and sequential fertility decisions.

Finally, the paper speaks to the economics of U.S. teen childbearing and its decline. [Kearney and Levine \(2012\)](#) provide a synthesis of the evidence and mechanisms, and related work quantifies the roles of improved contraceptive access and changing incentives (e.g., [Kearney and Levine, 2009, 2015](#)). [Di Nola et al. \(2025\)](#) develop a structural model of teenage risky sexual behavior with parental investments and welfare state institutions, but do not incorporate cognitive ability as a shifter of contraceptive effectiveness. More broadly, [Kearney and Levine \(2025\)](#) document that fertility has fallen across high-income countries through both rising childlessness and lower completed fertility, and [Doepke et al. \(2023\)](#) survey the new economics of fertility and show that the historically negative income–fertility relationship has flattened. The model developed in this paper helps explain one component of this decline: as new contraceptive technologies—particularly LARCs—have become widely available, they decouple effectiveness from user effort, reducing early births especially among the most disadvantaged women.

3 Data and Motivating Evidence

This section introduces the data and presents new reduced-form evidence on the relationship between cognitive ability and fertility outcomes. I first describe the survey and sample

²See [Heckman and Mosso \(2014\)](#) for a survey; see also [Almlund et al. \(2011\)](#) and [Cunha and Heckman \(2007\)](#).

construction. I then document that early births among low-ability women are disproportionately unintended, pointing to failures in fertility control rather than preferences for early childbearing. Finally, I show that a substantial ability gradient in birth hazard persists within education groups and within contraceptive method, consistent with a per-use effectiveness channel that motivates the fertility-control mechanism in the structural model.

3.1 Data Description

The data come from the National Longitudinal Survey of Youth 1979 (NLSY79), which follows a nationally representative cohort of individuals born between 1957 and 1964 who were ages 14–22 at the initial interview in 1979. The survey provides detailed longitudinal information on schooling, labor market outcomes (employment, hours, and earnings), marital status and partnership histories, and fertility (pregnancies and births). Because respondents are now in their late sixties, the NLSY79 provides near-complete observation of their life cycles—reproductive years, working lives, and retirement transitions—making it uniquely suited to study fertility timing over the life cycle. Cognitive ability is proxied by the Armed Forces Qualification Test (AFQT), a composite cognitive ability measure derived from the Armed Services Vocational Aptitude Battery (ASVAB) administered early in the panel. The AFQT is widely used as a measure of cognitive skills in economics (e.g., [Heckman et al., 2006](#); [Neal and Johnson, 1996](#)) and captures reasoning ability. I treat invalid and nonresponse codes as missing and exclude women with missing AFQT. After applying these restrictions, the working sample contains 5,634 women with valid AFQT scores and complete fertility histories.

The reduced-form analysis in this section uses seven NLSY79 survey waves spanning the prime fertile years: 1982, 1984, 1985, 1986, 1988, 1990, and 1992. These are the years in which the survey collects information on both contraceptive method use and pregnancy wantedness, and in which the cohort is aged 18–35—the period over which the vast majority of first births occur.

A central feature of the empirical design is the restriction to women *not trying to conceive*. At each survey wave, the NLSY79 asks respondents directly whether they want to become pregnant. I classify a woman–wave observation as “not trying to conceive” if the respon-

dent does not report actively trying to become pregnant at the survey date. Specifically, I exclude observations where the respondent reports that she stopped using contraception in order to become pregnant ($Q9-78 = 0$ and $Q9-81 = 1$ or $Q9-80 = 1$). This restriction removes planned conceptions from the sample, so that the birth hazard captures unintended or mistimed pregnancies—the margin relevant to the model’s contraceptive-effort decision. The outcome variable—contraceptive failure—is constructed from child birth-date records rather than from the NLSY79 self-reported contraceptive-failure variable ($Q9-79$). For each woman $\times$ wave observation, I impose a fixed 12-month window running from July of year t through June of year $t + 1$ and flag a birth if any child’s birth date falls within this window. The 12-month length corresponds approximately to the time required for a conception to result in a live birth, providing a window long enough to capture pregnancies occurring shortly after the interview. Because NLSY79 interviews were fielded primarily between late January and mid-August, the July–June window ensures that the contraceptive method reported at the wave- t interview precedes the outcome window. For the within-method analysis, I further classify women by the contraceptive method reported at each survey wave. Methods are grouped into three tiers ordered by typical-use efficacy: oral contraceptives (pill), barrier methods (condom, diaphragm, foam, jelly, suppository, sponge, cervical cap), and behavioral methods (rhythm, withdrawal, natural family planning). When a woman reports multiple methods simultaneously, she is assigned to the most effective tier (pill $>$ barrier $>$ behavioral), ensuring that each woman–wave observation maps to exactly one method category. Long-acting reversible contraception (LARC) and sterilization are excluded: IUD uptake was negligible in the NLSY79 cohort following the Dalkon Shield recall, and sterilization removes a woman from the risk pool irrespective of ability. Additional details on variable definitions and cleaning conventions are provided in Appendix [OA.1](#).

3.2 Wantedness and the Nature of the Ability Gap

Before testing specific channels, I document a key fact about the *nature* of early births across ability groups. Figure [1](#) plots the fraction of births classified as unintended—mistimed or unwanted—by AFQT quartile, restricting to women not actively trying to conceive (as defined in Section [3.1](#)). The fraction of births classified as unintended is highest among women in

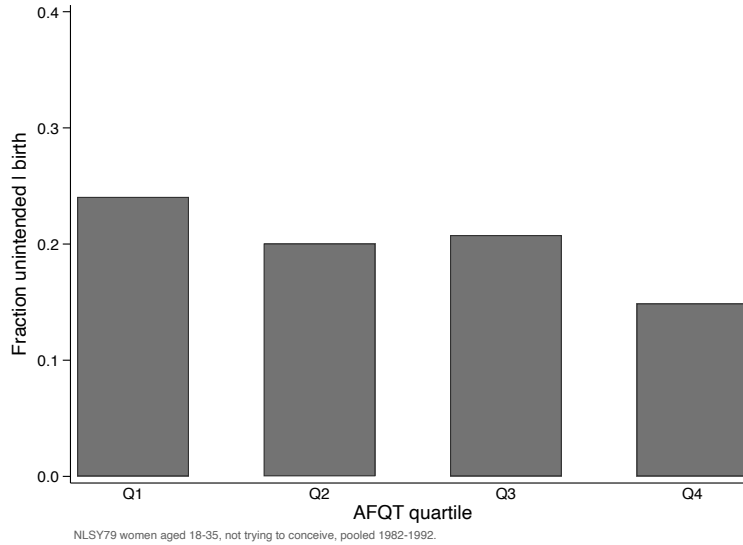


Figure 1. Fraction of births classified as unintended by AFQT quartile. Sample: NLSY79 women not actively trying to conceive who had a birth. Unintended = mistimed or unwanted (NLSY79 wantedness Q9-81, categories 2–4).

the lowest quartile and declines with ability. Low-ability women are more likely to report their births as unintended, consistent with a role for fertility-control capacity in addition to preferences. This pattern suggests that the ability gap in early fertility is unlikely to reflect heterogeneous preferences for early childbearing alone, and instead points toward differences in the capacity to control fertility timing.

3.3 Ability and Birth Hazard: Within-Education and Within-Method Evidence

The wantedness evidence raises a natural question: does cognitive ability affect birth hazard independently, or does the ability gap simply reflect differences in educational attainment and contraceptive method choice? I address this in three steps. First, I estimate birth hazards separately within education groups to assess whether education sorting accounts for the gradient. I then examine contraceptive method selection by ability and show that it runs counter to the gradient. Finally—the main test—I condition on the contraceptive method and show that a substantial ability gap in birth rates persists within method, consistent with a per-use effectiveness channel.

Is the ability gap an artifact of education sorting? AFQT predicts educational attainment, and education affects fertility through opportunity costs. A concern is that the ability gradient in births merely reflects education differences. To test this, I estimate a linear probability model for the probability of a birth *separately within each education group*:

$$B_{it} = \alpha_e + \sum_{q=2}^4 \beta_{eq} \mathbb{1}[\text{AFQT}_i = q] + \mathbf{x}'_{it} \gamma_e + \delta_t + \varepsilon_{it}, \quad \text{estimated within education } e, \quad (1)$$

where $B_{it} \in \{0, 1\}$ equals one if woman i had a live birth in the 12-month window following snapshot wave t , $\mathbb{1}[\text{AFQT}_i = q]$ are AFQT quartile indicators (quartile 1 omitted), the control vector $\mathbf{x}_{it}$ includes parity at the snapshot year (0/1/2+ prior live births), and δ_t are snapshot-year fixed effects. Standard errors are clustered at the respondent level. The sample is all NLSY79 women aged 18–35 not trying to conceive, pooled across the seven survey waves described in Section 3.1. The key coefficients are the β_{eq} : if AFQT matters only through education, then $\beta_{eq} = 0$ within each group. Figure 2 plots the LPM-predicted birth probability by AFQT quartile separately for each education group, and Appendix Table OA.5 reports the corresponding regression coefficients.

The ability gradient holds within every education group. Among high-school dropouts, the gradient is steepest: women in the fourth AFQT quartile are 5.4 percentage points less likely to have a birth than women in the first quartile ($p < 0.01$), and the third-quartile coefficient is -3.6 pp ($p < 0.05$). Among high-school graduates, the Q4 coefficient is -2.0 pp ($p < 0.01$) and the Q3 coefficient is -1.8 pp ($p < 0.01$). Among college graduates, the gradient is present (-2.1 pp for Q3, $p < 0.05$) but the Q4 estimate is smaller and imprecise (-0.9 pp), consistent with higher-ability women within the college group having more homogeneous fertility-control capacity. Panel B of Appendix Table OA.5 shows that the pattern is similar for unintended births: among high-school dropouts, the Q4 coefficient is -0.5 pp and not statistically significant, reflecting the small number of high-ability women who fail to complete high school, but among high-school graduates and college graduates, Q4 women are 1.6 and 3.1 percentage points less likely to report an unintended birth ($p < 0.01$), respectively. Education sorting therefore cannot account for the ability gradient: within every education group, higher-AFQT women experience fewer births, and the gap is concentrated

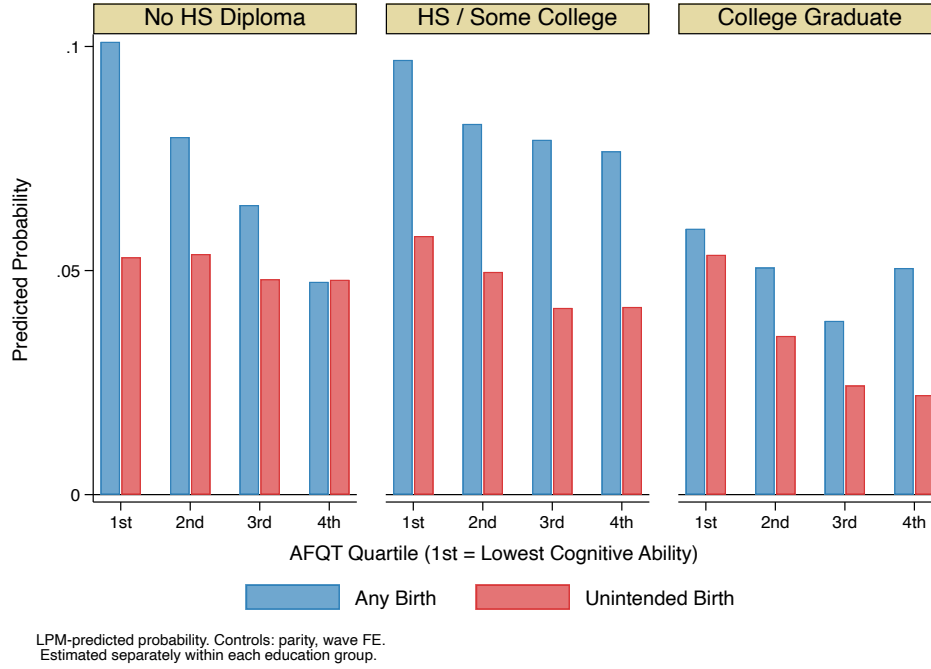


Figure 2. Birth probability by AFQT quartile and education group, unconditional on contraceptive method. Denominator: NLSY79 women aged 18–35 not actively trying to conceive, pooled across seven survey waves. Blue bars: any live birth in the 12-month window. Red bars: births reported as unwanted or mistimed. LPM-predicted probabilities from Equation (1), evaluated at AFQT quartile means; controls set at sample averages within each education group.

in unintended conceptions.

Cognitive ability and birth hazard within contraceptive method. A central concern is that the ability gradient in birth hazard may reflect method selection rather than per-use contraceptive performance. If high-AFQT women disproportionately adopt oral contraceptives while low-AFQT women rely on rhythm or withdrawal, the aggregate ability gradient in birth hazard is consistent with identical per-use effectiveness across ability groups and merely captures compositional differences in technology adoption. I address this concern in two steps. First, I document the method-selection margin directly and show that it exists but runs *counter* to the gradient. Second—the main test—I condition on the contraceptive method reported at each NLSY79 interview and ask whether cognitive ability still predicts the probability of a live birth in the following year.

Method selection by ability. To characterize how contraceptive method choice varies with cognitive ability, I estimate a multinomial logit on NLSY79 women reporting reversible contraceptive use, pooled across the seven survey waves (1982–1992). The dependent variable is the method group $m \in \{1, 2, 3\}$ (pill, barrier, behavioral, as defined in Section 3.1), with the pill as the base outcome:

$$\Pr(m_{it} = j \mid Q_i, \mathbf{x}_{it}) = \frac{\exp(\mathbf{z}_{it}'\boldsymbol{\delta}_j)}{1 + \sum_{k=2}^3 \exp(\mathbf{z}_{it}'\boldsymbol{\delta}_k)}, \quad j \in \{2, 3\}, \quad (2)$$

where $\mathbf{z}_{it}$ includes AFQT quartile indicators, education group, parity, and snapshot-year fixed effects. Standard errors are clustered at the respondent level. Figure 3 plots the predicted method-choice probabilities by AFQT quartile, averaged over the sample distribution of controls; Appendix Table OA.6 reports the predicted probabilities.

Among women using reversible contraception, those in the bottom AFQT quartile have a predicted probability of using oral contraceptives of 75.7%, compared with 60.1% for the top quartile; the corresponding barrier-method probabilities are 16.5% and 31.5% (Figure 3).³ Behavioral method use is roughly constant across quartiles at around 8%. This selection pattern is noteworthy because it runs *counter* to the standard prediction that higher-ability women choose methods with lower typical-use failure rates. If anything, method selection works against the ability gradient in birth hazard: low-ability women concentrate on the pill—which has lower typical-use failure rates than barriers—yet still experience higher birth rates. This makes the case for a per-use effectiveness channel even stronger.

³This pattern may appear to contradict studies using NSFG data that find pill use increasing in education (e.g., Sundaram et al., 2017). The key difference is that the sample here conditions on women actively using reversible contraception to avoid a pregnancy—not all women of reproductive age. This conditioning selects on the decision to contracept, so the method-choice distribution reflects preferences *among active users* rather than the population-wide mix that includes non-users. Since non-use is more prevalent among low-ability women, the unconditional population distribution overweights the non-use margin and understates pill use at the bottom of the ability distribution. A supply-side factor reinforces this pattern: low-income women disproportionately accessed contraception through Title X-funded clinics, which primarily dispensed oral contraceptives on-site (Mosher, 1990; Piccinino and Mosher, 1998).

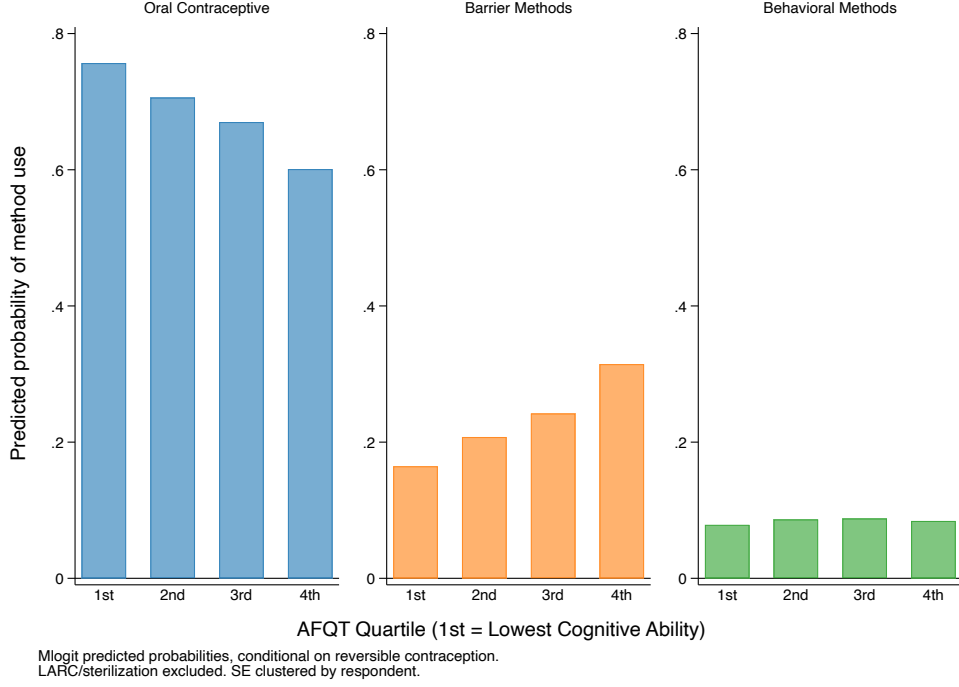


Figure 3. Predicted method-choice probability by AFQT quartile. Multinomial logit estimated on NLSY79 women reporting reversible contraceptive use across seven survey waves (1982–1992); base outcome = pill. Predicted probabilities averaged over the sample distribution of education, parity, and wave. LARC/sterilization excluded.

Within-method birth hazard. For each method group $m \in \{\text{pill, barrier, behavioral}\}$, I estimate the linear probability model

$$B_{it} = \alpha_m + \sum_{q=2}^4 \beta_{mq} \mathbb{1}[Q_i = q] + \mathbf{x}'_{it} \boldsymbol{\gamma}_m + \delta_t + \varepsilon_{it}, \quad (3)$$

where $B_{it} \in \{0, 1\}$ equals one if woman i had a live birth in the 12-month window following snapshot wave t , $Q_i \in \{1, 2, 3, 4\}$ is her AFQT quartile (quartile 1 omitted), and the control vector $\mathbf{x}_{it}$ includes education-group indicators at the snapshot year (less than high school, high school graduate, college or more), parity at the snapshot year (0/1/2+ prior live births), and a window-length indicator. Snapshot-year fixed effects δ_t absorb cohort and age-at-wave effects. Standard errors are clustered at the respondent level. A pooled specification adds method-group fixed effects. The identifying restriction is that if the ability gradient operates purely through method selection, fixing the method group eliminates the ability channel, and

Table 1. Within-Method Birth Probability by AFQT Quartile (LPM, 1982–1992)

	(1) Pooled	(2) Pill	(3) Barrier	(4) Behavioral
AFQT Q2	−0.019*** (0.006)	−0.020*** (0.006)	−0.015 (0.015)	−0.016 (0.026)
AFQT Q3	−0.022*** (0.006)	−0.023*** (0.006)	−0.031** (0.015)	0.003 (0.030)
AFQT Q4	−0.023*** (0.006)	−0.028*** (0.006)	−0.015 (0.016)	−0.017 (0.034)
<i>N</i>	16,263	11,406	3,879	978

Notes: LPM. Unit: woman $\times$ snapshot wave (1982–1992). Outcome: any live birth within a fixed 12-month window (July of t through June of $t + 1$) among women using contraception at t , excluding births from explicitly planned pregnancies. Controls: education group, parity, and snapshot-year FE. Pooled column adds method-group FE. SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

$\beta_{mq} = 0$ for all m and q . Figure 4 plots the LPM-predicted birth probability by AFQT quartile within each method group; Table 1 reports the corresponding regression coefficients. The gradient does not disappear within method. In the pooled specification, women in the second AFQT quartile are 1.9 percentage points less likely to have a birth than women in the first quartile ($p < 0.01$), women in the third quartile are 2.2 percentage points less likely ($p < 0.01$), and women in the fourth quartile are 2.3 percentage points less likely ($p < 0.01$).

Within oral contraceptive users, the gradient is steeper: the fourth-quartile coefficient is −2.8 percentage points ($p < 0.01$). This pattern is noteworthy because pill compliance involves a narrow, well-defined behavioral margin—daily ingestion—where cognitive ability can affect consistency even within a homogeneous technology. Within barrier-method users, the gradient is also present: the second-quartile coefficient is −1.5 pp, the third-quartile coefficient is −3.1 pp ($p < 0.05$), and the fourth-quartile coefficient is −1.5 pp, though only the Q3 estimate is statistically significant. For behavioral methods, none of the AFQT coefficients are statistically significant.

Several features of the design make these estimates conservative. The hierarchy assignment places dual users in the most effective tier, concentrating careful users in the pill group and working against finding a gradient within it. Any within-window method switching generates classical measurement error that attenuates the gradient toward zero.

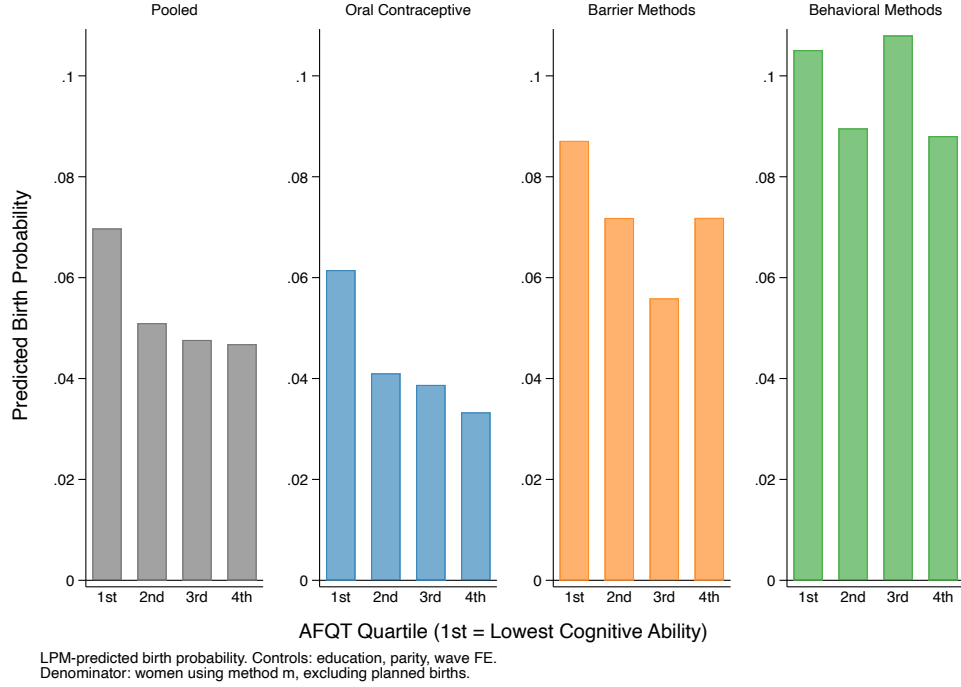


Figure 4. LPM-predicted birth probability by AFQT quartile within each method group. Bars plot LPM predicted probabilities evaluated at each AFQT quartile, averaged over the sample distribution of controls (education group and parity at the snapshot year). Pooled specification adds method-group fixed effects. Denominator: NLSY79 women reporting use of method m at the snapshot-year interview; sample pools seven cohorts ($t \in \{1982, 1984, 1985, 1986, 1988, 1990, 1992\}$), ages 18–35. Outcome: any live birth in a fixed 12-month window (July t to June $t + 1$), excluding explicitly planned pregnancies. Snapshot-year FEs included in all specifications.

Unconditional birth hazard by ability. The within-method test conditions on the method to isolate the per-use effectiveness margin. A complementary question is whether the *unconditional* ability gradient in births—combining both the method-selection and per-use margins—is quantitatively important. I estimate a version of equation (3) on the full population of NLSY79 women aged 18–35 who were not actively trying to conceive, pooled across the same survey waves, without conditioning on the method used. The outcome is again a live birth in the fixed 12-month window. In a separate specification, I restrict the outcome to births that the woman reports as unwanted or mistimed, isolating conceptions that represent failures of fertility control.

Table 2 reports the results. The unconditional birth gradient is declining in AFQT: Q4

Table 2. Birth Probability Unconditional on Contraceptive Method (LPM, 1982–1992)

	(1) Any birth	(2) Unintended birth
AFQT Q2	−0.015*** (0.004)	−0.007** (0.003)
AFQT Q3	−0.020*** (0.004)	−0.016*** (0.003)
AFQT Q4	−0.017*** (0.005)	−0.017*** (0.003)
<i>N</i>	40,617	40,617

Notes: LPM. Denominator: all NLSY79 women aged 18–35 not actively trying to conceive, pooled across seven snapshot cohorts ($t \in \{1982, 1984, 1985, 1986, 1988, 1990, 1992\}$). Unconditional on method choice; includes users and non-users. Column 1: any live birth in a fixed 12-month window (July of t through June of $t + 1$). Column 2: any birth reported as unwanted or mistimed. Controls: education group, parity, and snapshot-year FE. SE clustered by respondent. Reference: AFQT Q1.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

women are 1.7 percentage points less likely to have a birth than Q1 women ($p < 0.01$). The unintended-birth column confirms an identical gradient (-1.7 pp, $p < 0.01$), indicating that the gap reflects differential contraceptive effectiveness rather than differential birth planning. Notably, the within-method Q4 gradient (-2.3 pp) *exceeds* the unconditional gradient (-1.7 pp), confirming that method selection compresses rather than generates the ability gradient. The results are consistent with a genuine per-use effectiveness channel: higher-ability women implement a chosen method more consistently and correctly. The structural model developed in Section 5 formalizes this channel through an ability-dependent effectiveness parameter.

4 Life-Cycle Facts

The reduced-form evidence in Section 3 establishes that the ability gradient in birth rates reflects a per-use effectiveness channel rather than education sorting or method selection. I now document the broader set of life-cycle patterns that the structural model must match. I present the empirical regularities—fertility timing, educational attainment, marriage, and labor-market outcomes by ability group—that serve as estimation targets and discipline the

Table 3. Fertility Timing and Outcomes by Ability Quartile

	Ability Quartile			
	Q1 (Low)	Q2	Q3	Q4 (High)
<i>Panel A: Conditional First-Birth Probability by Age Bin (%)</i>				
14–17	28	16	9	3
18–21	49	38	25	16
22–29	54	53	46	45
<i>Panel B: Age at First Birth and Completed Fertility</i>				
Age at first child	20.14	21.66	23.45	25.56
Married at first birth	0.38	0.56	0.72	0.84
At least one child by age 40	0.87	0.82	0.74	0.72

Notes: Panel A reports conditional first-birth probabilities; denominator is women childless at the start of each age bin. Panel B reports mean age at first birth, share married at first birth, and share with at least one child by age 40.

model’s key channels.⁴

Table 3 reports conditional first-birth probabilities by age bin and AFQT quartile, computed among women childless at the start of each bin. The ability gradient in early first births is steep and attenuates with age: at ages 14–17, 28% of women in the lowest quartile have a first birth compared with 3% in the highest quartile; at 18–21 the gap is 49% versus 16%; by 22–29 it narrows to 54% versus 45%. Mean age at first birth differs by more than five years across quartiles (20.1 to 25.6), and women in the highest quartile are far more likely to be married at their first birth (0.84 vs. 0.38). Yet the share with at least one child by age 40 varies only modestly—from 0.87 in the lowest quartile to 0.72 in the highest—so the pattern reflects predominantly *postponement* rather than avoidance of motherhood. This combination of large early gradients with convergence in completed fertility is a key target for the structural model.

Cognitive ability also predicts educational attainment strongly. Table 4 shows that the relationship is monotone: women in the highest quartile are far less likely to drop out of high school (1% vs. 29%) and far more likely to complete college (52% vs. 4%). Because education affects both the opportunity costs of childbearing and the effectiveness of fertility control (Rosenzweig and Schultz, 1989), these ability–education gradients could in principle fully account for the ability–fertility gradient. The distribution of first births by education closely tracks this sorting: by age 21, roughly 74% of high-school dropouts have had a first

⁴All variables and sample restrictions are defined in Appendix OA.1. Additional descriptive tables on birth timing, marriage, and labor-market outcomes are collected in Online Appendix OA.3.

Table 4. Educational Attainment by Cognitive Ability Quartile

Education outcome	Cognitive Ability (AFQT) Quartile				
	Q1 (Low)	Q2	Q3	Q4 (High)	All
HS dropout	29	9	2	1	10
HS graduate	68	80	75	47	68
College attendance	11	25	41	67	36
College graduate	4	11	23	52	22

Notes: Sample includes women from the NLSY79. Education measured as highest degree completed; college attendance includes those who attended between ages 18–22. Cognitive ability measured using AFQT percentile scores. Entries report column percentages (share of women in each quartile with indicated education level).

birth versus 11% of college graduates (Appendix Table OA.1). Whether the ability gradient operates entirely through education or also through an independent effectiveness channel is the central question the model addresses—and the reduced-form evidence in Section 3 suggests the latter.

The ability gradient extends to labor-market outcomes and marriage patterns that amplify the costs of early childbearing. Higher-ability women face steeper returns to experience—57% log wage growth after five years in the highest quartile versus 24% in the lowest—and accumulate substantially more work experience by any given age, implying larger opportunity costs from career interruptions associated with early births (Appendix Table OA.4). Premarital childbearing is in turn associated with lower marriage rates and lower spousal earnings, especially for less educated women (Appendix Tables OA.2 and OA.3), reinforcing the economic penalty from early births. These patterns collectively discipline the model’s opportunity-cost channel: the wage process, experience accumulation, and marriage market must generate ability-graded incentives to delay childbearing that are quantitatively consistent with the data before any additional effectiveness channel is invoked.

5 Model

I develop a dynamic life-cycle model of fertility timing with imperfect fertility control, in the tradition of models that jointly endogenize schooling, labor supply, marriage, and fertility (Adda et al., 2017; Attanasio et al., 2008; Ejrnæs and Jørgensen, 2020; Keane and Wolpin, 2010). The key innovation is a fertility-control technology in which cognitive ability shifts how

effectively contraceptive effort translates into reduced birth risk. The reduced-form evidence in Section 3 points to a per-use effectiveness channel that the model formalizes by allowing ability to shift the mapping from contraceptive effort to realized birth risk, consistent with the large gaps between perfect-use and typical-use failure rates documented in the medical literature (Black et al., 2010; Trussell, 2011). Prior structural models with imperfect fertility control (Choi, 2017; Ejrnæs and Jørgensen, 2020) do not allow innate ability to shift this mapping (see Section 2).

Costs and benefits of childbearing. Before turning to the formal structure, it is useful to preview how the model captures the economic trade-offs around fertility timing. Each woman can have at most one child. This assumption simplifies the state space but captures the key margin—the timing of the *first* birth—which drives the ability gradient documented in Section 3. A birth delivers two sources of utility: a direct “warm glow” from becoming a parent and utility from child quality that depends on parental monetary investment. Against these benefits, a birth imposes three types of costs. First, a *goods cost*: the child raises the household’s consumption-equivalence scale, so each unit of effective consumption requires more expenditure, and a one-time child investment absorbs resources directly. Second, an *opportunity cost*: a birth during the schooling years raises the psychic cost of continuing education, potentially foreclosing the college option. Even after schooling is complete, a birth raises the disutility of working, reducing labor supply and slowing experience accumulation—a channel that depresses future wages. Third, a *fertility-control cost*: avoiding a birth requires contraceptive effort, which is costly and imperfectly effective.

Although the child resides with the household for only one period (so parental investment is a one-time choice), assets provide an intertemporal margin: households can save in anticipation of a birth, building a buffer that relaxes the resource constraint in the birth period and allows greater child investment. At the same time, a no-borrowing constraint prevents households from borrowing against future income to finance child investment, so the timing of the birth relative to asset accumulation matters. Women who give birth early—before they have accumulated savings or completed education—face a tighter resource constraint and invest less in their children, reinforcing the welfare costs of early fertility. A natural concern with the one-period structure is that it could artificially incentivize delay by rewarding asset ac-

cumulation before birth. Three features of the model limit this force. First, the child-quality technology exhibits diminishing returns to investment, so the marginal gain from additional savings falls quickly. Second, the direct utility from parenthood is received immediately upon birth and is discounted the longer the household waits, creating an incentive to have children sooner. Third, the finite fertility window (ages 14–37) imposes a hard deadline, so excessive delay risks involuntary childlessness.

Because ability raises wages and returns to experience, high-ability women face larger opportunity costs of a birth, giving them stronger incentives to delay. The model also allows education to improve the baseline effectiveness of fertility control—capturing the idea that schooling raises contraceptive knowledge and correct use (Rosenzweig and Schultz, 1989)—so that college-educated women face lower conception risk at any given level of effort. However, the reduced-form evidence in Section 3 shows that opportunity costs and education-related effectiveness alone cannot account for the ability gradient: a substantial gap in birth rates persists within education groups and within contraceptive method. The model therefore incorporates an additional channel—ability-dependent contraceptive effectiveness—that allows cognitive skills to shift the mapping from effort to birth risk beyond what education provides. The nested specification test in Section 7 confirms that shutting down this channel significantly worsens model fit. Importantly, all women in the model value children—the preference for parenthood is common across types—so the ability gradient reflects heterogeneous *timing incentives* and *capacity to control timing*, not heterogeneous preferences for motherhood.

Time is discrete, with each period representing four years. Women enter the model at age 14 with cognitive ability θ and initial assets $a_1 = 0$. They remain fertile through ages 14–37 and can work until age 61. From ages 62 to 78, households are retired and receive Social Security income that depends on educational attainment. The unit of decision-making is the household: before marriage, it is a single-adult unit, and after marriage, it is a two-adult unit that pools income and makes joint decisions. Marriage is absorbing (no divorce).

Contraception is modeled in reduced form as effort that is costly and imperfect. The model abstracts from income uncertainty to focus on the joint determination of fertility timing, schooling, work experience, and marriage.

5.1 State variables, choices, and timing

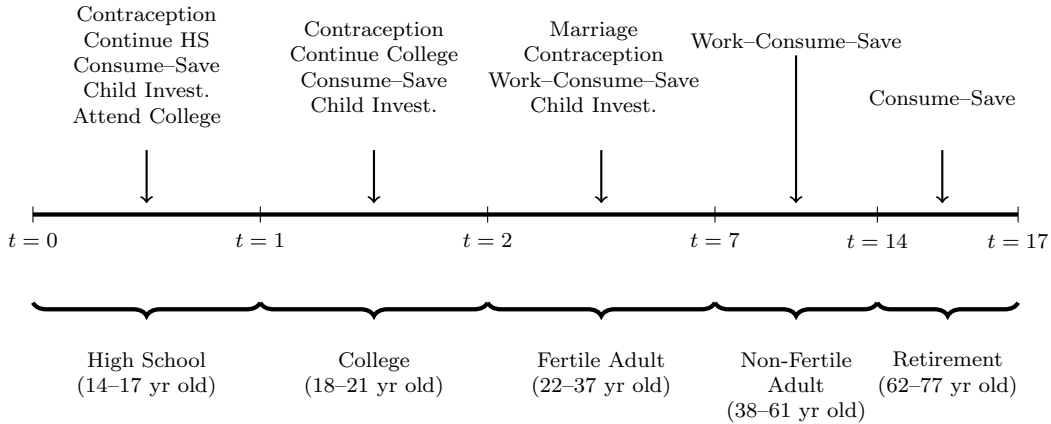
The household state at the beginning of period t is

$$\Omega_{it} = \{a_t, \theta_i, e_t, x_t, m_t, k_t, m_k\},$$

where a_t denotes assets; $\theta_i \in \{1, 2, 3, 4\}$ is the cognitive-ability quartile (1 lowest, 4 highest); $e_t \in \{HSD, HS, C\}$ is education attainment/status; x_t is accumulated labor-market experience; $m_t \in \{0, 1\}$ is marital status; $k_t \in \{1, 2, 3\}$ records childbearing status (1: no prior birth, 2: first birth occurs in period t , 3: first birth occurred in an earlier period); and $m_k \in \{0, 1\}$ records marital status at the first birth (only relevant when $k_t \neq 1$).

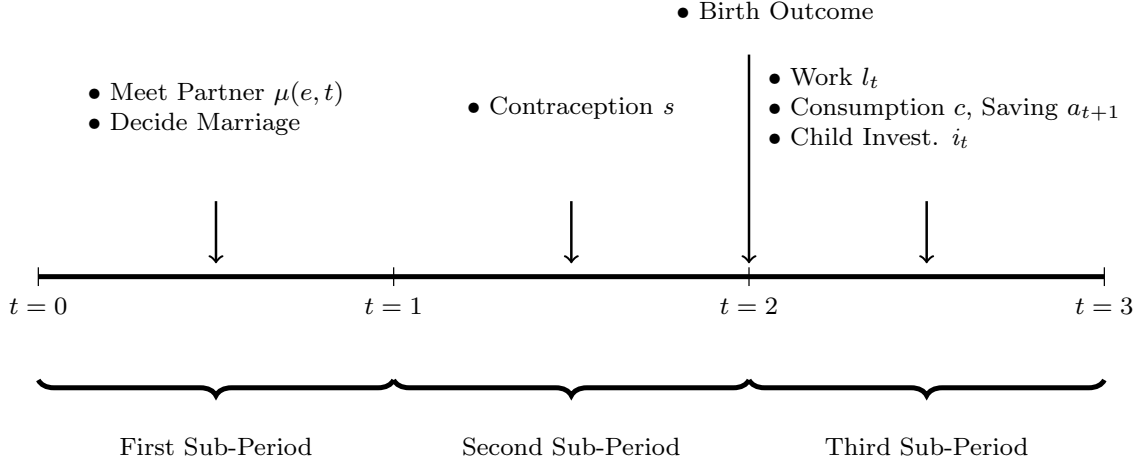
Within-period timing depends on life stage. In fertile ages, single women may meet and accept a partner, childless women choose contraception effort s_t and then face stochastic conception, and households choose labor supply l_t , consumption c_t , saving a_{t+1} , and (if a birth occurs) child investment i_t . In schooling periods, the schooling continuation decision occurs after the fertility outcome. After fertility ends, the problem reduces to a labor-savings problem, and in retirement labor supply is fixed at zero. Figure 5 and Figure 6 summarize the period mapping and within-period sequencing.

Figure 5. Women Attending College Life Cycle



Notes: The figure describes women's life cycle. The life cycle is divided into four stages: (i) teen, (ii) college age, (iii) young adult, and (iv) rest of life. Above the timeline, we show women's decisions in each period.

Figure 6. Childless Women Between Ages 22–37: Within-Period Timing



Notes: Each period is divided into three sub-periods: (i) marriage (if single), (ii) contraception (if childless and fertile), and (iii) labor supply, consumption–saving, and (if a birth occurs) child investment.

In the next subsection, I describe the key Bellman equations that characterize household decisions over the life cycle. Online Appendix [OA.2.1](#) provides the complete set of Bellman equations by life stage (teen, college, young adult, post-fertile, and retirement) and the associated within-period sequencing.

5.2 Dynamic Household Problem and Value Functions

This section presents the recursive household problem and highlights the main Bellman equations. The household makes a sequence of interrelated discrete and continuous decisions—schooling and college entry, marriage, work, saving and consumption, contraceptive effort while fertile and childless, and child investment upon a first birth. To keep the exposition transparent, I organize the recursion into four building blocks that correspond to the within-period timing: (i) marriage, which determines whether resources are pooled; (ii) contraception, which determines first-birth risk when childless; (iii) the working-stage labor–consumption–saving problem, which depends on whether a newborn arrives; and (iv) college entry at the end of adolescence.

Marriage. Marriage plays two economic roles in the model. First, it is a resource-pooling device: married households have access to spousal earnings, which relaxes the budget constraint and lowers the consumption cost of a birth. Second, marriage affects the opportunity

cost of early childbearing through the child-status channel: a married woman who gives birth receives income from her spouse, so the earnings she forgoes by reducing labor supply are partially offset. As a result, the value of avoiding a birth—and hence the incentive to exert contraceptive effort—is lower for married women than for single women with the same ability and education, all else equal. Marriage is endogenous but absorbing (no divorce), so the timing of marriage relative to the first birth is a key determinant of fertility incentives.

If single ($m_t = 0$), the woman meets a potential husband with probability $\mu(e_t, t)$.⁵ Conditional on meeting, she compares continuation values under marriage and singlehood. Let $\Omega_{it}(m)$ denote the state with m_t set to $m \in \{0, 1\}$. Then

$$V_t^1(\Omega_{it}) = \begin{cases} \mu(e_t, t) \max\{V_t^2(\Omega_{it}(1)), V_t^2(\Omega_{it}(0))\} + (1 - \mu(e_t, t)) V_t^2(\Omega_{it}(0)), & \text{if } m_t = 0, \\ V_t^2(\Omega_{it}), & \text{if } m_t = 1. \end{cases}$$

Contraception and first-birth risk. Only childless women choose contraceptive effort, i.e. when $k_t = 1$ and $t \leq T_F$. Let $p_t(\theta_i, e_t, s_t)$ denote the probability of a first birth in period t , decreasing in contraception effort and depending on age, ability, and education. Then

$$V_t^2(\Omega_{it}) = \max_{s_t \geq 0} \left\{ -\phi_s s_t + p_t(\theta_i, e_t, s_t) V_t^{3,j=2}(\Omega_{it}) + (1 - p_t(\theta_i, e_t, s_t)) V_t^{3,j=1}(\Omega_{it}) \right\}.$$

If $k_t \neq 1$ (a first birth already occurred in t or in the past), the household skips contraception:

$$V_t^2(\Omega_{it}) = V_t^{3,j=3}(\Omega_{it}).$$

Working-stage problem with and without a newborn. In fertile ages ($t \leq T_F$), after the fertility realization, the household solves a labor–consumption–saving problem that depends on whether a first birth occurs in period t . Let j index the fertility/child-status outcome: $j = 2$ if a first birth occurs in t (newborn present), $j = 1$ if no birth occurs and the woman remains childless, and $j = 3$ if the woman had a birth in a previous period. The discrete labor choice $l_t \in \{0, 1\}$ is subject to Type-I extreme value shocks. Two parameters govern the disutility of working. The baseline component $\psi_l(t, e)$ varies by age group and

⁵Throughout, I use superscripts on the value function (V^1, V^2, V^3) to index within-period sub-stages; these do not denote time periods.

education level: the age dependence is specified over three broad life stages (ages 14–26, 30–50, and 54–62), and the education dependence captures sorting into occupations with different flexibility and time demands. A second component, $\psi_{lk}(e)$, is an additional disutility of working that applies only when the child is present ($j = 2$). This parameter varies by education and captures the child penalty on labor supply: having a young child raises the cost of working, which leads women to reduce labor supply and forgo experience accumulation, depressing lifetime earnings (Adda et al., 2017). Because $\psi_{lk}(e)$ varies by education, the model allows the child penalty to differ across education groups. Conditional on (Ω_{it}, j) , the ex-ante value for the working stage is

$$V_t^{3,j}(\Omega_{it}) = \mathbb{E}_\varepsilon \left[\max_{l \in \{0,1\}} \{v_t^{3,j}(\Omega_{it}, l) + \sigma_l \varepsilon_t(l)\} \right].$$

If a first birth occurs in t ($j = 2$), the household chooses labor supply, consumption, saving, and one-time child investment. The disutility of working includes both the baseline component and the child penalty:

$$\begin{aligned} v_t^{3,j=2}(\Omega_{it}, l) = & \max_{a_{t+1} \geq 0, c_t \geq 0, i_t \geq 0} \left\{ u(c_t) + [\psi_l(t, e) + \psi_{lk}(e)] \cdot \mathbb{1}_{\{l=1\}} + u_k(i_t) \right. \\ & \left. + \beta \mathbb{E}[V_{t+1}(\Omega_{i,t+1}) \mid \Omega_{it}, l, a_{t+1}, j = 2] \right\} \\ \text{s.t. } & \phi_c(m_t, 1) c_t + a_{t+1} = (1 + r) a_t + y_t(\Omega_{it}, l) - i_t, \\ & x_{t+1} = x_t + \mathbb{1}_{\{l=1\}}. \end{aligned}$$

If no birth occurs ($j = 1$) or the woman is an “older” mother without the child present ($j = 3$), investment is absent, the equivalence scale depends only on marital status, and the child penalty $\psi_{lk}(e)$ does not apply:

$$\begin{aligned} v_t^{3,j}(\Omega_{it}, l) = & \max_{a_{t+1} \geq 0, c_t \geq 0} \left\{ u(c_t) + \psi_l(t, e) \mathbb{1}_{\{l=1\}} + \beta \mathbb{E}[V_{t+1}(\Omega_{i,t+1}) \mid \Omega_{it}, l, a_{t+1}, j] \right\} \\ \text{s.t. } & \phi_c(m_t, 0) c_t + a_{t+1} = (1 + r) a_t + y_t(\Omega_{it}, l), \\ & x_{t+1} = x_t + \mathbb{1}_{\{l=1\}}. \end{aligned}$$

College. At the end of $t = 1$, teens who complete high school ($d = HSG$) draw a Type-I extreme value shock and choose whether to enroll in college at $t = 2$, $d_C \in \{C, NC\}$. Let $v_2^1(\cdot)$ denote the beginning-of-period value at $t = 2$ given the education choice. College enrollment entails a non-pecuniary (“psychic”) entry cost, $\kappa_C(\theta, j)$, capturing the disutility of attending college—e.g., effort costs, adjustment costs, and other non-monetary barriers—which may vary with ability θ and with birth status j . Then

$$V_2^{CD,j}(\Omega_{i2}) = \max_{d_C \in \{C, NC\}} \{v_2^1(\Omega_{i2}; d_C) - \kappa_C(\theta, j) + \sigma_C \varepsilon_2(d_C)\}.$$

Only teens who complete high school face the college-entry decision.

5.3 Preferences, technologies and transfer system

I choose functional forms that are flexible enough to match the joint distribution of fertility timing, schooling, marriage, and labor supply.

Preferences over effective consumption. Utility is CRRA over effective consumption:

$$u(c_t) = \frac{c_t^{1-\rho}}{1-\rho},$$

where ρ is the coefficient of relative risk aversion. Household composition affects the expenditure needed to attain a given c_t . I implement this through an equivalence scale in the budget constraint:

$$\phi_c(m_t, k_t) c_t + a_{t+1} = (1+r)a_t + y_t - \mathbb{1}_{\{k_t=2\}} i_t,$$

so c_t is what enters utility and $\phi_c(m_t, k_t)c_t$ is the required expenditure. I parameterize

$$\phi_c(m_t, k_t) = 1 + \omega_m \mathbb{1}_{\{m_t=1\}} + \omega_{ch} \mathbb{1}_{\{k_t=2\}},$$

where $\omega_m \geq 0$ captures additional needs in a two-adult household and $\omega_{ch} \geq 0$ captures additional needs when a child is present in the household (i.e., a birth occurs in the current period under the one-period-child assumption).

Preferences over child quality. If a first birth occurs, parents choose a one-time monetary investment i_t that increases child “quality.” Parental altruism enters as utility from child outcomes:

$$u^k(i_t) = \omega_0 + \omega_1 i_t^{\omega_2},$$

where ω_0 is baseline utility from having a child, ω_1 scales the marginal value of investment, and $\omega_2 \in (0, 1)$ imposes diminishing returns and ensures an interior investment choice.

College psychic cost. College choices are disciplined by a student allowance w_C , tuition TC , and an ability-dependent psychic cost. Following the structural education literature, I model the psychic cost of college attendance as

$$\kappa_c(\theta, k_t) = \frac{\xi_c}{\theta^{\omega_c}} + \mathbb{1}_{\{k_t > 0\}} \phi_{kbac},$$

where the first term captures the standard finding that psychic costs decline with cognitive ability, and the second term captures the additional burden of attending college while raising a child from a prior birth.⁶ This formulation allows early motherhood to impose lasting costs on human capital accumulation: women who gave birth as teenagers face higher utility costs of subsequent college attendance, reflecting the practical difficulties of balancing childcare with academic demands. I allow continuation (graduate vs. dropout) to be differentially costly when a child is present via an additional cost. High-school continuation/dropout is modeled analogously through a cost wedge that can increase when a birth occurs in the high-school period.

Progressive taxes and transfers. To approximate the U.S. tax-and-transfer system, I adopt the parametric schedule in [Daruich and Fernández \(2024\)](#). Let $\tilde{y}^0$ denote gross annual household income before taxes and transfers (labor earnings, spousal earnings if married,

⁶Psychic (non-pecuniary) costs or tastes for schooling are a standard ingredient in structural models of educational choice (e.g., [Abbott et al., 2019](#); [Keane and Wolpin, 2010](#)). These frameworks typically allow the utility cost of attending school to be heterogeneous (often interpreted as effort/adjustment costs), and they emphasize that unmeasured monetary costs can be observationally equivalent to psychic costs ([Keane and Wolpin, 2010](#)). Consistent with this modeling approach, a large empirical literature shows that both cognitive and noncognitive skills are strong predictors of schooling attainment ([Heckman et al., 2006](#)), and structural estimates often find substantial heterogeneity in the instantaneous utility cost of attending school that is related to latent “school ability”.

schooling allowances when enrolled, or Social Security in retirement). Disposable annual income is

$$\tilde{y} = \lambda(y^0)^{1-\tau} + T(m_t),$$

so the corresponding net-tax function is $\mathcal{T}(\tilde{y}^0, m_t) = \tilde{y}^0 - \tilde{y}$. Progressivity ($\tau > 0$) reduces the sensitivity of after-tax resources to gross income, while $T(m_t)$ captures a reduced-form transfer floor that varies by marital status.

5.4 Fertility control technology

Fertility is stochastic and can be controlled imperfectly through contraceptive effort $s \geq 0$. For a childless woman, the probability of conceiving in model period t depends on age (through an age-group index g), education e , and effort s , while cognitive ability θ affects how effectively effort reduces conception risk.⁷ The specification is motivated by the large gap between “typical use” and “perfect use” failure rates for contraceptive methods, driven primarily by inconsistent or incorrect use (Trussell, 2011), and by the fact that these gaps vary systematically with socioeconomic characteristics (Black et al., 2010; Rosenzweig and Schultz, 1989).

To simplify notation, write $g = g(t)$ and suppress the time index. Let $\lambda_{ge} > 0$ denote the baseline odds of conception for age group g and education e (i.e., risk absent contraceptive effort), and let $\eta_{\theta g} > 0$ capture how ability shifts the effectiveness of effort. The conception probability is

$$p_{ge}(\theta, s) = \left[\lambda_{\max} \cdot \frac{\lambda_{ge} \exp(-\eta_{\theta g} s)}{1 + \lambda_{ge} \exp(-\eta_{\theta g} s)} \right]_{\lambda_{\min}}^{\lambda_{\max}}, \quad (4)$$

where $[x]_{\lambda_{\min}}^{\lambda_{\max}} \equiv \min\{\lambda_{\max}, \max\{\lambda_{\min}, x\}\}$ truncates the risk to lie in $[\lambda_{\min}, \lambda_{\max}]$. I restrict $\eta_{\theta g}$ to be weakly increasing in ability quartiles (implemented via nonnegative increments). The bounds $(\lambda_{\min}, \lambda_{\max})$ ensure imperfect control even at high effort and rule out deterministic fecundity differences across groups.

The technology separates two roles. Baseline fecundity λ_{ge} shifts the level of conception risk at any effort level—a “risk shifter” that varies by age and education. The effectiveness

⁷Related life-cycle models with imperfect fertility control use logit-type mappings for conception risk; see, e.g., Choi (2017); Ejrnæs and Jørgensen (2020). The key distinction in my specification is that ability shifts the effectiveness of effort, not merely its cost or the baseline risk level.

parameter $\eta_{\theta g}$ governs how strongly effort reduces conception risk—a “technology shifter” that varies by ability and age. Ignoring the outer bounds, the comparative statics are

$$\frac{\partial p_{ge}}{\partial s} < 0, \quad \frac{\partial p_{ge}}{\partial \lambda_{ge}} > 0, \quad \frac{\partial p_{ge}}{\partial \eta_{\theta g}} < 0.$$

The third derivative is the source of within-education ability gradients: even if two women in the same education–age cell exert identical effort, the higher-ability woman achieves lower conception risk because her effort is more productive. Moreover, $\eta_{\theta g}$ scales the marginal benefit of effort: $|\partial p/\partial s|$ is increasing in $\eta_{\theta g}$ at low effort levels, so that high-ability women face a steeper risk–effort trade-off in the relevant range.

Figure 7 illustrates. Panel (a) shows conception probability as a function of effort for two ability types with the same baseline risk λ_{ge} . The high-ability type achieves lower conception probability at every positive effort level. Panel (b) shows the marginal benefit of effort. At low effort levels, the high-ability type has a higher marginal benefit, but the curves cross at intermediate levels—reflecting diminishing returns. High-ability women can achieve very low conception risk with moderate effort, leaving little room for further improvement; low-ability women still face substantial risk even at moderate effort, so their marginal benefit remains higher at high effort levels.

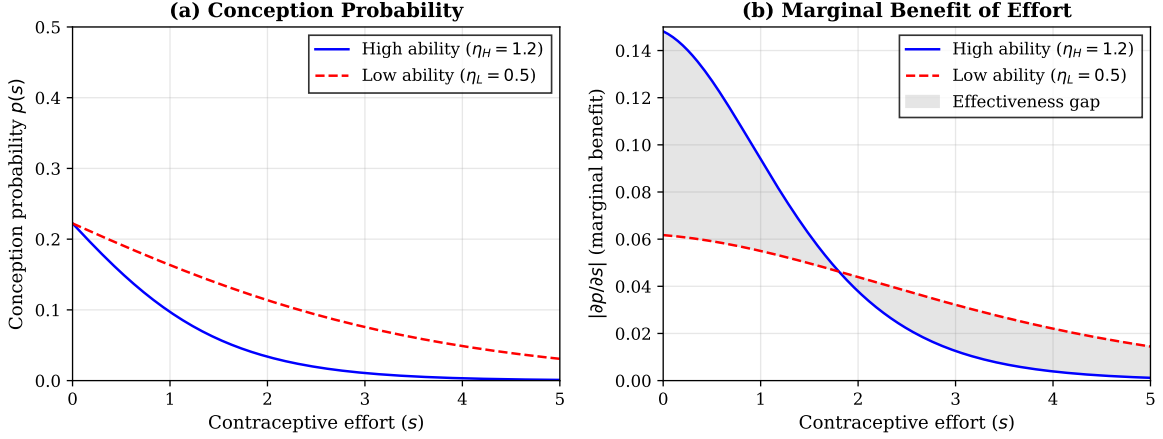
The household chooses effort by equating marginal cost to marginal benefit. The first-order condition is

$$\phi_s = \left| \frac{\partial p_{ge}(\theta, s)}{\partial s} \right| \times \underbrace{(V^{\text{no child}} - V^{\text{child}})}_{\text{value of avoiding birth}}, \quad (5)$$

where ϕ_s is the per-unit cost of effort. Optimal effort s^* is endogenous and varies with ability within every education–age cell, even holding ϕ_s fixed across types. Because the value of avoiding a birth depends on ability through wages, experience returns, and continuation values, two women in the same cell but different ability quartiles choose different effort levels. The model solves for the full equilibrium effort response of each type.

The cost parameter ϕ_s is a normalization: the data identify only the ratio $\eta_{\theta g}/\phi_s$, not each parameter separately. Appendix OA.9 derives the closed-form solution and shows that scaling $(\eta_{\theta g}, \phi_s) \rightarrow (k\eta_{\theta g}, k\phi_s)$ leaves the conception probability at the optimum invariant. Following standard practice (see, e.g., Seshadri and Zhou, 2022), I fix $\phi_s = 0.001$ and let $\eta_{\theta g}$

Figure 7. Conception Technology: Role of Ability-Dependent Effectiveness



Notes: Both panels assume identical baseline conception odds λ_{ge} across ability types. Panel (a): conception probability as a function of contraceptive effort. The high-ability type (solid blue, $\eta_H = 1.2$) achieves lower conception probability than the low-ability type (dashed red, $\eta_L = 0.5$) at any positive effort level. Panel (b): marginal benefit of effort, $|\partial p/\partial s|$. At low effort, the high-ability type has higher marginal benefit; at high effort, the curves cross and the low-ability type has higher marginal benefit. The shaded region shows effort levels where high-ability types have a marginal-benefit advantage.

absorb all ability-related variation in the effort–risk mapping.⁸

Ability therefore affects contraception through two channels. The *effectiveness channel* operates through $\eta_{\theta g}$: higher effectiveness raises $|\partial p/\partial s|$ at low effort levels, making each unit of effort more productive. The *incentive channel* operates through $V^{\text{no child}} - V^{\text{child}}$: higher ability raises wages and experience returns, increasing the value of avoiding a birth and encouraging greater effort. Both channels predict that high-ability women exert more effort and achieve lower conception risk, but they have different implications for identification. The incentive channel operates through opportunity costs that vary with education and can be disciplined using wage and employment data. The effectiveness channel generates ability gradients in birth hazards within education groups that opportunity costs alone cannot explain. Section 6.2 exploits this distinction to separately identify the two channels.

To see why higher wages predict later fertility, note that $V^{\text{no child}} - V^{\text{child}}$ is increasing in the woman’s wage and experience return: a birth raises the disutility of working through

⁸In Seshadri and Zhou’s (2022) formulation, the birth probability takes the logistic form $p(\kappa) = 2\exp(-p_c\kappa)/(1 + \exp(-p_c\kappa))$ and they normalize $p_c \equiv 1$, letting the cost function $\chi(h)$ vary by human capital. This is the mirror normalization: fixing effectiveness and letting cost vary. In the present model, I fix cost and let effectiveness vary. Both normalizations are observationally equivalent; the choice is dictated by the research question, which asks how effectiveness varies with ability.

the child penalty $\psi_{lk}(e)$, which reduces labor supply and forgone experience accumulation, and—if she is still in school—raises the psychic costs of continuing. Women with steeper wage profiles therefore have more to lose from an early birth, so they optimally exert more contraceptive effort and delay childbearing.

The parameter $\eta_{\theta g}$ is best interpreted as a *residual fertility-control channel*: because the model maps directly from effort to live births (rather than to conceptions and then to birth vs. abortion), $\eta_{\theta g}$ captures any ability-correlated factor that affects the mapping from intended contraceptive behavior to realized births, beyond what is explained by education, wages, and marriage. This includes differences in contraceptive compliance, planning capacity, partner negotiation, and potentially abortion access. The within-method analysis in Section 3.3 supports genuine effectiveness differences as a major component: cognitive ability predicts birth hazard even among women using the same method, and method selection runs counter to the gradient (low-ability women disproportionately use the pill). The contribution of the paper is to show that this composite residual channel is quantitatively large and cannot be absorbed by standard opportunity-cost mechanisms.

The model nests the opportunity-cost explanation as a special case. If $\eta_{\theta g} = \eta_g$ for all θ —effectiveness does not vary with ability—then the only channel through which ability affects fertility timing is the incentive channel. In this restricted model, ability gradients in fertility should disappear after conditioning on education, wages, and experience profiles. The nested specification test in Section 7 formally evaluates this restriction.

6 Estimation

This section describes the estimation strategy and the identification of the model’s key mechanisms. Parameters are disciplined in three steps: (i) a set of externally calibrated parameters, (ii) an earnings process estimated outside the structural model, and (iii) the remaining structural parameters estimated internally using the Simulated Method of Moments. I then discuss how the targeted moments identify the education and opportunity-cost channels separately from ability-driven heterogeneity in effective fertility control.

Table 5. Externally Set Parameters

Parameter	Value	Source / Interpretation
Discount factor β_a	0.959 (annual)	Standard value (Adda et al., 2017)
Risk aversion ρ	1.98	CRRA curvature (Adda et al., 2017)
Risk-free rate r_a	0.04 (annual)	Real return (Adda et al., 2017)
College tuition TC	\$10,200	Annual tuition, 2016 \$ (Vandenbroucke, 2023)
Tax progressivity τ	0.18	(Daruich and Fernández, 2024)
Tax scale λ	0.85	
Transfer floor T_S (single)	\$8,634	Annual floor, 2016 \$ (Daruich and Fernández, 2024)
Transfer floor T_C (couple)	\$12,943	
Conception bounds $(\underline{\lambda}, \bar{\lambda})$	(0.05, 0.80)	Imperfect control bounds (Trussell, 2004)
Contraception cost ϕ_s	0.001	Normalization

Notes: Monetary values in 2016 dollars per year. Annual flows converted to four-year model periods as described in text.

6.1 Externally Set Parameters and Earnings Process

Externally set parameters. Table 5 reports parameters fixed outside SMM. I discipline (i) preferences and financial conditions using standard values from the structural life-cycle literature, (ii) policy and institutional objects (tuition, taxes, transfers) using established calibrations, and (iii) biological constraints by imposing bounds on conception probabilities that rule out both perfect control and deterministic fecundity.

Externally estimated earnings process. A key input to the model is the earnings process. I estimate reduced-form earnings profiles in the NLSY79 and use the fitted values to parameterize the model’s deterministic component of earnings as a function of observed states. Specifically, I predict annual real wage-and-salary earnings and treat the fitted profiles as the earnings opportunities faced by women and husbands in each model period. Estimating the earnings process outside the structural model ensures that wage–experience profiles by ability are disciplined by labor-market data alone, preventing the optimizer from distorting these profiles to improve fertility fit (Eisenhauer et al., 2015). Appendix OA.6 provides full details on the estimation sample and specification.

Women’s earnings. Let $\tilde{w}_t^f$ denote predicted annual earnings for women. Earnings depend flexibly on age, education, experience, cognitive-ability quartile, and interactions. To allow earnings to vary systematically with family formation, I also include reduced-form indicators

for marriage and nonmarital first birth:

$$\tilde{w}_t^f = X_t^f \hat{\beta}^f,$$

where X_t^f includes age and age-squared, education and ability indicators, experience and interactions (education $\times$ experience, ability $\times$ experience, education $\times$ ability), and family-formation indicators.

Husbands' earnings. Husbands' earnings are modeled as a reduced-form function of the wife's observed characteristics and marital status at childbirth (capturing assortative mating and marriage selection):

$$\tilde{w}_t^h = X_t^h \hat{\beta}^h,$$

where X_t^h includes age (and a quadratic), the wife's education, and interactions with an indicator for whether the first birth occurs out of wedlock.

6.2 Estimation and Identification

Estimation. I estimate the model by Simulated Method of Moments (SMM). SMM is well-suited for dynamic life-cycle models with discrete choices, unobserved taste shocks, and nonlinear state transitions (Gourieroux et al., 1993; McFadden, 1989; Pakes and Pollard, 1989). As emphasized by Eisenhauer et al. (2015), the finite-sample properties of SMM depend critically on the choice of moments; I therefore include both static moments (levels and shares at given ages) and dynamic moments (transitions and hazards) to sharpen identification of the model's life-cycle dynamics. I choose parameters Θ to minimize the distance between an empirical moment vector $m^{data} \in \mathbb{R}^{118}$ and its model analogue $m^{sim}(\Theta)$:

$$\hat{\Theta} = \arg \min_{\Theta} (m^{sim}(\Theta) - m^{data})' W (m^{sim}(\Theta) - m^{data}),$$

where W is diagonal and each moment is scaled by its empirical magnitude to keep the criterion approximately comparable across outcomes (with a small floor to avoid division by zero).

Moment blocks. The 118 targets are organized into blocks aligned with the model mechanisms. The first block contains 12 moments capturing schooling and early fertility: high school dropout rates with and without a first birth by ages 14–17, college attendance conditional on a first birth by ages 14–17, college attendance by ability quartile, and college graduation conditional on a first birth by ages 18–21. Child investment contributes 2 moments measuring relative investment ratios by mother’s education. Fertility timing is targeted through 35 moments, including the ratio of first- to fourth-quartile birth rates across ages 14–38 and birth rates by age bin and ability quartile. Marriage patterns contribute 17 moments (share married by age bin and education), labor supply contributes 36 moments (employment rates by age bin and education), and contraception use contributes 18 moments (use rates by age bin and education).

All parameters are disciplined by the joint fit across blocks through the model’s cross-equation restrictions; the block structure is a guide to the main sources of identifying variation rather than a claim of one-to-one identification. This approach follows the standard practice in structural life-cycle estimation of organizing moments by the margins they primarily discipline while recognizing that identification ultimately relies on the full set of cross-equation restrictions implied by the model ([Adda et al., 2017](#); [Keane and Wolpin, 1997, 2010](#)).⁹

Identification: separating opportunity costs from fertility control. Cognitive ability affects fertility timing through (i) opportunity costs (schooling, wages, experience accumulation), (ii) marriage-market incentives and spousal resources, and (iii) fertility control. Identification leverages the life-cycle structure: the same latent ability type that matches schooling, labor supply, and marriage outcomes must also rationalize fertility timing, limiting the scope for reallocating fit across margins without degrading non-fertility moments. This logic parallels the identification strategy in [Keane and Wolpin \(2010\)](#), where unobserved heterogeneity (“types”) must simultaneously explain education, work, marriage, fertility, and welfare participation—any reallocation of explanatory burden across margins degrades fit on the other margins.

⁹Although the model is written in terms of a latent conception risk, the empirical targets are defined on first live births; the estimated “conception technology” should therefore be interpreted as a reduced-form birth-producing hazard that matches birth-based hazards in the data.

Key restrictions and mapping to moments. Identification of the fertility-control channel relies on three restrictions that prevent within-education teen birth gradients from being absorbed elsewhere. First, the earnings process is estimated outside the structural SMM step using wage data, so wage–experience profiles by ability are not chosen to improve fertility fit. Second, schooling-cost parameters and education-stage taste-shock scales are disciplined primarily by schooling outcomes—including schooling moments conditional on early fertility—ensuring the model matches the joint distribution of education and ability before attributing residual fertility gradients to fertility control. Third, baseline birth risk $\lambda_h(g, e)$ is common within age–education cells, so within-cell ability gradients in first-birth hazards load on the ability shifter in fertility control rather than on baseline risk differences.¹⁰

Identifying fertility-control parameters. Conditional on the externally set wage environment and the disciplined schooling/marriage/labor-supply blocks, the fertility technology parameters are disciplined by the joint behavior of (a) first-birth hazards by age and ability and (b) contraception use by age and education. In the model, $\lambda_h(g, e)$ shifts baseline risk within an age–education cell, ϕ_s shifts the cost of contraceptive effort (and thus average use), and $\eta_{\theta, g}$ shifts the effectiveness with which a given effort level reduces realized risk. The key identifying variation for $\eta_{\theta, g}$ is therefore the residual within-education ability gradient in early birth hazards once schooling, wages, marriage, and average contraception use are pinned down.

Intuition for identification of $\eta_{\theta, g}$. To build intuition, consider two women with the same education and age but different cognitive ability. Both face the same baseline conception risk λ_{ge} and the same effort cost ϕ_s . However, the high-ability woman faces higher opportunity costs of a birth (higher wages, steeper experience returns), so the value of avoiding a birth, $V^{\text{no child}} - V^{\text{child}}$, is larger for her. The model therefore predicts that the high-ability woman *optimally exerts more contraceptive effort* even without any effectiveness channel—this is the standard opportunity-cost mechanism, and it generates within-education ability gradients in

¹⁰As shown in Section 5, the effort cost ϕ_s is normalized: the FOC quadratic is homogeneous of degree one in $(\eta_{\theta, g}, \phi_s)$, so conception probabilities depend only on their ratio. Importantly, this normalization does *not* restrict effort to be constant across ability groups within an education–age cell. Optimal effort s^* varies across ability quartiles because the value of avoiding a birth, $V^{\text{no child}} - V^{\text{child}}$, depends on ability through wages and continuation values (see Equation 5).

both effort and birth hazards purely through ΔV .

The question is whether this incentive channel *alone* can account for the observed ability gradient in early births. If it can, then restricting $\eta_{\theta g} = \eta_g$ (no ability dependence in effectiveness) should not degrade model fit. But if—after the model has fully optimized effort by each ability type within each education–age cell to match wages, schooling, marriage, and contraception use—there remain systematic ability gradients in birth hazards that the opportunity-cost channel cannot explain, this residual variation identifies $\eta_{\theta g}$. The nested specification test in Section 7.2 shows exactly this: the restricted model, despite allowing effort to respond to ability-specific opportunity costs, cannot match the within-education fertility gradient without ability-dependent effectiveness.

Overidentification and specification checks. The model is overidentified: 118 empirical moments discipline 60 estimated structural parameters (with the earnings process estimated externally), implying 58 overidentifying restrictions. Overidentification is used as discipline rather than as a formal test of a single restriction: I assess whether the model matches fertility hazards without degrading contraception, schooling, marriage, and labor-supply moments. Following Keane and Wolpin (2007), I assess model fit using within-sample comparisons between simulated and empirical moments, and fit improvement across nested model variants.

Nested specification test. A key contribution of this paper is to formally test whether the standard opportunity-cost model—in which ability affects fertility only through schooling, wages, and marriage incentives—can account for the observed ability gradient in early fertility. To implement this test, I estimate a restricted model that imposes $\eta_{\theta, g} = \eta_g$ for all θ : effectiveness does not vary with ability, so the only channel through which ability affects fertility is the incentive channel. Comparing the unrestricted and restricted models provides a direct test of whether ability-dependent fertility control is quantitatively necessary to explain the data.

The restricted model is nested within the unrestricted model, so any deterioration in fit is informative about the empirical relevance of the fertility-control channel. Section 7.2 shows that restricting contraceptive effectiveness to be equal across ability groups nearly doubles the total SSE relative to the full model, with the sharpest deterioration on the birth-and-ability

moments, confirming that this channel cannot be absorbed by opportunity-cost mechanisms alone.

7 Results

Three main findings emerge from the estimated model. First, the model accounts for the sharp ability gradient in early fertility while simultaneously fitting education, marriage, labor supply, and contraception profiles (Section 7.1). Second, this pattern cannot be explained by schooling choices and wage-based opportunity costs alone. Nested specification tests show that restricting contraceptive effectiveness to be equal across ability groups ($\eta = 1$) causes the model to underpredict the ability–fertility gradient by a factor of five; only when effectiveness is allowed to vary by ability does the model match both the steep gradient in birth hazards and the relatively flat gradient in contraception use, providing direct evidence that ability-dependent fertility control is quantitatively necessary (Section 7.2). Third, a policy reducing contraception costs by 7.8%—sufficient to lower teen births by approximately 10%—generates the largest welfare gains for low-ability women (10% consumption-equivalent variation), with gains declining monotonically as ability increases (Section 8.1).

7.1 Model fit

This section evaluates how well the estimated model matches the targeted moments. The estimation uses 118 moments organized into five blocks: fertility timing by ability, education outcomes, marriage, labor supply, and contraception use. I discuss fit for each block in turn, highlighting where the model succeeds and where it falls short.

Throughout the figures, solid lines with circular markers denote model predictions; dashed lines with square markers denote NLSY79 data moments.

7.1.1 Fertility timing by cognitive ability

Figure 8 (panel (a)) compares the cumulative fraction of women who have had a first birth by age and ability quartile. Throughout, Q1 denotes the lowest cognitive ability quartile and Q4 the highest. The model reproduces the sharp ability gradient in the onset of motherhood. Before age 22, where the gradient is steepest and most informative about fertility-control

incentives, the model closely tracks the data for Q3 and Q4; for Q1 and Q2 it slightly underpredicts the cumulative fraction (model: 62.5% and 47.8%, respectively; data: 69.1% and 54.0%). This gradient persists through the twenties: by age 30, over 80% of Q1 women have had a first birth, while Q4 women continue accumulating first births into their thirties, a pattern the model captures well. At older ages (30+), the model slightly overpredicts the ability gradient—predicting somewhat too little convergence across ability groups. For example, the data Q1–Q4 gap narrows to 13.7 percentage points by age 42, while the model gap is 21.6 percentage points.

7.1.2 Schooling and child-related outcomes

Table 6 evaluates whether the model captures the joint distribution of schooling attainment and early fertility. The model successfully reproduces several key patterns. First, it matches the strong association between a teen birth and high school dropout: among women without a first birth by age 14, the dropout rate is 7.0% in the data and 6.8% in the model, while among those with a first birth by age 14, the dropout rate rises to 29% in the data and 29.7% in the model. The model closely reproduces this gap, and correctly captures the qualitative pattern that an early birth is strongly associated with school leaving.

Second, the model captures the steep ability gradient in college attendance. In the data, attendance rises monotonically from 11% in Q1 to 67% in Q4; the model reproduces this gradient with rates of 11.3%, 30.3%, 42.2%, and 48.9% for Q1 through Q4, respectively. The fit is excellent for Q1 and Q3, but the model underpredicts attendance for Q4 (49% vs. 67%). This shortfall is consistent with the model’s abstraction from parental resources, financial aid, and credit constraints—factors that covary strongly with measured ability and disproportionately boost college-going for high-ability youth from advantaged backgrounds.

Third, the model reproduces the negative association between early birth and college outcomes. College attendance conditional on no first birth by age 14 is 41% in the data and 40.8% in the model; conditional on a first birth, it falls to 8% in the data and 7.2% in the model. Similarly, college graduation conditional on attendance and no first birth by age 18 is 62% in the data and 61.5% in the model; conditional on a first birth, it falls to 26% in the data and 25.6% in the model. The model closely reproduces graduation rates for both

groups and accurately predicts the severe penalty of an early birth on college completion.

Finally, child-investment moments are disciplined using external evidence on expenditure gradients from [Caucutt and Lochner \(2020\)](#). The model reasonably matches the college-versus-dropout ratio (3.81 vs. 4.6) and the high-school-versus-dropout ratio (1.97 vs. 1.2).

Table 6. Education Moments: Model vs. Data

Moment	Data	Model	Moment	Data	Model
<i>High School Dropout (Age 14)</i>			<i>College Attendance</i>		
HS Dropout (No Birth by 14)	0.070	0.068	College Attend (No Preg. at 14)	0.410	0.408
HS Dropout (Birth by 14)	0.290	0.297	College Attend (Preg. at 14)	0.080	0.072
<i>College Attendance (Ability)</i>			<i>College Graduation (Given Attendance)</i>		
College Attend (Ability Q1)	0.110	0.113	College Grad (No Preg. at 18)	0.620	0.615
College Attend (Ability Q2)	0.250	0.303	College Grad (Preg. at 18)	0.260	0.256
College Attend (Ability Q3)	0.410	0.422	<i>Child Investment (Relative to HSD)</i>		
College Attend (Ability Q4)	0.670	0.489	Child Inv: HS/HSD Ratio	1.20	1.97
			Child Inv: College/HSD Ratio	4.60	3.81

Notes: Data moments from NLSY79. “Birth” indicates first birth occurred by the specified age. Child investment ratios (expenditure on children relative to high school dropouts) from [Caucutt and Lochner \(2020\)](#).

7.1.3 Labor-market profiles: participation and experience

Figure 8 (panels (b) and (c)) displays labor-force participation and accumulated work experience by age and education.

Labor-force participation (panel b). The model reproduces the education gradient in participation: college graduates participate at rates of 65–75% during prime working ages, high school graduates at 50–56%, and high school dropouts at 20–26%. The model also captures the life-cycle pattern—rising participation in the twenties, a plateau through the fifties, and decline after age 58. The fit is tightest for college graduates and high school dropouts; for high school graduates, the model slightly underpredicts participation at young ages (before age 25) and slightly overpredicts at ages 38–50.

Work experience (panel c). Because experience is a state variable that accumulates with labor supply, fitting experience profiles provides an indirect validation of the model’s dynamic structure. The model closely tracks experience accumulation for all three education groups: by age 55, college graduates have accumulated roughly 22 years of experience, high school

graduates 18–20 years, and dropouts 8–10 years. The slopes are well-matched, indicating that the model correctly captures the rate at which women accumulate experience over the life cycle. This fit is important because experience feeds into wages through the estimated returns to experience, and wages in turn affect the opportunity cost of childbearing.

7.1.4 Marriage: levels and education gradient

Panel (d) displays marriage rates by age and education. The model successfully reproduces the broad life-cycle pattern of marriage: rates rise with age across all education groups and largely converge by the early thirties, consistent with the data. By ages 30–38, the model tracks the data closely for college graduates and high school dropouts (e.g., age 38: college model 84.8% vs. data 80.7%; HSD model 69.1% vs. data 74.4%).

Two more limited discrepancies arise at younger ages and for high school graduates. First, the model somewhat overpredicts marriage rates for high school graduates at older ages—by age 38, the model predicts 96.3% compared to 81.4% in the data. Second, the model slightly overpredicts marriage for college graduates at ages 22–26 (model: 35.7%–64.1% vs. data: 30.8%–56.7%), consistent with the delayed entry of college women into the marriage market; this gap closes by age 30 and is negligible thereafter.

These discrepancies have a natural interpretation. The model features perfect assortative matching on education and a marital surplus that depends solely on partner earnings and consumption smoothing. It does not capture non-pecuniary dimensions of match quality—such as shared values, lifestyle compatibility, or complementarities in home production—nor the insurance value of dual-earner households.

7.1.5 Contraception use

Panel (e) displays contraception use by age and education. The model matches the education gradient in contraception use: college-educated women report higher use than high school graduates, who in turn report higher use than dropouts. The model also captures the declining age profile—contraception use falls from roughly 73%–88% at ages 17–22 to 20%–48% by the late thirties—reflecting the transition from fertility postponement to completed fertility.

For college graduates, the model predicts universal contraception use at ages 18–30, while

the data shows rates of 74%–88%; the model substantially overpredicts college contraception use at young ages. A first consideration mitigates this concern: birth rates among college women are very low in both the data and the model, so the few births that do occur are fully consistent with the residual conception probability embedded in the fertility-control technology. Even at maximum contraceptive effort, the model retains a 5% baseline failure rate reflecting method failure, so the small number of college births is compatible with near-universal use without requiring any mismatch in birth levels.

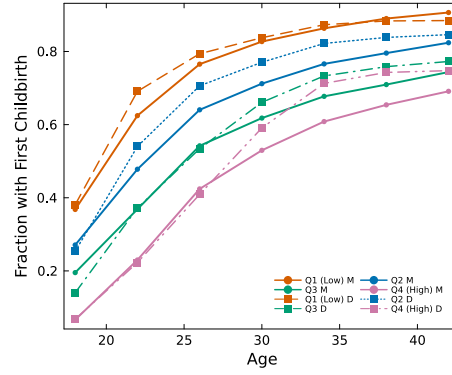
A second consideration concerns the mapping between the model and the data. In the model, any strictly positive effort $s > 0$ is recorded as using contraception, but women in the data may report use only when effort exceeds some higher perceived threshold—for instance, when using a prescription method consistently rather than relying on low-effort alternatives. Under this interpretation, a college woman whom the model assigns low but positive s would be coded as a user in the model but a non-user in the survey. Crucially, the model predicts substantial heterogeneity in effort among college women even when all are recorded as users, so the apparent overprediction of the use rate partly reflects a difference in thresholds rather than a fundamental mismatch. The fit on college birth rates—the outcome directly governed by the effort choice—remains close to the data, which is the margin that matters for the fertility-control channel identified in the paper.

7.2 Nested Specification Test: Is Ability-Dependent Fertility Control Necessary?

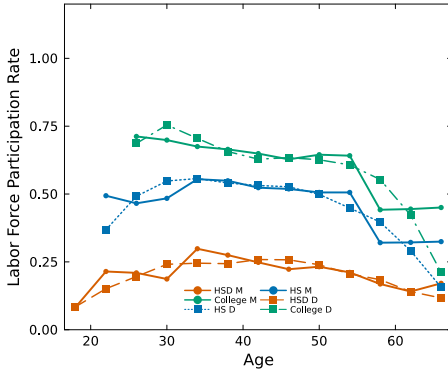
The model allows cognitive ability to affect fertility timing through two channels: (i) opportunity costs, operating through schooling choices and wage profiles, and (ii) fertility-control effectiveness, operating through the parameter $\eta_{\theta g}$ that governs how effort maps into conception risk. A natural question is whether the second channel is empirically necessary, or whether the standard opportunity-cost channel alone can explain the ability gradient in fertility timing. To address this question, I estimate three nested specifications that progressively relax restrictions on the fertility-control technology:

- (1) **Baseline (Age only):** Baseline conception risk λ_g varies by age group g only, and the effectiveness parameter is normalized to $\eta = 1$ for all individuals. Any ability gradient

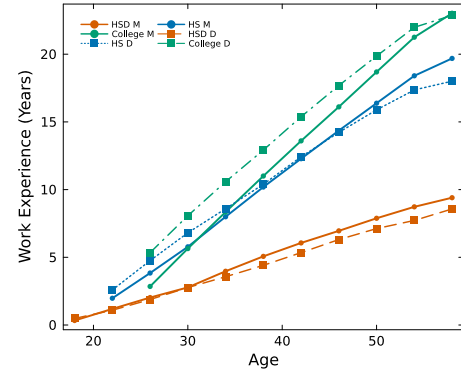
Figure 8. Model Fit: Fertility Timing, Labor Market, Marriage, and Contraception



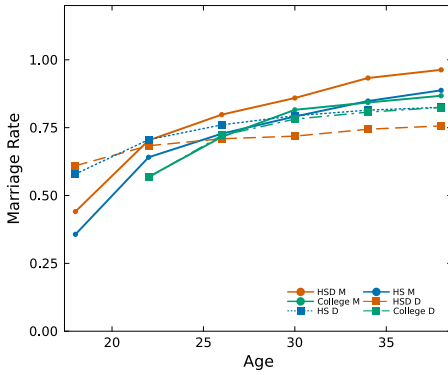
(a) Cumulative first births by age and ability quartile



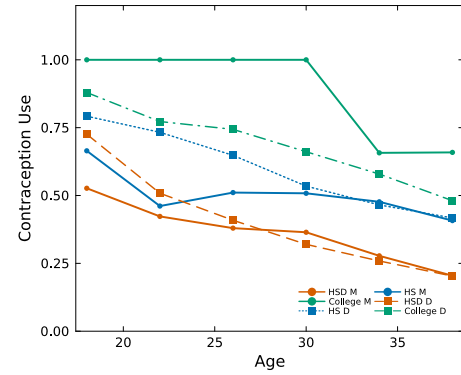
(b) Labor-force participation by education



(c) Accumulated work experience by education



(d) Marriage rates by education



(e) Contraception use by education

Notes: Panel (a) reports the cumulative fraction of women who have had a first birth by each age, separately by AFQT ability quartile (Q1 = lowest, Q4 = highest). Panels (b)–(e) report outcomes by age and completed education (HSD = high school dropout, HS = high school graduate, College = college graduate). Solid lines with circular markers show model predictions; dashed lines with square markers show NLSY79 data moments. “M” denotes model; “D” denotes data.

in fertility timing must arise entirely from differences in schooling choices, wages, and marriage prospects.

(2) Baseline + Education Heterogeneity: Baseline conception risk λ_{ge} varies by age and education, but effectiveness remains $\eta = 1$ for all individuals. Ability gradients can arise from opportunity costs and from sorting into education groups with different baseline risk, but not from within-education differences in how effectively effort reduces conception risk.

(3) Full Model (+ Ability in Effectiveness): Baseline risk λ_{ge} varies by age and education, and the effectiveness parameter $\eta_{\theta g}$ is allowed to vary by ability quartile. This specification allows ability to directly shift how effort translates into reduced conception risk, generating within-education ability gradients in birth hazards even when observed contraception use is similar.

The key restriction in specifications (1) and (2) is $\eta = 1$ for all individuals: a unit of contraceptive effort produces the same reduction in conception risk regardless of cognitive ability. Specification (3) relaxes this restriction, allowing high-ability individuals to achieve greater risk reduction per unit of effort ($\eta_{\theta,H} > \eta_{\theta,L}$). If the opportunity-cost channel alone can explain the data, specifications (1) and (2) should fit the fertility-timing moments as well as (3). If, instead, the data require ability-dependent effectiveness, specification (3) should substantially outperform (1) and (2)—particularly on the within-education ability gradient in birth hazards.

To assess model fit, I use the normalized sum of squared errors (SSE),

$$\text{SSE}(\hat{\vartheta}) = \sum_i \left(\frac{m_i - m_i(\hat{\vartheta})}{m_i} \right)^2,$$

where m_i are empirical moments and $m_i(\hat{\vartheta})$ are their model counterparts under parameter vector $\hat{\vartheta}$. I decompose the total SSE into contributions from each moment block to diagnose which margins drive the fit improvement.

Results. Table 7¹¹ reports SSE decompositions across the three specifications. Three patterns emerge.

First, adding education heterogeneity in baseline risk (column 2) yields only modest improvement in total fit (11%) and improves fit on the contraception-use moments (SSE falls from 0.65 to 0.56). However, it worsens fit on the birth-and-ability block (SSE rises from 1.32 to 1.64, a 24% deterioration). This occurs because, with η fixed at 1, the model can generate some ability gradient through sorting into education groups with different baseline risk, but this is insufficient to match the steep within-education gradient in teen birth hazards.

Second, allowing η to vary by ability (column 3) generates a substantial improvement in total fit (51% reduction in SSE relative to baseline). The improvement is especially broad-based: the birth-and-ability block improves by 78%, the labor-market block improves by 64%, and the education outcomes block improves by 48%. The contraception-use block worsens relative to baseline (SSE rises from 0.65 to 0.95, a 46% deterioration), reflecting that the model now attributes a larger share of the ability gradient to effectiveness differences rather than effort differences.

Third, the bottom panel of Table 7 reports the model-implied correlation between first-birth probability and ability by age group. The data show a strong negative correlation (approximately -0.26 for ages 14–17 and -0.27 for ages 18–21), indicating that high-ability women are substantially less likely to have a first birth at young ages. Specifications (1) and (2) overpredict the magnitude of this gradient: the implied correlations are -0.34 and -0.35 for ages 14–17, respectively, and -0.31 and -0.32 for ages 18–21. Despite generating correlations of the right sign, the restricted models achieve this through mechanisms that distort the birth-and-ability SSE block, as shown in Panel A. The full model (specification 3) closely replicates the data correlations (-0.26 for ages 14–17 and -0.29 for ages 18–21) while simultaneously delivering the largest overall reduction in SSE.

The key finding is that fixing $\eta = 1$ —that is, restricting contraceptive effectiveness to be the same across ability groups—prevents the model from jointly matching the observed ability gradient in fertility timing and the overall structure of birth hazards across moment blocks. Specifications (1) and (2) can produce correlations between ability and birth probability

¹¹For details on the estimated parameters, consult Appendix OA.8.

of the right sign, but do so at the cost of large misfit on the birth-and-ability moment block (SSE of 1.32–1.64 vs. 0.30 in the full model) and total SSE nearly twice as large as specification (3). Only when η is allowed to vary by ability does the model achieve the substantial improvement in overall fit—a 51% reduction in total SSE—that identifies ability-dependent fertility-control effectiveness as a quantitatively important and distinct channel from the standard opportunity-cost mechanism.

Consolidating the identification argument. The case for a genuine effectiveness channel rests on three complementary layers of evidence. *First*, the nested specification test establishes *existence*: the opportunity-cost-only model, despite allowing effort to respond optimally to ability-specific wages and continuation values, cannot match the within-education fertility gradient (SSE on the birth–ability block rises from 0.30 to 1.32–1.64). *Second*, the descriptive evidence in Section 3.3 provides *model-free support*: women with the same education, age, and contraceptive method but different AFQT have significantly different birth hazards, and the within-method Q4 gradient (−2.3 pp) exceeds the unconditional gradient (−1.7 pp). This pattern cannot be rationalized by differential method choice or differential incentives to exert effort among women already using the same method—it points to genuine per-use effectiveness differences. *Third*, the policy counterfactuals in Section 8.1 provide *robustness to interpretation*: LARC programs, which bypass user effort entirely, target the effectiveness channel directly regardless of the exact composition of $\eta_{\theta g}$. The finding that these programs generate the largest welfare gains for Q1 women is robust to whether $\eta_{\theta g}$ primarily reflects cognitive adherence, planning capacity, partner dynamics, or some combination thereof—because LARCs eliminate the scope for all user-dependent components simultaneously.

8 Counterfactual Experiments and Policy Implications

8.1 Welfare from a Cost-Reduction Policy

The estimated model implies substantial heterogeneity in fertility-control frictions across women. A central implication of the ability-dependent effectiveness parameter $\eta_{\theta g}$ is that policies reducing early births should generate differential welfare gains across groups. To

Table 7. Nested Specification Test: Model Fit Decomposition

	Specification		
	(1) Baseline ($\eta = 1$)	(2) + Education Het. in λ ($\eta = 1$)	(3) Full Model (+ Ability in η)
<i>Panel A: Sum of Squared Errors by Moment Block</i>			
Total SSE	6.82	6.06	3.34
Birth $\times$ Ability	1.32	1.64	0.30
Education outcomes	0.70	0.79	0.37
Marriage	0.78	0.81	0.66
Labor supply	2.93	2.39	1.06
Contraception use	0.65	0.56	0.95
<i>Panel B: Fit Improvement Relative to Baseline (%)</i>			
Total	—	11	51
Birth $\times$ Ability	—	−24	78
Education outcomes	—	−12	48
Marriage	—	−4	15
Labor supply	—	18	64
Contraception use	—	14	−46
<i>Panel C: Corr. of First-Birth Probability with Ability</i>			
Ages 14–17 (<i>Data</i> : −0.26)	−0.34	−0.35	−0.26
Ages 18–21 (<i>Data</i> : −0.27)	−0.31	−0.32	−0.29
Ages 22–29 (<i>Data</i> : −0.07)	−0.18	−0.19	−0.23
Ages 14–29 (<i>Data</i> : −0.24)	−0.18	−0.19	−0.23

Notes: Columns (1)–(3) compare three nested specifications. (1) λ varies by age only; $\eta = 1$. (2) λ varies by age and education; $\eta = 1$. (3) λ varies by age and education; η varies by ability quartile. Panel A reports SSE by moment block: $SSE = \sum_i [(m_i - \hat{m}_i)/m_i]^2$. Panel B reports $100 \times (1 - SSE_j/SSE_1)$ for $j \in \{2, 3\}$; negative values indicate worse fit. Panel C reports the model-implied correlation between first-birth probability and ability quartile; data moments are italicized.

quantify these effects in economically interpretable units, I evaluate a policy counterfactual that reduces early births (defined as a first birth before age 22) by 10%—a target anchored in observable outcomes rather than arbitrary parameter perturbations. The model then reveals the contraception-cost reduction required to achieve this target and the resulting welfare gains by group.

Welfare is reported in consumption-equivalent variation (CEV): the permanent proportional increase in consumption that would make agents indifferent between the baseline and counterfactual environments. For group g , I compute

$$\tau_g = 100 \left[\left(1 + \frac{\bar{U}_g^{cf} - \bar{U}_g^{base}}{\bar{W}_g^{c,base}} \right)^{\frac{1}{1-\rho}} - 1 \right],$$

Table 8. Behavioral Responses and Welfare Gains from Reducing Early Births

	Contraception		Effort		Early Preg.			
Ability	Base	Policy	Base	Policy	Base	Policy	Δ	CEV
<i>Panel A: Contraception and Fertility (Ages 14–22)</i>								
Q1 (Low)	69.2	72.0	1.39	1.81	58.9	51.2	−7.7	+10
Q2	76.9	78.6	1.87	2.09	45.5	41.1	−4.4	+4
Q3	80.9	82.1	1.94	2.09	36.1	33.1	−3.0	+3
Q4 (High)	87.1	87.3	1.33	1.36	22.5	22.1	−0.4	+0
All	78.6	80.0	1.63	1.84	40.7	36.8	−3.9	+4
	College		HS Dropout					
Ability	Base	Policy	Base	Policy				
<i>Panel B: Education at Age 22 (%)</i>								
Q1 (Low)	11.4	12.5	24.6	23.2				
Q2	20.8	21.6	11.9	10.7				
Q3	23.7	24.5	8.5	7.8				
Q4 (High)	26.8	26.8	4.1	3.9				

Notes: Policy reduces contraception cost ϕ_s by 7.8%, lowering early births from 40.7% to 36.8%. Panel A: “Contraception” = extensive margin (% using any); “Effort” = intensive margin (0–1 scale); “Early Birth” = ever had a first birth before age 22 (%); Δ = change in pp. CEV = consumption-equivalent variation (%), measuring welfare gain as permanent increase in consumption.

where $\bar{U}_g^{base}$ and $\bar{U}_g^{cf}$ denote average lifetime utility in baseline and counterfactual simulations, respectively, and $\bar{W}_g^{c,base}$ is average discounted consumption-utility in baseline. A value of $\tau_g = 2$ indicates that the policy is worth a 2% permanent increase in consumption for that group.

Table 8 reports the behavioral responses and welfare gains by ability quartile. The policy targets an approximate 10% reduction in early births (from 40.7% to 36.8%), achieved through a 7.8% decrease in the contraception cost parameter ϕ_s . The results reveal considerable heterogeneity: welfare gains are largest for Q1 women (+10% CEV) and decline monotonically across the ability distribution, reaching essentially zero for Q4 (+0% CEV), with intermediate gains for Q2 (+4%) and Q3 (+3%).

Figure 9 summarizes the CEV results by ability and education. The welfare pattern is monotonically decreasing in ability: gains are largest for Q1 (+10% CEV) and fall to zero for Q4 (+0% CEV). By education, high school dropouts gain the most (+10%), followed by high school graduates (+4%) and college graduates (+1%). This pattern reflects the interaction between baseline early birth risk, behavioral responsiveness to cost reductions, and the opportunity cost of an early first birth, and carries important policy implications.

Why do Q1 women gain the most? The large welfare gain for Q1 women reflects three reinforcing channels. First, Q1 women face the highest baseline early birth rate (58.9%), creating substantial scope for welfare improvement from any meaningful fertility reduction. Second, Q1 women exhibit the largest behavioral response to the cost reduction: the extensive margin of contraception rises 2.8 percentage points (69.2% to 72.0%) and effort intensity increases by 30% (1.39 to 1.81), generating a 7.7 percentage point reduction in early births—the largest of any group. Third, reduced early births generate meaningful educational gains for Q1 women: college attendance rises from 11.4% to 12.5% and high school dropout rates fall from 24.6% to 23.2%. The compounded lifetime value of averted early births—through improved education, higher wages, and better consumption paths—is large precisely because Q1 women face substantial penalties from early childbearing, both in terms of foregone schooling and reduced lifetime earnings.

Why do Q4 women gain so little? The near-zero welfare gain for Q4 women is equally instructive. Three features of the model combine to produce this result.

First, the behavioral response channel is attenuated. Table 8 shows that Q4 women exhibit minimal changes in contraceptive behavior: the extensive margin rises only 0.2 percentage points (87.1% to 87.3%) and effort intensity increases negligibly (1.33 to 1.36). The resulting reduction in early births is only 0.4 percentage points—compared to 7.7 pp for Q1.

Second, Q4 women already face a low baseline early birth rate (22.5%) and high contraceptive use (87.1%). Their fertility timing is already well-aligned with human capital investment, so the marginal welfare value of further reductions in early births is small.

Third, the absence of educational gains reinforces the welfare stagnation. College attendance for Q4 women remains unchanged at 26.8% under the policy, reflecting that their educational trajectories are not constrained by fertility timing at the margin. The ability-dependent effectiveness parameter $\eta_{\theta g}$ implies that Q4 women already translate contraceptive effort into fertility control very efficiently; a reduction in the cost of effort generates little additional behavioral change for this group.

Policy implications. Policies that reduce the utility cost of contraception disproportionately benefit the most disadvantaged women. In the model, ϕ_s captures the disutility cost of

exerting contraceptive effort—the psychic, time, and hassle costs of consistent method use—rather than monetary expenditure. A 7.8% reduction in this utility cost generates the largest welfare gains for Q1 women—who face the highest baseline early birth rates—and essentially no gain for Q4 women, who are already near their optimal fertility path. The same policy has diminishing returns up the ability distribution: by the time women reach Q4, baseline early birth risk is low enough and existing contraceptive use is high enough that cost reductions generate no meaningful behavioral or welfare response.

This finding aligns with evidence that typical-use failure rates for user-dependent methods vary sharply by socioeconomic status, even within method (Sundaram et al., 2017), and with Demographic and Health Surveys evidence that the poorest women face higher failure-related unintended birth risk (Bradley et al., 2019). The large perfect–typical use gap for pills versus the much smaller gap for IUDs Trussell et al. (2018) highlights the user-effort margin captured by the model. Consistent with this mechanism, the age gradient in unintended birth risk is pronounced for pills/patch/ring but not for IUDs/implants (Winner et al., 2012).

In the language of the model, these patterns reflect the mechanism captured by $\eta_{\theta g}$: user-dependent methods require consistent effort that translates differentially into reduced conception risk across women, while LARCs decouple effectiveness from user effort. For Q4 women, further reductions in effort costs generate negligible gains because they already use contraception effectively. For Q1 women, the gains from cost reductions are large precisely because they have the most room to improve their fertility outcomes. Effective interventions for the most disadvantaged women can therefore work through two complementary channels: reducing the utility cost of effort (lowering ϕ_s , as in this counterfactual) or shifting the technology of contraception toward methods where the gap between perfect and typical use is negligible. Evidence from the latter channel supports the model’s predictions: Colorado’s Family Planning Initiative, which expanded LARC access, reduced teen birth rates by approximately 6% overall and by larger margins in higher-poverty counties (Lindo and Packham, 2017); in the Contraceptive CHOICE Project, early birth rates among participants who received no-cost LARCs with comprehensive counseling were approximately 79% lower than national rates among sexually experienced young women (Secura et al., 2014). Both channels operate most powerfully at the bottom of the ability distribution, consistent with

the model’s prediction that Q1 women are the primary beneficiaries of contraception-cost policies.

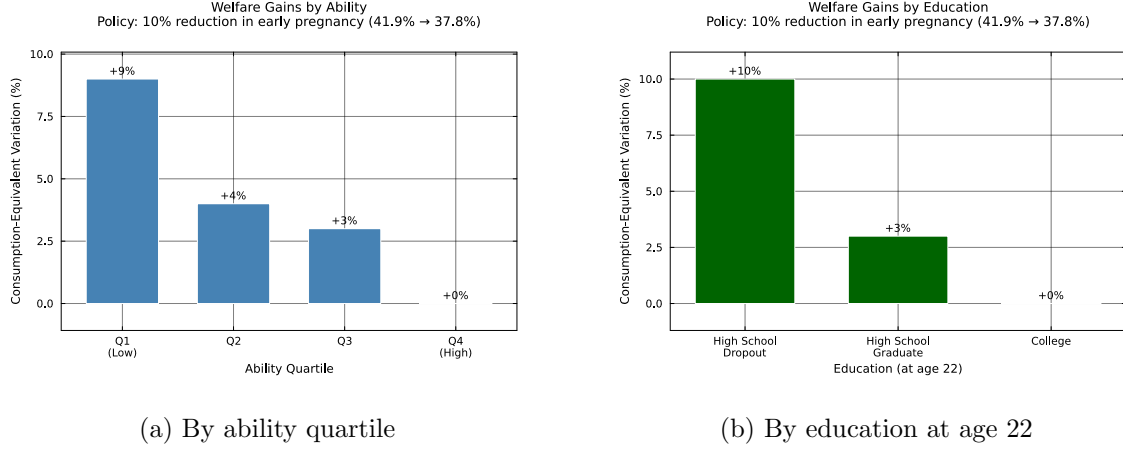


Figure 9. Consumption-Equivalent Welfare Gains from a 10% Reduction in Early Births

Notes: Each bar reports group-average CEV (percent). Panel (a): Q1 (lowest ability) through Q4 (highest). Panel (b): HSD = high school dropout, HS = high school graduate, College = some college or more. Education measured at age 22. Early birth is defined as a first birth before age 22.

8.2 Decomposing the Teen Fertility–Schooling Gradient

Teen motherhood is strongly associated with low educational attainment, but this correlation is difficult to interpret causally. Adolescents who become teen mothers differ systematically from their peers along many dimensions—family background, prior achievement, discount rates, and expectations—so naive comparisons may overstate the causal impact of childbearing on education. Consistent with this concern, [Hotz et al. \(2005\)](#) use miscarriages as an instrument and find limited long-run effects of teen births on completed schooling. Similarly, [Levine and Painter \(2003\)](#) use propensity-score matching and conclude that a substantial share of the raw association reflects selection rather than causation, though meaningful negative effects on college attendance remain.

This paper takes a different approach. Rather than attempting to identify a single “causal effect of teen childbearing,” I use the estimated structural model to decompose the teen fertility–schooling gradient into distinct channels and ask: when we observe that early childbearers have low educational attainment, how much reflects barriers to effective fertility control versus how much reflects low returns to schooling that both depress education and

raise early fertility?

This decomposition matters for policy. If the gradient is primarily driven by fertility-control frictions, then improving access to contraception should reduce teen births and raise schooling through the channel documented in Section 8.1. If instead the gradient reflects low schooling returns, then contraception interventions may reduce teen births without substantially affecting educational attainment—and the appropriate policy response is to address the underlying barriers to education directly.

8.2.1 Counterfactual Design

I conduct three counterfactual experiments that isolate specific channels by equalizing ability-related margins while holding other features of the environment fixed:

1. **Equalize fertility-control environment:** Give all women the contraception parameters (baseline risk λ_{ge} , effectiveness $\eta_{\theta g}$, and effort costs ϕ_s) of the highest-ability group, holding fixed college costs and wage profiles. This experiment asks: how much of the fertility–schooling gradient would close if low-ability women faced the same fertility-control technology as high-ability women?
2. **Equalize schooling and earnings opportunities:** Give all women the college cost schedule and wage profile of the highest-ability group, holding fixed the fertility-control environment. This experiment asks: how much would teen fertility fall if low-ability women faced the same returns to education and labor market opportunities as high-ability women?
3. **Equalize both margins:** Simultaneously equalize fertility control, college costs, and wage profiles. This experiment captures the total effect and allows assessment of complementarities between channels.

8.2.2 Results

Table 9 reports percentage changes in key outcomes relative to the baseline economy. Three findings emerge.

Table 9. Counterfactual Experiments: Decomposing the Fertility–Schooling Gradient

Outcome	Counterfactual		
	(1) Equalize Fertility Control	(2) Equalize Educ./Wages	(3) Equalize Both
<i>Panel A: Percentage Change Relative to Baseline</i>			
College attendance	+14.6	+13.0	+31.4
First births by age 18	−50.2	+12.7	−55.2
First births by age 22	−36.4	+7.1	−42.0

Notes: Each column equalizes parameters to highest-ability quartile values. (1) Fertility-control parameters (λ_{ge} , $\eta_{\theta g}$, ϕ_s). (2) College costs and wage profile. (3) Both fertility-control and education/wage parameters. “First births by age X ” = cumulative share with first birth by age X .

Fertility-control frictions are a first-order driver of early fertility. Column 1 shows that equalizing the fertility-control environment to the high-ability level reduces first births by age 18 by 50.2% and first births by age 22 by 36.4%. These are large effects: half of teen births in the baseline economy would not occur if low-ability women had access to the same effective fertility control as their high-ability peers. This finding corroborates the welfare results in Section 8.1 and confirms that ability-dependent effectiveness is quantitatively important for early fertility outcomes.

Improved fertility control generates meaningful spillovers to education. The same counterfactual (Column 1) increases college attendance by 14.6% relative to baseline. This spillover arises endogenously: when early births are avoided, women who would otherwise have dropped out or foregone college now find it optimal to continue their education. The mechanism operates through the career-cost channel—avoiding early childbirth preserves the option to accumulate human capital and experience—and through the direct effect of not having child-rearing responsibilities during the college years.

Schooling opportunities raise education but do not reduce early fertility. Column 2 isolates the education-and-earnings channel. Equalizing college costs and wage profiles to the high-ability level increases college attendance by 13.0%—roughly comparable to the effect of equalizing fertility control alone. Notably, however, this intervention does not reduce early fertility: first births by age 18 rise by 12.7% and first births by age 22 rise by 7.1%. The intuition is that higher returns to schooling attract women toward college investment, but for

women in the lower ability quartiles whose binding constraint is contraceptive effectiveness rather than foregone wages, equalizing schooling opportunities without improving fertility control cannot prevent early conception. When college-going becomes more attractive but the fertility-control technology remains unchanged, the net effect on early births is slightly positive.

Complementarities. Column 3 combines both channels. When fertility control and schooling opportunities are simultaneously equalized, college attendance rises by 31.4%, first births by age 18 fall by 55.2%, and first births by age 22 fall by 42.0%. The reduction in early fertility is driven almost entirely by the fertility-control channel, since equalizing schooling alone slightly increases early births (Column 2). For college attendance, however, the channels are complementary: the combined gain (31.4%) substantially exceeds each channel in isolation (14.6% and 13.0%, respectively). The intuition is as follows: reducing early births increases the pool of women who can benefit from improved schooling opportunities, while improved schooling opportunities raise the value of avoiding early births. Together these forces amplify the educational response beyond what either policy achieves alone.

8.2.3 Policy Implications

These counterfactuals have three implications for policy design.

First, interventions that improve access to effective fertility control—such as subsidized long-acting reversible contraceptives (LARCs), improved sex education, or expanded access to family planning services—can generate large reductions in teen births. The 50.2% reduction in first births by age 18 under the fertility-control counterfactual is comparable in order of magnitude to the effects documented in evaluations of LARC programs: the Contraceptive CHOICE Project found that teen birth rates among participants were roughly 78% lower than national rates for sexually experienced teens (Secura et al., 2014). Because the NLSY79 cohort’s reproductive years largely predate the diffusion of LARCs, the model’s counterfactuals should be interpreted as quantifying the scope for effectiveness-shifting interventions in an environment dominated by user-dependent methods. In a contemporary environment where LARCs are more widely available, the residual ability gradient in contraceptive effectiveness may be smaller; however, typical-use failure rates for pills and condoms remain substantial

and socioeconomically graded (Sundaram et al., 2017), and user-dependent methods continue to account for the majority of contraceptive use in the U.S.

A caveat is warranted when comparing these magnitudes. The LARC evaluations are not randomized experiments: women who enroll in programs offering free contraception may be more motivated to avoid pregnancy than the general population, so the estimated program effects likely reflect both the true causal effect of LARCs and positive selection into take-up. My counterfactual, by contrast, holds preferences fixed and varies only the fertility-control technology, isolating the technology channel. That my estimate (50.2%) is smaller than the CHOICE Project’s (78%) is consistent with this interpretation: the CHOICE estimate includes positive selection, while mine does not. Despite these methodological differences, the comparison is informative because LARCs operate through the same mechanism my model emphasizes: they largely eliminate the scope for user error by providing contraception that does not require daily compliance or correct use. In the language of my model, LARCs raise η —the effectiveness with which effort translates into reduced conception risk—toward its upper bound. The CHOICE Project’s dramatic reductions in unintended pregnancy thus provide external validation that the effectiveness channel is quantitatively important, even if the precise magnitudes are not directly comparable.

Second, reducing teen births do generate meaningful increases in educational attainment, but the spillover is partial. The 14.6% increase in college attendance under the fertility-control counterfactual implies that roughly one-quarter to one-third of the fertility–schooling gradient reflects a causal effect of childbearing on education. The remainder reflects selection: women who become teen mothers would have had lower educational attainment even absent the birth, because they face higher schooling costs and lower returns.

Third, the most effective strategy for jointly improving fertility and education outcomes is to address both margins simultaneously. Policies that only target fertility control will reduce teen births substantially but leave underlying educational barriers in place. Policies that only improve schooling opportunities will raise attainment but, without complementary improvements in fertility control, may produce little reduction—or even a slight increase—in early fertility. Bundled interventions—for example, comprehensive programs that combine contraception access with educational support, mentoring, or college preparation—have the

potential to achieve the largest joint gains.

9 Conclusion

This paper asks whether the standard economic channels emphasized in life-cycle models—schooling choices and wage-based opportunity costs—can fully explain why women with higher cognitive ability delay first births. In a nationally representative U.S. cohort (NLSY79), the data show a steep negative relationship between adolescent cognitive ability and early fertility that attenuates with age: at ages 14–17, 28% of women in the lowest AFQT quartile have a first birth compared with 3% in the highest; by ages 22–29 the gap narrows sharply. These patterns coexist with strong ability gradients in schooling attainment, marriage, and labor-market returns, motivating a framework in which these outcomes are jointly determined.

A sequence of reduced-form exercises establishes that the ability gradient cannot be attributed entirely to education sorting or method selection. Within education groups, higher-ability women have significantly lower birth hazards: among high-school graduates, Q4 women are 2.0 pp less likely to give birth in a 12-month window than Q1 women ($p < 0.01$), and the gradient is even steeper for unintended births among college graduates (3.1 pp, $p < 0.01$). A multinomial logit for method choice shows that higher-ability women sort toward more effective methods—the predicted probability of pill use falls from 76% in Q1 to 60% in Q4, with a corresponding rise in barrier methods. Yet conditioning on the method actually used does not eliminate the gradient: within each method group, higher-ability women still have lower birth rates, and the within-method ability gradient exceeds the unconditional one. This pattern points to a per-use effectiveness channel that operates beyond both education and method selection.

To interpret these findings, I develop and estimate a dynamic life-cycle model in which women make decisions over schooling, marriage, fertility, labor supply, savings, and contraceptive effort. The key innovation is a fertility-control technology in which education shifts baseline conception risk—capturing the idea that schooling raises contraceptive knowledge and correct use—while cognitive ability shifts how effectively contraceptive effort translates into reduced birth risk. The model is estimated by the simulated method of moments to

jointly match fertility timing by ability, education outcomes, marriage profiles, labor supply, and contraception use.

The estimated model delivers three main results. First, it accounts for the sharp ability gradient in early first-birth hazards and the subsequent convergence with age, consistent with postponement rather than avoidance of motherhood. Second, opportunity costs alone cannot rationalize the data: nested specification tests show that allowing cognitive ability to directly shift fertility-control effectiveness is necessary to match the joint set of moments. Restricting contraceptive effectiveness to be equal across ability groups causes the model to underpredict the ability–fertility gradient by a factor of five. Third, differences in effective fertility control generate heterogeneous welfare effects: policies that improve contraceptive effectiveness produce the largest welfare gains for Q1 women—the most disadvantaged group—with gains declining monotonically to near zero for Q4 women.

The counterfactual experiments yield three policy lessons. First, equalizing fertility-control effectiveness to high-ability levels reduces first births before age 18 by 50% and raises college attendance by 15%, but equalizing schooling opportunities alone does *not* reduce early fertility—the binding constraint is effective fertility control, not educational incentives. Second, combining both channels generates the largest joint gains (college attendance up 31%, early births down 55%). Third, cost-reduction policies generate sharply heterogeneous welfare effects: a 10% reduction in teen fertility yields welfare gains of approximately 10% of lifetime consumption for Q1 women but near-zero for Q4 women. Technology-shifting interventions such as LARC programs—which decouple effectiveness from user effort—could generate even larger gains for the most disadvantaged women (Section 8.1).

The paper contributes a quantified mechanism linking cognitive ability to fertility timing through heterogeneity in effective fertility control, disciplined by a model that jointly matches fertility, schooling, marriage, labor supply, and contraception profiles. The reduced-form evidence—an ability gradient that survives within education groups and within contraceptive method—combined with the structural estimation establishes that the binding constraint for disadvantaged women is the capacity to control fertility, not the incentive to do so. Interventions targeting this margin generate the largest welfare gains precisely for the most disadvantaged group.

References

- Brant Abbott, Giovanni Gallipoli, Costas Meghir, and Giovanni L Violante. Education policy and intergenerational transfers in equilibrium. *Journal of Political Economy*, 127(6):2569–2624, 2019.
- Jérôme Adda, Christian Dustmann, and Katrien Stevens. The career costs of children. *Journal of Political Economy*, 125(2):293–337, 2017.
- Anna Aizer, Paul J. Devereux, and Kjell G. Salvanes. Grandparents, moms, or dads? why children of teen mothers do worse in life. *Journal of Human Resources*, 57(6):2012–2047, 2022.
- Mathilde Almlund, Angela L. Duckworth, James J. Heckman, and Tim Kautz. Personality psychology and economics. In Eric A. Hanushek, Stephen Machin, and Ludger Woessmann, editors, *Handbook of the Economics of Education*, volume 4, pages 1–181. Elsevier, 2011. doi: 10.3386/w16822.
- Diego Amador. The consequences of abortion and contraception policies on young women’s reproductive choices, schooling and labor supply. Technical Report 2017-43, Universidad de los Andes, Facultad de Economía, CEDE, June 2017. URL <https://ssrn.com/abstract=2987367>.
- Orazio Attanasio, Hamish Low, and Virginia Sánchez-Marcos. Explaining changes in female labor supply in a life-cycle model. *American Economic Review*, 98(4):1517–1552, 2008.
- Martha J. Bailey. More power to the pill: The impact of contraceptive freedom on women’s life cycle labor supply. *Quarterly Journal of Economics*, 121(1):289–320, 2006.
- Martha J Bailey, Vanessa Wanner Lang, Alexa Prettyman, Iris Vrioni, Lea J Bart, Daniel Eisenberg, Paula Fomby, Jennifer Barber, and Vanessa Dalton. How costs limit contraceptive use among low-income women in the u.s.: A randomized control trial. Working Paper 31397, National Bureau of Economic Research, June 2023. URL <http://www.nber.org/papers/w31397>.
- Gary S Becker. An economic analysis of fertility. In *Demographic and economic change in developed countries*, pages 209–240. Columbia University Press, 1960.
- Gary S Becker. *A treatise on the family: Enlarged edition*. Harvard university press, 1991.
- Gary S Becker and H Gregg Lewis. On the interaction between the quantity and quality of children. *Journal of political Economy*, 81(2, Part 2):S279–S288, 1973.
- Yoram Ben-Porath. Fertility response to child mortality: micro data from israel. *Journal of Political Economy*, 84(4, Part 2):S163–S178, 1976.
- Kirsten I Black, Surabhi Gupta, Andrew Rassi, and Ali Kubba. Why do women experience untimed pregnancies? a review of contraceptive failure rates. *Best Practice & Research Clinical Obstetrics & Gynaecology*, 24(4):443–455, 2010.

- Sarah E K Bradley, Chelsea B Polis, Akinrinola Bankole, and Trevor Croft. Global contraceptive failure rates: Who is most at risk? *Studies in Family Planning*, 50(1):3–24, 2019.
- Stephen G Bronars and Jeff Grogger. The economic consequences of unwed motherhood: Using twin births as a natural experiment. *The American Economic Review*, pages 1141–1156, 1994.
- Elizabeth M Caucutt and Lance Lochner. Early and late human capital investments, borrowing constraints, and the family. *Journal of Political Economy*, 128(3):1065–1147, 2020.
- Elizabeth M. Caucutt, Nezih Guner, and John A. Knowles. Why do women wait? matching, wage inequality, and the incentives for fertility delay. *Review of Economic Dynamics*, 5(4): 815–855, 2002. doi: 10.1006/redy.2002.0196.
- Sekyu Choi. Fertility risk in the life cycle. *International economic review*, 58(1):237–259, 2017.
- Flavio Cunha and James J. Heckman. The technology of skill formation. *American Economic Review*, 97(2):31–47, 2007. doi: 10.1257/aer.97.2.31.
- Diego Daruich and Raquel Fernández. Universal basic income: A dynamic assessment. *American Economic Review*, 114(1):38–88, 2024.
- Alessandro Di Nola, Georgi Kocharkov, Jan Mellert, and Haomin Wang. Teenage childbearing and the welfare state. *Macroeconomic Dynamics*, 29:e31, 2025.
- Matthias Doepke, Anne Hannusch, Fabian Kindermann, and Michèle Tertilt. The economics of fertility: A new era. In *Handbook of the Economics of the Family*, volume 1, pages 151–233. Elsevier, 2023. doi: 10.1016/bs.hefam.2023.01.003.
- Philipp Eisenhauer, James J. Heckman, and Stefano Mosso. Estimation of dynamic discrete choice models by maximum likelihood and the simulated method of moments. *International Economic Review*, 56(2):331–357, 2015.
- Mette Ejrnæs and Thomas H. Jørgensen. Family planning in a life-cycle model with income risk. *Journal of Applied Econometrics*, 35(5):567–586, 2020. doi: 10.1002/jae.2766.
- Eduardo Fe, David Gill, and Victoria Prowse. Cognitive skills, strategic sophistication, and life outcomes. *Journal of Political Economy*, 130(10):2643–2704, 2022.
- Lawrence B Finer and Mia R Zolna. Declines in unintended pregnancy in the united states, 2008–2011. *New England journal of medicine*, 374(9):843–852, 2016.
- Margherita Fort, Nicole Schneeweis, and Rudolf Winter-Ebmer. Is education always reducing fertility? evidence from compulsory schooling reforms. *Economic Journal*, 126(595):1823–1855, 2016.
- Marco Francesconi. A joint dynamic model of fertility and work of married women. *Journal of Labor Economics*, 20(2):336–380, 2002.

- Claudia Goldin and Lawrence F Katz. The power of the pill: Oral contraceptives and women's career and marriage decisions. *Journal of political Economy*, 110(4):730–770, 2002.
- Christian Gourieroux, Alain Monfort, and Eric Renault. Indirect inference. *Journal of Applied Econometrics*, 8(S):S85–S118, 1993. doi: 10.1002/jae.3950080507.
- James J Heckman and Stefano Mosso. The economics of human development and social mobility. *Annu. Rev. Econ.*, 6(1):689–733, 2014.
- James J. Heckman and James R. Walker. The relationship between wages and income and the timing and spacing of births: Evidence from swedish longitudinal data. *Econometrica*, 58(6):1411–1441, 1990.
- James J. Heckman, Jora Stixrud, and Sergio Urzua. The effects of cognitive and noncognitive abilities on labor market outcomes and social behavior. *Journal of Labor Economics*, 24(3):411–482, 2006.
- V. Joseph Hotz and Robert A. Miller. An empirical analysis of life cycle fertility and female labor supply. *Econometrica*, 56(1):91–118, 1988.
- V Joseph Hotz, Susan Williams McElroy, and Seth G Sanders. Teenage childbearing and its life cycle consequences exploiting a natural experiment. *Journal of Human Resources*, 40(3):683–715, 2005.
- Fedor Iskhakov, Thomas H Jørgensen, John Rust, and Bertel Schjerning. The endogenous grid method for discrete-continuous dynamic choice models with (or without) taste shocks. *Quantitative Economics*, 8(2):317–365, 2017.
- Rachel K. Jones and Jenna Jerman. Population group abortion rates and lifetime incidence of abortion: United States, 2008–2014. *American Journal of Public Health*, 107(12):1904–1909, 2017.
- Michael P. Keane and Kenneth I. Wolpin. The career decisions of young men. *Journal of Political Economy*, 105(3):473–522, 1997.
- Michael P. Keane and Kenneth I. Wolpin. Exploring the usefulness of a nonrandom holdout sample for model validation: Welfare effects on female behavior. *International Economic Review*, 48(4):1351–1378, 2007.
- Michael P. Keane and Kenneth I. Wolpin. The role of labor and marriage markets, preference heterogeneity, and the welfare system in the life cycle decisions of black, hispanic, and white women. *International Economic Review*, 51(3):851–892, 2010.
- Melissa S Kearney and Phillip B Levine. Subsidized contraception, fertility, and sexual behavior. *The review of Economics and Statistics*, 91(1):137–151, 2009.
- Melissa S Kearney and Phillip B Levine. Why is the teen birth rate in the united states so high and why does it matter? *Journal of Economic Perspectives*, 26(2):141–166, 2012.
- Melissa S. Kearney and Phillip B. Levine. Investigating recent trends in the u.s. teen birth rate. *Journal of Health Economics*, 41:15–29, 2015. doi: 10.1016/j.jhealeco.2015.01.003.

- Melissa S. Kearney and Phillip B. Levine. Why is fertility so low in high income countries? Working Paper 33989, National Bureau of Economic Research, 2025. Forthcoming, *Journal of Economic Literature*.
- Melissa S. Kearney, Phillip B. Levine, and Luke Pardue. The puzzle of falling US birth rates since the Great Recession. *Journal of Economic Perspectives*, 36(1):151–176, 2022.
- David I Levine and Gary Painter. The schooling costs of teenage out-of-wedlock childbearing: Analysis with a within-school propensity-score-matching estimator. *Review of Economics and Statistics*, 85(4):884–900, 2003.
- Jason M Lindo and Analisa Packham. How much can expanding access to long-acting reversible contraceptives reduce teen birth rates? *American Economic Journal: Economic Policy*, 9(3):348–376, 2017.
- Justin McCrary and Heather Royer. The effect of female education on fertility and infant health: Evidence from school entry policies using exact date of birth. *American Economic Review*, 101(1):158–195, 2011.
- Daniel McFadden. A method of simulated moments for estimation of discrete response models without numerical integration. *Econometrica*, 57(5):995–1026, 1989. doi: 10.2307/1913621.
- Amalia R Miller. Motherhood delay and the human capital of the next generation. *American Economic Review*, 99(2):154–158, 2009. doi: 10.1257/aer.99.2.154.
- William D Mosher. Contraceptive practice in the United States, 1982–1988. *Family Planning Perspectives*, 22(5):198–205, 1990.
- Kelly Musick, Paula England, Sarah Edgington, and Nicole Kangas. Education differences in intended and unintended fertility. *Social Forces*, 88(2):543–572, 2009.
- Derek A. Neal and William R. Johnson. The role of premarket factors in black–white wage differences. *Journal of Political Economy*, 104(5):869–895, 1996. doi: 10.1086/262045.
- Milagros Onofri, Inés Berniell, Raquel Fernández, and Azul Menduñña. Understanding Latin America’s fertility decline: Age, education, and cohort dynamics. SSRN Working Paper 6275390, 2026.
- Ariel Pakes and David Pollard. Simulation and the asymptotics of optimization estimators. *Econometrica*, 57(5):1027–1057, 1989. doi: 10.2307/1913622.
- Linda J Piccinino and William D Mosher. Trends in contraceptive use in the United States: 1982–1995. *Family Planning Perspectives*, 30(1):4–10, 46, 1998.
- Ferdinando Regalia, José-Víctor Ríos-Rull, and Jacob Short. What accounts for the increase in the number of single households? Manuscript, 2019. URL <https://economics.sas.upenn.edu/system/files/rrs2019-singlehh.pdf>.
- Mark R Rosenzweig and T Paul Schultz. Schooling, information and nonmarket productivity: Contraceptive use and its effectiveness. *International Economic Review*, 30(2):457–477, 1989.

- Gina M Secura, Tessa Madden, Colleen McNicholas, Jennifer Mullersman, Christina M Buckel, Qihong Zhao, and Jeffrey F Peipert. Provision of no-cost, long-acting contraception and teenage pregnancy. *New England Journal of Medicine*, 371(14):1316–1323, 2014.
- Ananth Seshadri and Anson Zhou. Intergenerational mobility begins before birth. *Journal of Monetary Economics*, 2022.
- Michelle Sheran. The career and family choices of women: A dynamic analysis of labor force participation, schooling, marriage, and fertility decisions. *Review of Economic Dynamics*, 10(3):367–399, 2007.
- Aparna Sundaram, Barbara Vaughan, Kathryn Kost, Akinrinola Bankole, Lawrence Finer, Susheela Singh, and James Trussell. Contraceptive failure in the united states: Estimates from the 2006–2010 national survey of family growth. *Contraception*, 95(2):150–158, 2017.
- James Trussell. Contraceptive failure in the united states. *Contraception*, 70(2):89–96, 2004.
- James Trussell. Contraceptive failure in the united states. *Contraception*, 83(5):397–404, 2011.
- James Trussell, Abigail R A Aiken, Elizabeth Micks, and Katherine A Guthrie. Contraceptive efficacy. In Robert A Hatcher, Anita L Nelson, James Trussell, Carrie Cwiak, Patricia Cason, Michael S Policar, Alison Edelman, Abigail R A Aiken, Jeanne Marrazzo, and Deborah Kowal, editors, *Contraceptive Technology*. Ayer Company Publishers, New York, 21st edition, 2018.
- Wilbert Van der Klaauw. Female labour supply and marital status decisions: A life-cycle model. *The Review of Economic Studies*, 63(2):199–235, 1996.
- Guillaume Vandenbroucke. The return on investing in a college education. *The Federal Reserve Bank of St. Louis*, <https://www.stlouisfed.org/publications/regional-economist/2023/mar/return-investing-college-education>, pages 379–400, 2023.
- Robert J. Willis. A new approach to the economic theory of fertility behavior. *Journal of Political Economy*, 81(2, Part 2):S14–S64, 1973. doi: 10.1086/260152.
- Brooke Winner, Jeffrey F Peipert, Qihong Zhao, Christina Buckel, Tessa Madden, Jenifer E Allsworth, and Gina M Secura. Effectiveness of long-acting reversible contraception. *New England Journal of Medicine*, 366(21):1998–2007, 2012.
- Kenneth I Wolpin. An estimable dynamic stochastic model of fertility and child mortality. *Journal of Political economy*, 92(5):852–874, 1984.
- Michael A. Woodley of Menie, Jonatan Sanger, and Gerhard Meisenberg. No relationship between abortion numbers and maternal cognitive ability. *Personality and Individual Differences*, 104:47–52, 2017.

Appendices

Online Appendix (Not for Publication)

OA.1 Data Construction and Variable Definitions

This appendix documents the data cleaning and construction of the variables used in the paper.

OA.1.1 Data source and cohort coverage

The NLSY79 follows a nationally representative cohort of people born 1957–1964 who were ages 14–22 at the first interview in 1979. Interviews are annual from 1979–1994 and biennial thereafter, with rich topical modules covering schooling, labor market outcomes, family formation, and fertility.

OA.1.2 Panel structure and alignment to model time

I construct an annual panel indexed by individual i and calendar year t and compute age at interview as:

$$\text{Age}_{it} = t - \text{BirthYear}_i.$$

Birth year is inferred from the respondent’s age at baseline and used to impute age in years when this variable is missing.

Because the structural model uses four-year periods starting at age 14, I map annual observations into four-year periods:

$$\text{AgeBin}_{it} = 14 + 4 \left\lfloor \frac{\text{Age}_{it} - 14}{4} \right\rfloor,$$

yielding periods 14–17, 18–21, 22–25, ... When a model object is defined at the period level (e.g., employment, experience, fertility hazard), I aggregate annual measures within the period using the rules described below.

OA.1.3 Global cleaning conventions and special codes

NLSY variables commonly use negative values to encode nonresponse and survey routing (e.g., refusal, don't know, valid skip, non-interview). I apply the following conventions before constructing analysis variables:

1. **Invalid / nonresponse codes:** Values < 0 are treated as missing unless they have a structural interpretation (e.g., “no spouse” for spouse income).
2. **Structural zeros:** Variables that are economically meaningful zeros (e.g., spouse income when no partner is present) are explicitly set to 0 rather than missing.
3. **Deflation:** Nominal dollar amounts are converted to real 2016 dollars using CPI-based deflators merged by calendar year.

OA.1.4 Cognitive ability

I use the AFQT measure available in the NLSY79 created score files. Observations with invalid AFQT codes (negative values) are dropped. I then form within-sample quartiles of the AFQT distribution ($q \in \{1, 2, 3, 4\}$), which serve as the ability measure used throughout the empirical moments and wage estimation.

OA.1.5 Education

Education is measured as highest grade completed (HGC) and mapped into three mutually exclusive groups:

$$\text{HSD} : \text{HGC} < 12, \quad \text{HSG} : 12 \leq \text{HGC} < 16, \quad \text{COL} : \text{HGC} \geq 16.$$

To reduce spurious year-to-year reporting variation, I use the individual-specific maximum of HGC observed over the panel.

Additionally, I construct an indicator for college attendance between ages 18 and 22, defined as

$$\mathbf{1} \{ \exists t \text{ s.t. } 18 \leq \text{Age}_{it} \leq 22 \text{ and } \text{HGC}_{it} > 12 \},$$

i.e., equal to one if the respondent reports completing more than 12 years of schooling at any interview conducted when she is ages 18–22, and zero otherwise.

OA.1.6 Fertility and pregnancy histories

First birth timing. I use the created child-birth-date variables for the first child (month/year) to define:

$$\text{AgeAtFirstBirth}_i = \text{BirthYearChild1}_i - \text{BirthYear}_i,$$

setting $\text{AgeAtFirstBirth}_i = 99$ for women with no recorded birth in the observation window.

Wantedness and contraception. To discipline moments on pregnancy intentions and contraceptive behavior, I construct pregnancy-level indicators using the fertility and contraception modules and then aggregate them to the model’s four-year periods.

(i) *Wantedness.* For each pregnancy p of woman i , let $\text{Wanted}_{ip} \in \{0, 1\}$ indicate whether the respondent reports that the pregnancy was wanted at the time of conception.¹²

(ii) *Contraception at conception.* For each pregnancy p , define

$$\text{NoContraception}_{ip} \equiv \mathbb{1}\{\text{no contraceptive method reported at the time of conception}\},$$

where “contraceptive method” includes any reported method (e.g., pill, condom, IUD, or rhythm/withdrawal). Invalid/nonresponse codes are set to missing.

(iii) *Contraceptive method type (Q9-65 battery).* The within-method analysis in Section 3.3 uses the Q9-65 battery, which records the specific contraceptive methods a woman reports currently using at each survey wave. Methods are classified into three groups ordered by typical-use efficacy:

- *Oral contraceptives (pill):* Q9-65 method index 001.
- *Barrier methods:* condom (002), foam (003), jelly/cream (004), suppository (005), diaphragm (006), sponge, cervical cap.

¹²When the survey distinguishes mistimed from unwanted pregnancies, I code $\text{Wanted}_{ip} = 0$ for both categories and report robustness separating the two. Responses coded as “don’t know”, “refused”, or “valid skip” are treated as missing.

- *Behavioral methods*: rhythm/calendar (012), withdrawal (013), natural family planning (011).

IUD (008), female sterilization (009), and male sterilization (010) are excluded: IUD uptake was negligible in the NLSY79 cohort following the Dalkon Shield recall, and sterilization removes a woman from the risk pool irrespective of ability. When a woman reports multiple methods simultaneously, she is assigned to the most effective tier (pill > barrier > behavioral). Women who report abstinence (015) as their *only* method are excluded from the behavioral group, since they have no biological exposure; women reporting abstinence combined with rhythm or withdrawal remain, as they have a fallible backup method.

The within-method birth outcome is constructed from child birth-date records rather than from the self-reported contraceptive-failure variable (Q9-79). For each woman $\times$ snapshot-wave observation, a fixed 12-month window (July of year t through June of year $t + 1$) is imposed, and a birth is flagged if any child’s birth date falls within this window. Births associated with explicitly wanted new pregnancies (effort category 3: Q9-78 = 0 and Q9-81 = 1 or Q9-80 = 1) are excluded so that the outcome captures the model’s birth hazard $\pi(e, \eta, \varphi)$. This approach avoids the restrictive definition embedded in Q9-79, which classifies most imperfect-use failures as “deliberate cessation” and yields annual failure rates an order of magnitude below typical-use benchmarks.

Mapping to the model. The model features a period-level contraception choice that applies to women who are at risk of conception. Accordingly, the targeted moments are constructed as at-risk non-use rates:

$$\Pr(\text{NoContraception}_{it} = 1 \mid \text{AtRisk}_{it} = 1, \text{period } b, \theta_i, \text{Educ}_{it}),$$

where $\text{AtRisk}_{it} = 1$ indicates that the woman is fertile and has not yet had a first birth. In the model, $\text{AtRisk}_{it} = 1$ corresponds to periods in which the woman is in the fertile stage and first birth has not yet occurred, so the model-implied moments are computed over the same at-risk set.

OA.1.7 Marriage and partner outcomes

Marital status. Marital status is defined annually using marriage start/end dates:

$$\text{Married}_{it} = \mathbb{1}\{t \in [\text{MarriageStart}_i, \text{MarriageEnd}_i)\},$$

treating an open-ended marriage (missing end date with a valid start date) as ongoing.

Partner earnings and work. Partner wage-and-salary income is taken from spouse/partner earnings modules when available. “No spouse” codes are set to 0; invalid negative codes are treated as missing. Partner weeks worked and hours worked are used for partner employment definitions in the wage-process estimation.

OA.1.8 Labor market outcomes: hours, earnings, employment, experience

Annual hours. Annual hours are constructed from the Work History and Weekly Status files, yielding (i) total annual hours and (ii) annual weeks worked. These variables are documented in the NLSY79 topical guides.

Annual earnings. I use annual wage-and-salary income (respondent and spouse/partner) deflated to 2016 dollars.

Interpolation and internal consistency checks. Because annual earnings can exhibit missingness and occasional spurious zeros in years with positive hours, I implement two consistency checks before estimation and aggregation: (1) set annual earnings to 0 when annual hours are 0; (2) treat earnings as missing in years with very low hours (but nonzero) when earnings are recorded as zero, and linearly interpolate earnings over time within individual (only across years with valid neighboring information). This procedure is designed to reduce measurement-error spikes while preserving low earnings when corroborated by low hours.

Employment and experience. A woman-year is classified as employed if it satisfies: (i) at least 26 weeks worked; (ii) average weekly hours > 20 ; and (iii) real annual wage-and-salary income of at least \$10,500 (2016 dollars). I then define annual experience as $\text{ExpYear}_{it} = \mathbb{1}\{\text{employed}\}$ and cumulative experience as $\text{CumExp}_{it} = \sum_{\tau \leq t} \text{ExpYear}_{i\tau}$.

OA.2 The Model

OA.2.1 Environment, timing, and state space

Time is discrete in four-year periods. I index periods by $t \in \{1, \dots, T\}$, with decisions made for $t = 1, \dots, T - 1$ and terminal period $T = 17$ (age 78), in which agents consume all remaining resources and die. Fertility is feasible through ages 14–37, i.e. for $t \leq T_F = 6$. Women can work through age 61 and retire from age 62 onward. Each woman can have at most one child, and the child resides with the household for one period only (four years). Child investment is therefore a one-time choice made in the birth period.

Life-cycle mapping and within-period timing. Figure 5 maps periods to ages, and Figure 6 summarizes within-period sequencing during fertile working ages.

State variables. Let V_t^ℓ denote the value function in period t and within-period sub-stage $\ell \in \{1, 2, 3\}$; for non-fertile and retirement periods, there is a single stage and I suppress ℓ when convenient. The household state at the beginning of period t is

$$\Omega_{it} = \{a_t, \theta_i, e_t, x_t, m_t, k_t, m_k\},$$

where:

- a_t denotes assets;
- $\theta_i \in \{1, 2, 3, 4\}$ denotes the cognitive-ability quartile;
- $e_t \in \{\text{HSD}, \text{HSG}, \text{COL}\}$ denotes education attainment;
- x_t denotes accumulated labor-market experience (in four-year units);
- $m_t \in \{0, 1\}$ denotes marital status;
- $k_t \in \{1, 2, 3\}$ denotes child status, where $k_t = 1$ indicates the woman has never had a birth by period t ; $k_t = 2$ indicates a first birth occurs in period t (a child is present in t); $k_t = 3$ indicates the woman had a birth in an earlier period (mother, but child not present in t);

- $m_k \in \{0, 1\}$ records marital status at childbirth (relevant only if $k_t \neq 1$).

Controls. Choice variables are next-period assets $a_{t+1} \in [0, \bar{a}]$, consumption $c_t \geq 0$, female labor supply $l_t \in \{0, 1\}$,¹³ child investment $i_t \geq 0$ (only if $k_t = 2$), and contraceptive effort $s_t \geq 0$ (only in fertile periods when $k_t = 1$).

OA.2.2 Income, taxes/transfers, equivalence scales, and experience

Disposable resources. Gross annual household non-asset income is denoted $\tilde{y}_t^0(\Omega_{it}, l_t)$ and includes female earnings when working and spousal earnings when married. Disposable annual income is computed from gross income using a parsimonious approximation to the U.S. tax-and-transfer system:

$$y_t^a(\Omega_{it}, l_t) = \lambda(\tilde{y}_t^0(\Omega_{it}, l_t))^{1-\tau} + T(m_t),$$

following [Daruich and Fernández \(2024\)](#). Model-period resources aggregate annual resources:

$$y_t(\Omega_{it}, l_t) = 4 y_t^a(\Omega_{it}, l_t).$$

Budget constraint. The within-period budget constraint is

$$\phi_c(m_t, \mathbb{1}\{k_t = 2\}) c_t + a_{t+1} = (1 + r)a_t + y_t(\Omega_{it}, l_t) - \mathbb{1}\{k_t = 2\} i_t,$$

where $\phi_c(\cdot)$ is an equivalence scale that depends on household composition (marital status and the presence of a newborn).

Experience accumulation. Experience evolves according to

$$x_{t+1} = x_t + \mathbb{1}\{l_t = 1\}.$$

¹³I follow [Attanasio et al. \(2008\)](#) in modeling female labor supply.

OA.2.3 Discrete choices and taste shocks

Several stages feature discrete choices (e.g., labor supply, schooling continuation, college entry). Discrete alternatives are subject to i.i.d. Type-I extreme value taste shocks. For a generic discrete choice $d \in \mathcal{D}$ with shocks $\varepsilon_t(d)$ and scale $\sigma_{\mathcal{D}}$, define the choice-specific value net of shocks as $v_t(\Omega, d)$. The ex-ante value is

$$V_t(\Omega) = \mathbb{E}_{\varepsilon} \left[\max_{d \in \mathcal{D}} \{v_t(\Omega, d) + \sigma_{\mathcal{D}} \varepsilon_t(d)\} \right] = \gamma \sigma_{\mathcal{D}} + \sigma_{\mathcal{D}} \log \sum_{d \in \mathcal{D}} \exp \left(\frac{v_t(\Omega, d)}{\sigma_{\mathcal{D}}} \right),$$

where γ is the Euler–Mascheroni constant.

OA.2.4 Retired households (ages 62–77; $t = 13$ –16)

From age 62 onward, the household is retired: female labor supply is fixed at zero and there are no schooling, fertility, marriage, or child-investment decisions. The only intertemporal choice is savings. Households receive Social Security benefits that depend on education and marital status. Let $ss_t(e_t)$ denote the woman’s own benefit and $ss_t^h(e_t, m_k)$ denote the additional spousal benefit received when married.

For $t = 13, \dots, 16$, the retirement problem is

$$V_t(\Omega_{it}) = \max_{a_{t+1} \geq 0, c_t \geq 0} \left\{ u(c_t) + \beta V_{t+1}(\Omega_{i,t+1}) \right\},$$

$$\phi_c(m_t) c_t + a_{t+1} = (1 + r)a_t + y_t,$$

where gross annual non-asset income is

$$\tilde{y}_t^0 = ss_t(e_t) + \mathbb{1}_{\{m_t=1\}} ss_t^h(e_t, m_k),$$

and $y_t = 4y_t^a$ is disposable model-period income computed using the tax/transfer mapping in Subsection [OA.2.2](#). In the terminal period $t = T = 17$, agents consume all remaining resources and die.

OA.2.5 Working, non-fertile households (ages 38–61; $t = 7–12$)

After age 37 ($t \geq 7$), fertility risk is absent and no child is present under the one-period-child assumption. The household chooses whether the woman works, $l_t \in \{0, 1\}$, and chooses consumption and next-period assets. At the beginning of period t , the household draws taste shocks $\{\varepsilon_t(0), \varepsilon_t(1)\}$ for labor supply. Let $v_t(\Omega_{it}, l)$ denote the choice-specific value net of shocks. The ex-ante value is

$$V_t(\Omega_{it}) = \mathbb{E}_\varepsilon \left[\max_{l \in \{0, 1\}} \{v_t(\Omega_{it}, l) + \sigma_l \varepsilon_t(l)\} \right].$$

Conditional on l , the choice-specific problem is

$$\begin{aligned} v_t(\Omega_{it}, l) = & \max_{a_{t+1} \geq 0, c_t \geq 0} \left\{ u(c_t) + \psi_l \mathbb{1}_{\{l=1\}} + \beta \mathbb{E}[V_{t+1}(\Omega_{i,t+1}) \mid \Omega_{it}, l, a_{t+1}] \right\} \\ \text{s.t.} \quad & \phi_c(m_t) c_t + a_{t+1} = (1+r)a_t + y_t(\Omega_{it}, l), \\ & x_{t+1} = x_t + \mathbb{1}_{\{l=1\}}, \end{aligned}$$

where gross annual household income is

$$\tilde{y}_t^0(\Omega_{it}, l) = \mathbb{1}_{\{l=1\}} w(\Omega_{it}) + \mathbb{1}_{\{m_t=1\}} w^h(\Omega_{it}),$$

and disposable model-period resources are

$$y_t(\Omega_{it}, l) = 4 \left[\lambda (\tilde{y}_t^0(\Omega_{it}, l))^{1-\tau} + T(m_t) \right].$$

Here $w(\Omega_{it})$ denotes the woman's wage as a function of education and experience (and other state variables), and spousal labor income, $w^h(\Omega_{it})$, is received only when married.¹⁴

OA.2.6 Young adulthood (ages 22–37; $t = 3–6$)

In young adulthood, schooling is complete (e_t is fixed) and marriage-market and fertility risk are active until $t = T_F = 6$. Within each period $t \leq T_F$, decisions and uncertainty are ordered in three sub-stages.

¹⁴I model the husband's earnings as a reduced-form function of the woman's observed characteristics, as in Adda et al. (2017); Sheran (2007); Van der Klaauw (1996).

Sub-stage 3: labor supply, consumption–saving, and child investment. Let $j \in \{k, nk, ok\}$ index the fertility/child-status outcome in period t : $j = k$ if a first birth occurs in t (so $k_t = 2$); $j = nk$ if no birth occurs and the woman remains childless ($k_t = 1$); and $j = ok$ if the woman had a child in a previous period ($k_t = 3$). Conditional on (Ω_{it}, j) , the household chooses female labor supply, consumption, and savings; and it chooses child investment only when $j = k$. When a child is present ($j = k$), the disutility of working includes the child penalty $\psi_{lk}(e)$ in addition to the baseline $\psi_l(t, e)$. The choice-specific value function net of taste shocks is

$$\begin{aligned} v_t^{3,j}(\Omega_{it}, l) = & \max_{a_{t+1} \geq 0, c_t \geq 0, i_t \geq 0} \left\{ u(c_t) + [\psi_l(t, e) + \mathbb{1}_{\{j=k\}} \psi_{lk}(e)] \cdot \mathbb{1}_{\{l=1\}} \right. \\ & \left. + \mathbb{1}_{\{j=k\}} u_k(i_t) + \beta V_{t+1}^1(\Omega_{i,t+1}) \right\} \\ \text{s.t. } & \phi_c(m_t, \mathbb{1}_{\{j=k\}}) c_t + a_{t+1} = (1+r) a_t + y_t(\Omega_{it}, l) - \mathbb{1}_{\{j=k\}} i_t, \\ & x_{t+1} = x_t + \mathbb{1}_{\{l=1\}}. \end{aligned}$$

Investment enters only if a birth occurs ($j = k$). Because the child is present for one period only, this is the only period in which parents choose i_t .

Sub-stage 2: contraception and first-birth risk. Only childless women choose contraceptive effort, i.e. when $k_t = 1$ and $t \leq T_F$. Let $p_t(\theta_i, e_t, s_t)$ denote the probability of a first birth in period t , which is decreasing in s_t and depends on age, ability, and education. Then

$$V_t^2(\Omega_{it}) = \max_{s_t \geq 0} \left\{ -\phi_s s_t + p_t(\theta_i, e_t, s_t) V_t^{3,k}(\Omega_{it}) + (1 - p_t(\theta_i, e_t, s_t)) V_t^{3,nk}(\Omega_{it}) \right\}.$$

If $k_t \neq 1$ (a first birth already occurred earlier in t or in the past), the household skips contraception:

$$V_t^2(\Omega_{it}) = V_t^{3,ok}(\Omega_{it}).$$

Sub-stage 1: marriage. If single ($m_t = 0$), the woman meets a potential husband with probability $\mu(e_t, t)$. Conditional on meeting, she compares continuation values under marriage

and singlehood. Let $\Omega_{it}(m)$ denote the state with m_t set to $m \in \{0, 1\}$. Then

$$V_t^1(\Omega_{it}) = \begin{cases} \mu(e_t, t) \max\{V_t^2(\Omega_{it}(1)), V_t^2(\Omega_{it}(0))\} + (1 - \mu(e_t, t)) V_t^2(\Omega_{it}(0)), & \text{if } m_t = 0, \\ V_t^2(\Omega_{it}), & \text{if } m_t = 1, \end{cases}$$

and marriage is absorbing (no divorce).

Never having a child. In the last fertile period $t = T_F = 6$, I include a reduced-form utility shifter for remaining childless to match the observed mass of women who never have children:

$$V_6^{3,nk}(\Omega_{i6}) + \mathbb{1}_{\{k_6=1\}} \mu_0(e_6).$$

OA.2.7 College age (ages 18–21; $t = 2$)

Period $t = 2$ corresponds to ages 18–21 and is the period at which women can be in one of two education tracks.

- **Non-college track.** Women who do not enroll in college at $t = 2$ are already in the post-school environment: they participate in the labor market, face marriage-market risk if single, and (since they are still fertile and childless) choose contraception. At $t = 2$ they therefore follow the same within-period timing as in young adulthood.
- **College track.** Women who enroll in college at $t = 2$ do not work during this period. Instead, they receive a student allowance w_C and pay direct schooling costs TC . After observing the fertility outcome, they decide whether to remain in college (continue and graduate) or to drop out and enter the labor market immediately as a high school graduate. Having a child while enrolled in college raises the (psychic) cost of continuing by $\kappa_{k,C}$.

The remainder of this subsection describes the college track.

Sub-stage 3: consumption–saving and (if a birth occurs) child investment. Let $j \in \{k, nk\}$ denote the fertility outcome in $t = 2$. Conditional on the education decision $d \in \{G, CD\}$ from sub-stage 2 (continue/graduate vs. drop out), the within-period problem

differs because college students do not work in this period ($l_2 = 0$), while college dropouts choose labor supply as high-school graduates.

For $d = G$ (continue and graduate), the household solves

$$\begin{aligned} v_2^{3,j}(\Omega_{i2}; G) &= \max_{a_3 \geq 0, c_2 \geq 0, i_2 \geq 0} \left\{ u(c_2) + \mathbb{1}_{\{j=k\}} u_k(i_2) - \mathbb{1}_{\{j=k\}} \kappa_{k,C} + \beta V_3^1(\Omega_{i3}) \right\} \\ \text{s.t.} \quad & \phi_c(m_2, \mathbb{1}_{\{j=k\}}) c_2 + a_3 = (1+r)a_2 + (w_C - TC) - \mathbb{1}_{\{j=k\}} i_2. \end{aligned}$$

For $d = CD$ (drop out and work as a high school graduate), the household chooses labor supply $l_2 \in \{0, 1\}$ and solves

$$\begin{aligned} v_2^{3,j}(\Omega_{i2}; CD) &= \max_{l_2 \in \{0,1\}, a_3 \geq 0, c_2 \geq 0, i_2 \geq 0} \left\{ u(c_2) + \psi_l^j \mathbb{1}_{\{l_2=1\}} + \mathbb{1}_{\{j=k\}} u_k(i_2) + \beta V_3^1(\Omega_{i3}) \right\} \\ \text{s.t.} \quad & \phi_c(m_2, \mathbb{1}_{\{j=k\}}) c_2 + a_3 = (1+r)a_2 + y_2(\Omega_{i2}, l_2) - \mathbb{1}_{\{j=k\}} i_2, \\ & x_3 = x_2 + \mathbb{1}_{\{l_2=1\}}. \end{aligned}$$

In the dropout branch, disposable non-asset income is

$$y_2(\Omega_{i2}, l_2) = 4 \left[\lambda (\tilde{y}_2^0(\Omega_{i2}, l_2))^{1-\tau} + T(m_2) \right], \quad \tilde{y}_2^0(\Omega_{i2}, l_2) = \mathbb{1}_{\{l_2=1\}} w(\Omega_{i2}),$$

while in the college-student branch, disposable resources are given directly by the student allowance net of direct schooling costs, $w_C - TC$.¹⁵

Sub-stage 2: continue college vs. drop out. After observing j , college women choose $d \in \{G, CD\}$ with Type-I extreme value shocks. To match observed college dropouts, the model includes a graduation-specific cost that is incurred only upon completion. Let ϕ_G denote this graduation cost. Then

$$V_2^{2,j}(\Omega_{i2}) = \max_{d \in \{G, CD\}} \left\{ v_2^{3,j}(\Omega_{i2}; d) - \mathbb{1}_{\{d=G\}} \phi_G + \sigma_{CD} \varepsilon_2(d) \right\}.$$

¹⁵In school periods, the student allowance w_C (net of direct schooling costs TC) is treated as a non-taxable transfer-like resource in the model and therefore enters the budget constraint directly. The tax/transfer mapping is applied to labor-market income when working.

Sub-stage 1: contraception. At the start of $t = 2$, childless women in the college track choose s_2 :

$$V_2^1(\Omega_{i2}) = \max_{s_2 \geq 0} \left\{ -\phi_s s_2 + p_2(\theta_i, e_2, s_2) V_2^{2,k}(\Omega_{i2}) + (1 - p_2(\theta_i, e_2, s_2)) V_2^{2,nk}(\Omega_{i2}) \right\}.$$

OA.2.8 Teenage (ages 14–17; $t = 1$)

At $t = 1$, all women are in high school. The within-period timing is: (i) contraception, (ii) after observing the fertility outcome, continue high school vs. drop out, and (iii) consumption–saving (and child investment if a birth occurs). Teens who remain in school receive an allowance w_{HS} in sub-stage 3; dropouts enter the labor market immediately and begin accumulating experience.

Sub-stage 3: consumption–saving, child investment, and college entry at $t = 2$. Let $j \in \{k, nk\}$ denote the fertility outcome in $t = 1$. Conditional on the schooling decision $d \in \{\text{HSG}, \text{HSD}\}$ (stay and complete high school vs. drop out) from sub-stage 2, teens solve

$$\begin{aligned} v_1^{3,j}(\Omega_{i1}; d) = & \max_{a_2 \geq 0, c_1 \geq 0, i_1 \geq 0} \left\{ u(c_1) + \mathbb{1}_{\{j=k\}} u_k(i_1) - \mathbb{1}_{\{d=\text{HSG}\}} \mathbb{1}_{\{j=k\}} \kappa_{\text{HS}} \right. \\ & \left. + \beta \left[\mathbb{1}_{\{d=\text{HSG}\}} V_2^{CD,j}(\Omega_{i2}) + \mathbb{1}_{\{d=\text{HSD}\}} V_2^1(\Omega_{i2}) \right] \right\} \\ \text{s.t.} \quad & \phi_c(m_1, \mathbb{1}_{\{j=k\}}) c_1 + a_2 = (1+r)a_1 + y_1(\Omega_{i1}; d) - \mathbb{1}_{\{j=k\}} i_1, \\ & x_2 = x_1 + \mathbb{1}_{\{d=\text{HSD}\}}. \end{aligned}$$

Resources satisfy $y_1(\Omega_{i1}; d) = w_{HS}$ if $d = \text{HSG}$, while if $d = \text{HSD}$ the teen works as a dropout and

$$y_1(\Omega_{i1}; \text{HSD}) = 4 \left[\lambda (\tilde{y}_1^0(\Omega_{i1}))^{1-\tau} + T(m_1) \right], \quad \tilde{y}_1^0(\Omega_{i1}) = w(\Omega_{i1}).$$

At the end of $t = 1$, teens who complete high school ($d = \text{HSG}$) draw a Type-I extreme value shock and choose whether to enroll in college at $t = 2$, $d_C \in \{C, NC\}$. Let $v_2^1(\cdot)$ denote the beginning-of-period value at $t = 2$ given the education choice; then

$$V_2^{CD,j}(\Omega_{i2}) = \max_{d_C \in \{C, NC\}} \{v_2^1(\Omega_{i2}; d_C) - \kappa_C(\theta, j) + \sigma_C \varepsilon_2(d_C)\}.$$

Table OA.1. Conditional Distribution of Age at First Birth by Education Outcome

Age at first birth	Education Outcome		
	HS Dropout	HS Graduate	College Graduate
14–17	42	14	3
18–21	32	31	8
22–29	14	28	37

Notes: Entries report the share (%) of women in each education group whose first birth occurred in the indicated age bin (column percentages).

Only teens who complete high school face the college-entry decision.

Sub-stage 2: continue high school vs. drop out. After observing j , teens choose $d \in \{\text{HSG}, \text{HSD}\}$ with Type-I extreme value shocks:

$$V_1^{2,j}(\Omega_{i1}) = \max_{d \in \{\text{HSG}, \text{HSD}\}} \{v_1^{3,j}(\Omega_{i1}; d) + \sigma_{HS}\varepsilon_1(d)\}.$$

Sub-stage 1: contraception. At the start of $t = 1$, teens choose s_1 :

$$V_1^1(\Omega_{i1}) = \max_{s_1 \geq 0} \left\{ -\phi_s s_1 + p_1(\theta_i, e_1, s_1) V_1^{2,k}(\Omega_{i1}) + (1 - p_1(\theta_i, e_1, s_1)) V_1^{2,nk}(\Omega_{i1}) \right\}.$$

OA.3 Additional Descriptive Statistics

This appendix collects descriptive tables on birth timing, marriage, and labor-market outcomes that supplement Section 4. All variables and samples are defined in Appendix OA.1.

OA.3.1 Birth Timing and Education

Table OA.1 reports the share of women whose first birth occurs in each age bin, separately by completed education.¹⁶ Early motherhood is concentrated among less educated women: by ages 14–17, the first-birth share is 42% for high-school dropouts, compared with 14% for high-school graduates and 3% for college graduates. By age 21, roughly 74% of dropouts have had a first birth versus 11% of college graduates. More educated women shift first births into later ages, consistent with postponement along the education gradient.

¹⁶Entries are computed within education groups as shares of all women in the group. The table reports only the displayed age bins, so column totals need not sum to one; the omitted residual corresponds to first births after the last reported bin or no observed first birth by the end of the sample.

Table OA.2. Probability of Ever Marriage: Premarital Birth vs. No Premarital Birth

Age at first birth	Education Outcome		
	HS Dropout	HS Graduate	College Graduate
<i>Panel A: Premarital First Birth</i>			
14–17	67	81	83
18–21	58	75	72
22–29	46	59	69
<i>Panel B: No Premarital Birth</i>			
All ages	94	96	98

Notes: Panel A conditions on first birth occurring before first marriage; probabilities shown by age at first birth. Panel B conditions on no birth prior to first marriage (includes women who never marry). Entries report share (%) ever married during the survey window.

OA.3.2 Early Births and Marriage

Marriage shapes household resources, risk-sharing, and the incentives to invest in schooling and labor-market experience. A long tradition emphasizes that childbearing outside marriage can reduce subsequent marriage prospects by changing economic circumstances and the costs/returns to partner search (Becker, 1991).¹⁷

Table OA.2 reports the probability of ever marrying separately for women whose first birth occurs before first marriage (Panel A) and those with no premarital birth (Panel B). Within Panel A, ever-marriage rates are increasing in completed education and declining with the age at premarital first birth. Among women with no premarital birth (Panel B), ever-marriage rates are uniformly high (94%–98%). Premarital fertility is associated with lower marriage rates—especially for the least educated—consistent with a combination of selection and marriage-market penalties tied to premarital childbearing.

Table OA.3 reports average husbands’ annual wages by the woman’s education, age at first birth, and whether the first birth occurs out of wedlock, conditional on marrying. In most cells, women with an out-of-wedlock first birth marry lower-earning husbands. The spousal-earnings differential is largest for high-school graduates—about \$11,400 for ages 22–29 (55,143 vs. 43,719). These spousal-earnings penalties represent an additional cost of imperfect fertility control that the model incorporates through the marriage market.

¹⁷Bronars and Grogger (1994) document that women with unplanned births are less likely to be married when their children are young.

Table OA.3. Average Husband Wage by Education and Women’s Childbearing Status at Marriage

Age at first birth	HS Dropout		HS Graduate		College Graduate	
	Out-wed.	In-wed.	Out-wed.	In-wed.	Out-wed.	In-wed.
14–17	35,089	34,563	—	—	—	—
18–21	35,806	39,064	44,602	46,000	—	—
22–29	33,622	35,806	43,719	55,143	66,025	73,628

Notes: Husband’s average annual wage (2016 dollars) by woman’s education, age at first birth, and whether first birth occurred out of wedlock. Sample restricted to married women; spouse-years require $\geq 2,000$ hours worked and $\geq \$2.50/\text{hour}$. Empty cells indicate insufficient observations.

OA.3.3 Education, Experience, and Labor-Market Outcomes

Table OA.4 summarizes wage levels, wage growth, and experience accumulation by ability and age; all wage statistics are computed among employed women, using the definitions stated in Appendix OA.1.

Panel A shows that the wage gap between quartiles 4 and 1 widens from about \$3,354 at age 20 to more than \$35,000 at age 40. Panel B shows that returns to experience are substantially steeper at higher ability levels: after 5 years, log wage growth is 24% in quartile 1 versus 57% in quartile 4. Panel C documents that higher-ability women accumulate more work experience by a given age (1.85 vs. 3.99 years at age 25). Panel D reports monotonically increasing annualized wage growth (2.69% to 4.33%). Panel E summarizes labor-market dynamics around the first birth: following maternity-related gaps, mean log wage changes are weak or negative for lower-ability women and modestly positive for higher-ability women. Time out of the labor force following the first birth is increasing in ability (0.31 to 0.60 years), suggesting that high-ability women face both steeper returns to continuous experience and larger career costs from early childbearing.

OA.4 Within-Education Birth Hazard Regressions

Table OA.5 reports the estimated coefficients from the within-education linear probability model specified in equation (1). The regression is estimated separately within each education group; Figure 2 in the main text plots the corresponding LPM-predicted probabilities.

Table OA.4. Descriptive Statistics by Ability: Labor Market Outcomes

Outcome	Ability Quartile			
	Q1 (Low)	Q2	Q3	Q4 (High)
<i>Panel A: Wage at Given Age (2016 \$, workers only)</i>				
Age 20	19,688	21,554	22,811	23,042
Age 25	23,954	27,850	32,250	38,412
Age 30	27,778	33,689	38,978	49,126
Age 40	30,382	40,112	46,392	65,713
<i>Panel B: Return to Experience (log wage growth, %)</i>				
5 years potential exp.	24	35	45	57
10 years potential exp.	39	52	65	78
15 years potential exp.	47	61	75	90
<i>Panel C: Cumulative Work Experience (years)</i>				
At age 25	1.85	3.20	3.85	3.99
At age 30	3.60	6.04	7.20	7.51
At age 40	7.94	12.65	14.64	14.44
<i>Panel D: Annualized Log Wage Growth Rate (%)</i>				
Average	2.69	3.12	3.53	4.33
<i>Panel E: Wage Changes Around Non-Work Spells</i>				
$\Delta \log \text{ wage after 1-year gap}$	0.02	-0.01	0.00	0.04
$\Delta \log \text{ wage after 3-year gap}$	-0.02	-0.03	0.05	0.05
$\Delta \log \text{ wage after 5-year gap}$	-0.01	-0.12	0.02	0.07
Time out after first child (years)	0.31	0.50	0.56	0.60

Notes: “Work” defined as ≥ 20 hours/week for ≥ 26 weeks at $\geq$ minimum wage. Panel A: average wages for workers at each age. Panel B: $\ln w_{t+x} - \ln w_t$ where t is first working year; potential experience cumulates only working years. Panel C: cumulative years meeting work definition. Panel E: log wage change between last working year and 1/3/5 years after gap; “time out” = weeks not working in 5 years post-birth $\div 52$.

OA.5 Method-Selection Multinomial Logit

Table OA.6 reports predicted method-choice probabilities from the multinomial logit specified in equation (2). Predicted probabilities are averaged over the sample distribution of education, parity, and wave. The pill-use probability declines from 75.7% in the lowest AFQT quartile to 60.1% in the highest, while the barrier-method probability rises from 16.5% to 31.5%. Behavioral method use is roughly constant across quartiles. Figure 3 in the main text plots these probabilities graphically.

OA.6 Wage Process Estimation

This appendix describes how I estimate the wage profiles used to parameterize earnings opportunities in the structural model. The goal is to recover flexible conditional mean earnings profiles by age, education, cognitive ability, and experience, separately for women and (when

Table OA.5. Within-Education Birth Probability by AFQT Quartile (LPM, 1982–1992)

	(1) No HS Diploma	(2) HS / Some College	(3) College
<i>Panel A: Any birth</i>			
AFQT Q2	−0.021** (0.010)	−0.014** (0.006)	−0.009 (0.010)
AFQT Q3	−0.036** (0.014)	−0.018*** (0.006)	−0.021** (0.009)
AFQT Q4	−0.054*** (0.017)	−0.020*** (0.006)	−0.009 (0.009)
<i>Panel B: Unintended birth</i>			
AFQT Q2	0.001 (0.008)	−0.008* (0.004)	−0.018** (0.009)
AFQT Q3	−0.005 (0.011)	−0.016*** (0.004)	−0.029*** (0.008)
AFQT Q4	−0.005 (0.014)	−0.016*** (0.004)	−0.031*** (0.008)
Observations	5,970	24,838	9,809

Notes: LPM estimated separately within each education group. Unit: woman $\times$ snapshot wave (1982–1992). Panel A outcome: any live birth within a fixed 12-month window (July of t through June of $t + 1$). Panel B outcome: any birth reported as unwanted or mistimed. Sample: women not trying to conceive, unconditional on contraceptive method. Controls: parity and snapshot-year FE. SE clustered by respondent. Reference: AFQT Q1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.6. Predicted Method-Choice Probabilities by AFQT Quartile (1982–1992)

AFQT Quartile	P(Pill)	P(Barrier)	P(Behavioral)
Q1 (Lowest)	0.757	0.165	0.079
Q2	0.706	0.208	0.086
Q3	0.670	0.242	0.088
Q4 (Highest)	0.601	0.315	0.084
Observations	16,616		

Notes: Predicted probabilities from multinomial logit estimated on NLSY79 women reporting reversible contraceptive use, pooled across seven snapshot waves (1982–1992). Base outcome: pill. Probabilities averaged over the sample distribution of education group, parity, and wave. LARC and sterilization excluded.

relevant) husbands/partners.

OA.6.1 Wage measures and estimation samples

Women. Let w_{it} denote real annual wage-and-salary income (2016 dollars). The wage estimation sample includes woman-years that meet the employment definition in Appendix OA.1 (minimum weeks worked, minimum hours, and minimum annual earnings). The dependent variable is in levels (annual dollars), consistent with how the model is parameterized.

Husbands/partners. Let w_{it}^m denote partner annual wage-and-salary income (2016 dollars) and let Age_{it}^m denote the partner's age. The husband/partner wage estimation sample is restricted to years in which the woman is married and partner earnings exceed the same annual earnings threshold used for women. In the husband regressions, education is the woman's education, denoted $\text{Educ}_i^f \in \{\text{HSD}, \text{HSG}, \text{COL}\}$, matching the table panel "Wife's education." This specification captures assortative mating: wives' education proxies for husband characteristics through the marriage matching process, and the coefficient estimates reflect the quality of partners that women of different education levels tend to attract in the marriage market.

OA.6.2 Baseline specification: women

I estimate:

$$\begin{aligned}
 w_{it} = & \alpha_t + \beta_1 \text{Age}_{it}^f + \beta_2 \left(\text{Age}_{it}^f \right)^2 + \rho \text{CumExp}_{it} \\
 & + \sum_{e \in \{\text{HSG}, \text{COL}\}} \left(\gamma_e + \delta_e \text{CumExp}_{it} \right) \mathbb{1}\{\text{Educ}_i^f = e\} \\
 & + \sum_{q=2}^4 \left(\gamma_q + \delta_q \text{CumExp}_{it} \right) \mathbb{1}\{\text{Ability}_i = q\} \\
 & + \sum_{e \in \{\text{HSG}, \text{COL}\}} \sum_{q=2}^4 \left(\gamma_{eq} + \delta_{eq} \text{CumExp}_{it} \right) \mathbb{1}\{\text{Educ}_i^f = e\} \mathbb{1}\{\text{Ability}_i = q\} + \varepsilon_{it}.
 \end{aligned} \tag{OA.1}$$

where α_t are calendar-year fixed effects. The interaction structure $\text{CumExp} \times \text{Educ}^f \times \text{Ability}$ allows returns to experience to vary flexibly across education and cognitive-ability quartiles.

OA.6.3 Baseline specification: husbands/partners

For husbands/partners I estimate:

$$\begin{aligned}
w_{it}^m &= \alpha_t^m + \theta_{\text{HSG}} \mathbb{1}\{\text{Educ}_i^f = \text{HSG}\} \\
&+ \theta_{\text{COL}} \mathbb{1}\{\text{Educ}_i^f = \text{COL}\} + \beta_1^m \text{Age}_{it}^m + \beta_2^m (\text{Age}_{it}^m)^2 \\
&+ \sum_{e \in \{\text{HSG}, \text{COL}\}} \left(\beta_{1e}^m \text{Age}_{it}^m + \beta_{2e}^m (\text{Age}_{it}^m)^2 \right) \mathbb{1}\{\text{Educ}_i^f = e\} \\
&+ \lambda^m \mathbb{1}\{\text{MarryBeforeBirth}_i = 1\} + \sum_{e \in \{\text{HSG}, \text{COL}\}} \kappa_e^m \mathbb{1}\{\text{Educ}_i^f = e\} \mathbb{1}\{\text{MarryBeforeBirth}_i = 1\} + u_{it}.
\end{aligned} \tag{OA.2}$$

with year fixed effects α_t^m and an indicator $\mathbb{1}\{\text{MarryBeforeBirth}_i = 1\}$ capturing systematic differences in spouse earnings associated with marrying prior to first birth.

OA.6.4 Estimated parameters

Interpreting the ability gradient. Table OA.7 shows the estimated parameters of the women’s income process. The estimates reveal substantial wage premia associated with cognitive ability. The baseline ability effect (e.g., Q4 vs. Q1 for high school dropouts: \$3,078) reflects both direct productivity differences and indirect effects through sorting into higher-paying jobs. The college-ability interactions are particularly large (Q2–Q4 college graduates earn \$6,400–\$7,000 more than Q1 dropouts with the same experience), consistent with complementarity between cognitive ability and advanced education in the labor market.

Interpreting the education coefficients. Table OA.8 shows the estimated parameters of the husband’s income process. The negative main effects for wife’s education (-\$28,089 for HS grad, -\$106,082 for college) reflect the specification’s interaction structure. These coefficients capture husband earnings when the wife marries after first birth (the baseline) and when the husband is at age zero. The large positive age-education interactions (\$1,149/year for HS grad wives, \$5,490/year for college wives) indicate that more-educated women’s husbands experience steeper earnings growth. At typical ages (e.g., age 35), college-educated women’s husbands earn substantially more than dropout women’s husbands, consistent with positive

Table OA.7. Women's Earnings Process Estimates

Variable	Coef. (SE)	Variable	Coef. (SE)
<i>Experience</i>		<i>Education × ability</i>	
Cumulative experience	953*** (33)	HS grad × Q2	-50 (632)
		HS grad × Q3	1801* (1008)
<i>Education (base: HSD)</i>		HS grad × Q4	428 (1743)
HS graduate	1164*** (393)	College grad × Q2	6572*** (1186)
College graduate	2291** (904)	College grad × Q3	6423*** (1388)
<i>Education × experience</i>		College grad × Q4	6990*** (1951)
HS grad × exp	228*** (35)	<i>Educ × ability × exp</i>	
College grad × exp	916*** (77)	HS grad × Q2 × exp	134** (60)
<i>Ability (base: Q1)</i>		HS grad × Q3 × exp	-87 (82)
Ability Q2	1414** (564)	HS grad × Q4 × exp	540*** (93)
Ability Q3	1202 (963)	College grad × Q2 × exp	-254** (100)
Ability Q4	3078* (1710)	College grad × Q3 × exp	-12 (115)
<i>Ability × experience</i>		College grad × Q4 × exp	748*** (120)
Q2 × exp	-29 (55)	<i>Age profile</i>	
Q3 × exp	257*** (78)	Age	1444*** (118)
Q4 × exp	-170* (89)	Age ²	-22*** (2)
		Constant	-7617*** (1987)
Observations			94,156
Adjusted R^2			0.292

Notes: Robust standard errors in parentheses; year fixed effects included. Dependent variable is annual wage-and-salary income in 2016 dollars. Baselines: high school dropout (HSD) and ability quartile 1 (Q1). Coefficients and standard errors rounded to nearest dollar. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.8. Husband/Partner Earnings Process Estimates

Variable	Coef. (SE)	Variable	Coef. (SE)
<i>Wife's education (base: HSD)</i>		<i>Age profile</i>	
HS graduate	-28089*** (7298)	Age	794 (498)
College graduate	-106082*** (11760)	HS grad $\times$ age	1149*** (427)
		College grad $\times$ age	5490*** (668)
<i>Marriage timing (base: after birth)</i>		Age ²	0 (8)
Marry before birth	79 (1301)	HS grad $\times$ age ²	-7 (6)
<i>Education $\times$ marriage timing</i>		College grad $\times$ age ²	-48*** (9)
HS grad $\times$ marry before	13913*** (1414)	Constant	12431 (7908)
College grad $\times$ marry before	15275*** (2563)		
Observations			37,728
Adjusted R^2			0.159

Notes: Robust standard errors in parentheses; year fixed effects included. Dependent variable is husband's annual wage-and-salary income in 2016 dollars. Wife's education serves as proxy for husband characteristics through assortative mating. Baselines: high school dropout (HSD) and first marriage after first birth. Coefficients and standard errors rounded to nearest dollar. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

assortative mating. The marriage-timing interactions show that women who marry before first birth (a marker of deliberate family formation) have higher-earning husbands, with this premium larger for more-educated women.

OA.6.5 Retirement income process

The NLSY wage-and-salary measures do not capture the older-age components of retirement resources (Social Security benefits, employer pensions, and other transfers) because the survey population has not reached that age.

To ensure computational tractability, I model retirement income in reduced form as an education-specific replacement rate applied to pre-retirement earnings capacity, separately for women and husbands/partners.

Let T_R denote the first retirement period (the last N_{retired} model periods). For each education group $e \in \{\text{HSD}, \text{HSG}, \text{COL}\}$, I compute a baseline pre-retirement earnings level as the average predicted annual labor income in the final working period,

$$\bar{w}_e \equiv \mathbb{E}[\hat{w}_{it} \mid \text{Educ}_i = e, t = T_R - 1], \quad \bar{w}_e^m \equiv \mathbb{E}[\hat{w}_{it}^m \mid \text{Educ}_i = e, t = T_R - 1],$$

where expectations are taken over the model state distribution in that period (ability, accumulated experience, and other discrete states relevant for the wage grids).

In retirement periods $t \geq T_R$, labor income is replaced by a deterministic benefit level:

$$w_e^R = \phi_e \bar{w}_e, \quad (w^m)_e^R = \phi_e \bar{w}_e^m,$$

held constant over all retirement ages.

Social Security replacement rates. To discipline retirement income in the model, I calibrate education-specific replacement rates using Social Security Administration Office of the Chief Actuary replacement-rate statistics (first-year retired-worker benefits as a percent of wage-indexed career-average earnings). The model does not implement the statutory benefit formula (AIME/PIA) directly; instead, I use these statistics to discipline education-specific multipliers ϕ_e in a reduced-form retirement-income rule. In particular, ϕ_e should be

interpreted as a replacement rate relative to late-career earnings in the model, proxied by $\bar{w}_e$, rather than literally relative to wage-indexed career-average earnings.¹⁸

OA.6.6 Model inputs and aggregation to four-year periods

The estimated coefficients from the above regressions are used to generate predicted annual earnings paths by (age, education, ability quartile, cumulative experience). In the model, each period corresponds to four years; I therefore interpret the predicted annual earnings at the period’s representative age (the start-of-bin age) as the period-specific annual earnings opportunity, and update cumulative experience using the model-consistent experience accumulation rule.

For retirement periods, I do not predict the wage regressions. Instead, I replace labor earnings with an education-specific deterministic retirement-income level constructed from the pre-retirement predicted wage arrays, as described in Appendix [OA.6.5](#).

OA.7 Model Fit

OA.7.1 Targeted Moments

This appendix presents a detailed comparison between the empirical moments used to calibrate the model and their corresponding model-generated counterparts. The estimation procedure employs the Simulated Method of Moments (SMM), which minimizes the weighted distance between 118 empirical moments from the NLSY79 data and their model analogues.

The targeted moments are organized into six categories: (i) schooling and early fertility decisions, (ii) child investment, (iii) fertility timing by ability, (iv) marriage patterns by education, (v) labor force participation by education, and (vi) contraception use by education. This comprehensive set of moments disciplines the model’s ability to jointly capture the key life-cycle patterns that characterize women’s decisions regarding education, fertility, marriage, labor supply, and family planning.

Tables [OA.9–OA.13](#) report the data moments and model moments for each targeted

¹⁸Because $\bar{w}_e$ is a proxy for earnings capacity at the end of the working life rather than AIME, the mapping from the SSA tables into ϕ_e is an approximation that preserves the education gradient in replacement rates while maintaining a parsimonious retirement-income process.

Table OA.9. Model Fit: Schooling, Early Fertility, and Child Investment

Moment	Data	Model	Moment	Data	Model
<i>Panel A: HS Dropout by Birth Status at Age 14</i>					
No birth	0.070	0.068	Birth	0.290	0.300
<i>Panel B: College Attendance by Birth Status at Age 14</i>					
No birth	0.410	0.408	Birth	0.080	0.073
<i>Panel C: College Attendance by Ability Quartile at Age 18</i>					
Q1 (Low)	0.110	0.112	Q2	0.250	0.303
Q3	0.410	0.423	Q4 (High)	0.670	0.488
<i>Panel D: College Graduation by Birth Status at Age 18</i>					
No birth	0.620	0.615	Birth	0.260	0.251
<i>Panel E: Relative Child Investment by Education</i>					
HS Grad / HS Dropout	1.200	1.972	College Grad / HS Dropout	4.600	3.824
<i>Notes:</i> Data moments from NLSY79. Child investment ratios based on Caucutt and Lochner (2020) .					

Table OA.10. Model Fit: Fraction with Children by Ability Quartile and Age

	Q1 (Low)		Q2		Q3		Q4 (High)	
Age	Data	Model	Data	Model	Data	Model	Data	Model
18	0.381	0.370	0.254	0.271	0.140	0.195	0.068	0.066
22	0.691	0.625	0.540	0.478	0.372	0.368	0.223	0.230
26	0.794	0.766	0.706	0.640	0.534	0.541	0.409	0.424
30	0.838	0.828	0.770	0.712	0.661	0.617	0.591	0.530
34	0.873	0.864	0.822	0.766	0.733	0.676	0.713	0.608
38	0.884	0.890	0.838	0.795	0.759	0.708	0.743	0.654
42	0.885	0.907	0.846	0.824	0.773	0.743	0.748	0.691
<i>Notes:</i> Data moments from NLSY79. Ability quartiles based on AFQT scores. Ages represent end of 4-year periods.								

Table OA.11. Model Fit: Fraction Married by Education and Age

	HS Dropout		HS Graduate		College Grad	
Age	Data	Model	Data	Model	Data	Model
22	0.610	0.440	0.683	0.701	—	—
26	0.709	0.797	0.719	0.860	0.744	0.933
30	0.756	0.963	0.579	0.356	0.706	0.639
34	0.760	0.725	0.794	0.787	0.814	0.845
38	0.825	0.885	0.567	0.570	0.721	0.718
42	0.782	0.818	0.807	0.844	0.824	0.869
<i>Notes:</i> Data moments from NLSY79. College graduates enter marriage market at age 22. Ages represent end of 4-year periods.						

statistic. The model achieves a reasonable fit across all moment categories, capturing both the levels and the heterogeneity across education and ability groups.

Table OA.12. Model Fit: Labor Force Participation by Education and Age

Age	HS Dropout		HS Graduate		College Grad	
	Data	Model	Data	Model	Data	Model
18	0.084	0.082	—	—	—	—
22	0.151	0.215	0.196	0.211	—	—
26	0.241	0.188	0.245	0.297	0.243	0.275
30	0.259	0.248	0.258	0.227	0.239	0.233
34	0.206	0.214	0.184	0.168	0.140	0.140
38	0.117	0.170	0.368	0.495	0.492	0.468
42	0.548	0.488	0.557	0.554	0.541	0.548
46	0.532	0.523	0.527	0.517	0.499	0.503
50	0.448	0.506	0.397	0.318	0.291	0.318
54	0.157	0.322	0.686	0.713	0.754	0.698
58	0.705	0.675	0.657	0.664	0.628	0.649
62	0.634	0.629	0.626	0.645	0.607	0.642
66	0.554	0.445	0.422	0.445	0.211	0.451

Notes: Data moments from NLSY79. Labor market entry: HS dropouts age 14, HS graduates age 18, college graduates age 22. Ages represent end of 4-year periods.

Table OA.13. Model Fit: Contraception Use by Education and Age

Age	HS Dropout		HS Graduate		College Grad	
	Data	Model	Data	Model	Data	Model
22	0.726	0.529	0.792	0.665	0.880	1.000
26	0.508	0.425	0.733	0.462	0.773	1.000
30	0.409	0.377	0.648	0.515	0.744	1.000
34	0.320	0.365	0.535	0.513	0.661	1.000
38	0.259	0.278	0.465	0.463	0.579	0.658
42	0.203	0.205	0.417	0.412	0.481	0.660

Notes: Data moments from NLSY79. Contraception measured among sexually active, non-pregnant women. Ages represent end of 4-year periods.

OA.8 Estimated Structural Parameters

This appendix reports the structural parameters estimated by Simulated Method of Moments (SMM) across the three nested model specifications used in the fit decomposition of Section 7.2. The *Baseline* model imposes no heterogeneity in fertility risk across education or cognitive ability (42 parameters). The + *Educ. Het.* specification allows baseline conception odds $\lambda_h(g, e)$ to vary by education group (48 parameters). The + *Ab. Cont.* specification—the full model—additionally allows contraceptive effort effectiveness $\eta_{\theta, g}$ to vary by cognitive ability quartile and age group (60 parameters). “—” denotes parameters restricted to zero or not separately identified in that specification; $\Delta\eta$ increments are additive deviations from the Q1 baseline.

Table OA.14. Estimated Structural Parameters: Nested Model Specifications

Parameter	Description	Baseline	+ Educ. Het.	+ Ab. Cont.
<i>Panel A: Baseline Conception Odds $\lambda_h(g, e)$</i>				
$\lambda_{1,\text{HSD}}$	Ages 14–22, HSD	4.823	4.725	2.000
$\lambda_{1,\text{HS}}$	Ages 14–22, HS	–	1.403	1.417
$\lambda_{1,\text{Col}}$	Ages 14–22, College	–	1.192	0.172
$\lambda_{2,\text{HSD}}$	Ages 22–30, HSD	2.739	3.000	0.698
$\lambda_{2,\text{HS}}$	Ages 22–30, HS	–	1.037	1.104
$\lambda_{2,\text{Col}}$	Ages 22–30, College	–	0.393	1.243
$\lambda_{3,\text{HSD}}$	Ages 30–38, HSD	1.371	1.346	0.003
$\lambda_{3,\text{HS}}$	Ages 30–38, HS	–	0.544	0.550
$\lambda_{3,\text{Col}}$	Ages 30–38, College	–	0.674	0.297
<i>Panel B: Contraceptive Effectiveness $\eta_{\theta,g}$</i>				
$\eta_{Q1,g1}$	Q1, Ages 14–22	1.000	1.000	0.841
$\eta_{Q1,g2}$	Q1, Ages 22–30	1.000	1.000	7.916
$\eta_{Q1,g3}$	Q1, Ages 30–38	1.000	1.000	0.501
$\Delta\eta_{Q2,g1}$	Q2 incr., Ages 14–22	–	–	0.111
$\Delta\eta_{Q3,g1}$	Q3 incr., Ages 14–22	–	–	0.287
$\Delta\eta_{Q4,g1}$	Q4 incr., Ages 14–22	–	–	1.688
$\Delta\eta_{Q2,g2}$	Q2 incr., Ages 22–30	–	–	0.864
$\Delta\eta_{Q3,g2}$	Q3 incr., Ages 22–30	–	–	3.680
$\Delta\eta_{Q4,g2}$	Q4 incr., Ages 22–30	–	–	6.422
$\Delta\eta_{Q2,g3}$	Q2 incr., Ages 30–38	–	–	0.053
$\Delta\eta_{Q3,g3}$	Q3 incr., Ages 30–38	–	–	2.216
$\Delta\eta_{Q4,g3}$	Q4 incr., Ages 30–38	–	–	4.789
<i>Panel C: Marriage Meeting Probabilities $\mu(e, g)$</i>				
$\mu_{\text{HSD},g1}$	HSD, Ages 18–22	–0.232	–0.001	0.430
$\mu_{\text{HS},g1}$	HS, Ages 18–22	–0.188	–0.188	0.452
$\mu_{\text{Col},g1}$	College, Ages 22	–0.027	–0.034	0.590
$\mu_{\text{HSD},g2}$	HSD, Ages 26–30	0.416	0.191	0.305
$\mu_{\text{HS},g2}$	HS, Ages 26–30	–0.490	–0.461	0.234
$\mu_{\text{Col},g2}$	College, Ages 26–30	–0.415	–0.439	0.377
$\mu_{\text{HSD},g3}$	HSD, Ages 34–38	0.066	0.114	0.464
$\mu_{\text{HS},g3}$	HS, Ages 34–38	–0.470	–0.475	0.268
$\mu_{\text{Col},g3}$	College, Ages 34–38	0.451	0.442	0.132
<i>Panel D: Child Utility and Cost Parameters</i>				
ω_0	Utility baseline	–0.232	–0.001	–0.048
ω_1	Utility scale	0.416	0.191	0.166
ω_2	Utility curvature	0.066	0.114	0.019
ϕ_k^{HSD}	Cost shifter, HSD	–0.188	–0.188	–0.135
ϕ_k^{HS}	Cost shifter, HS	–0.490	–0.461	–0.387
ϕ_k^{BA}	Cost shifter, College	–0.470	–0.475	–0.484
ϕ_{grad}	Graduation cost	–0.027	–0.034	–0.475
ξ_{cf}	Cognitive college cost	–0.415	–0.439	–0.528
ω_{ch}	Cognitive convexity	0.312	0.298	1.298
<i>Panel E: Education Subsidies</i>				
	HS allowance	24.9	24.6	54.6
	College allowance	76.3	76.3	119.8
<i>Panel F: Labor Supply Parameters</i>				
ψ_ℓ (avg.)	Ages 14–26	0.408	0.417	–0.013
ψ_ℓ (avg.)	Ages 30–50	–0.010	–0.011	–0.007
ψ_ℓ (avg.)	Ages 54–62	–0.006	–0.007	–0.014
$\psi_{\ell k,1}$	Child penalty, HSD	–0.008	–0.008	–0.274
$\psi_{\ell k,2}$	Child penalty, HS	–0.878	–1.638	–0.656
$\psi_{\ell k,3}$	Child penalty, College	–0.869	–1.231	–0.000
$\phi_{nk,1}$	Terminal util., HSD	–0.790	–1.659	0.112
$\phi_{nk,2}$	Terminal util., HS	0.281	0.268	0.094
$\phi_{nk,3}$	Terminal util., College	0.221	0.251	0.347
<i>Panel G: Shock Standard Deviations</i>				
σ_ℓ	Labor supply	0.229	0.233	0.010
σ_c (avg.)	Child-related	0.173	0.176	0.239
# parameters	Estimated	42	48	60

Notes: HSD = high school dropout; HS = high school graduate; Col = college graduate. *Baseline*: no heterogeneity in fertility risk. *+ Educ. Het.*: adds education-specific $\lambda_h(g, e)$. *+ Ab. Cont.*: full model with ability-varying $\eta_{\theta,g}$. $\Delta\eta$ = additive increment from Q1 baseline. “–” = restricted or not identified.

OA.9 Optimal Contraceptive Effort: Derivation and Normalization

This appendix derives the first-order condition for optimal contraceptive effort in closed form and establishes the normalization result discussed in Section 5.

OA.9.1 The childless woman's contraception problem

The stage-2 objective for a childless woman is

$$V_t^{(2)} = \max_{s \geq 0} \left\{ p_t(s) V_t^{\text{child}} + (1 - p_t(s)) V_t^{\text{no child}} - \phi_s s \right\},$$

where V_t^{child} and $V_t^{\text{no child}}$ are the stage-3 continuation values with and without a birth. Let $\Delta V \equiv V_t^{\text{child}} - V_t^{\text{no child}} < 0$ (women who choose positive effort prefer to avoid a birth). The objective can be rewritten as

$$V_t^{(2)} = V_t^{\text{no child}} + \max_{s \geq 0} \left\{ p_t(s) \Delta V - \phi_s s \right\}.$$

The conception probability from Equation (4) is

$$p_t(s) = \lambda_{\max} \cdot \frac{\lambda_{ge} \exp(-\eta_{\theta g} s)}{1 + \lambda_{ge} \exp(-\eta_{\theta g} s)}.$$

Define $z \equiv \exp(-\eta_{\theta g} s)$, so that $p_t(s) = \lambda_{\max} \cdot \lambda_{ge} z / (1 + \lambda_{ge} z)$.

OA.9.2 First-order condition

Differentiating with respect to s :

$$\frac{\partial p_t}{\partial s} = -\eta_{\theta g} \lambda_{\max} \cdot \frac{\lambda_{ge} z}{(1 + \lambda_{ge} z)^2}.$$

The FOC $(\partial p_t / \partial s) \Delta V = \phi_s$ yields

$$-\eta_{\theta g} \lambda_{\max} \cdot \frac{\lambda_{ge} z}{(1 + \lambda_{ge} z)^2} \cdot \Delta V = \phi_s. \tag{OA.3}$$

OA.9.3 Reduction to a quadratic

Define the auxiliary variable $y \equiv 1 + \lambda_{ge} z$, so that $\lambda_{ge} z = y - 1$ and $(1 + \lambda_{ge} z)^2 = y^2$.

Substituting into (OA.3):

$$-\eta_{\theta g} \lambda_{\max} \cdot \frac{y - 1}{y^2} \cdot \Delta V = \phi_s.$$

Multiplying both sides by y^2 :

$$\phi_s y^2 + \eta_{\theta g} \lambda_{\max} \Delta V y - \eta_{\theta g} \lambda_{\max} \Delta V = 0.$$

This is a quadratic $a y^2 + b y + c = 0$ with

$$a = \phi_s, \quad b = \eta_{\theta g} \lambda_{\max} \Delta V, \quad c = -\eta_{\theta g} \lambda_{\max} \Delta V.$$

Since $\Delta V < 0$, we have $b < 0$ and $c > 0$. The discriminant is $b^2 - 4ac = b^2 + 4\phi_s \eta_{\theta g} \lambda_{\max} |\Delta V| > 0$, so both roots are real. The economically relevant root satisfies $y > 1$ (equivalently $s \geq 0$):

$$y^* = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

Effort is then recovered as $z^* = (y^* - 1)/\lambda_{ge}$ and $s^* = -\log(z^*)/\eta_{\theta g}$.

OA.9.4 Normalization: ϕ_s and $\eta_{\theta g}$ are not separately identified

Consider the rescaling $(\eta_{\theta g}, \phi_s) \rightarrow (k \eta_{\theta g}, k \phi_s)$ for any $k > 0$. Under the new parameters, the quadratic coefficients become $(k \phi_s, k \eta_{\theta g} \lambda_{\max} \Delta V, -k \eta_{\theta g} \lambda_{\max} \Delta V)$ —each multiplied by k . Dividing through by k , the quadratic is identical, so y^* is unchanged.

The recovered effort becomes

$$s_{\text{new}}^* = -\frac{\log(z^*)}{k \eta_{\theta g}} = \frac{s_{\text{old}}^*}{k},$$

so effort rescales. However, the product that enters the conception probability is

$$(k \eta_{\theta g}) \cdot s_{\text{new}}^* = k \eta_{\theta g} \cdot \frac{s_{\text{old}}^*}{k} = \eta_{\theta g} \cdot s_{\text{old}}^*,$$

which is invariant. Since $p_{ge}(\theta, s)$ depends on s only through $\eta_{\theta g} \cdot s$, the conception probability at the optimum—and therefore all observable implications of the model—are invariant to the rescaling. The data identify only the ratio $\eta_{\theta g}/\phi_s$, not each parameter separately. I therefore normalize $\phi_s = 0.001$ and let $\eta_{\theta g}$ absorb all ability-related variation in the effort–risk mapping.

OA.10 Computational Appendix

This appendix documents the numerical solution, simulation, and estimation procedures. The model is a finite-horizon discrete-time dynamic programming problem with both continuous (assets) and discrete (education, marital status, fertility) state variables. I solve the model by backward induction, using different numerical methods at different life stages depending on the structure of the within-period problem. The estimation uses the simulated method of moments (SMM), with a global optimization algorithm to search over the parameter space.

OA.10.1 State space, grids, and timing

Time is discrete in four-year periods, indexed by $t = 1, \dots, T$. The mapping from period to age is $\text{age}_t = 10 + 4t$, so $t = 1$ corresponds to age 14.

The individual state is

$$s_t \equiv (a_t, \theta, e_t, x_t, m_t, mk_t, k_t, t),$$

where a_t is assets at the beginning of t , θ is cognitive ability type (discrete), e_t is education (dropout / HS / college), x_t is experience (discrete, accumulated when working), m_t is marital status (single/married), mk_t is an indicator for whether the first birth occurred out of marriage, and k_t is child status. In the implementation, $k_t \in \{1, 2, 3\}$ corresponds to: no child; newborn in the current period; and older child in later periods.

The continuous state a_t is discretized on an exogenous grid $\mathcal{A} = \{a^1, \dots, a^{N_a}\}$ with cubic spacing:

$$a_j = a_{\min} + (a_{\max} - a_{\min}) \cdot (j/N_a)^3, \quad j = 0, \dots, N_a.$$

This concentrates grid points near the borrowing constraint where policy functions are steepest. Policy functions are stored on $\mathcal{A}$ and evaluated off-grid by linear interpolation in simu-

lation.

Within-period timing and sub-stages. The code solves a three-substage problem within each fertile working period:

1. **Stage 3:** Given marital status and realized fertility outcome (child/no child), the household chooses labor $l_t \in \{0, 1\}$, savings a_{t+1} , consumption c_t , and (if a newborn arrives) child investment i_t .
2. **Stage 2:** Prior to the fertility realization, the household chooses contraception effort s_t which governs birth probability; the stage-2 value integrates stage-3 values over the birth realization.
3. **Stage 1:** If single and eligible to meet, the household draws a meeting opportunity and chooses whether to marry; the stage-1 value integrates the stage-2 value over meeting opportunities and the marriage decision rule.

OA.10.2 Household problem and key first-order conditions

Preferences are CRRA in consumption, $u(c) = c^{1-\rho}/(1-\rho)$, where ρ is the coefficient of relative risk aversion for the woman. Per-adult-equivalent consumption is implemented via an equivalence-scale denominator

$$\text{den}(m_t, k_t) = 1 + \mathbf{1}\{m_t = \text{married}\}\phi_{ca} + \mathbf{1}\{k_t = 2\}\phi_{ck}.$$

Thus, the child-related equivalence-scale term ϕ_{ck} enters the budget constraint only in the birth period ($k_t = 2$), consistent with the “one-period child in the household” assumption. After the birth period, the state moves from $k_t = 2$ to $k_{t+1} = 3$ (child has left the household), so that $\mathbf{1}\{k_{t+1} = 2\} = 0$ in all subsequent periods.

Let y_t denote disposable (post-tax/post-transfer) income in period t (four-year total). Gross income is transformed by a progressive tax-transfer function:

$$y_t = \tau(\text{gross}_t, m_t) = \lambda \cdot \text{gross}_t^{1-\tau} + T_{m_t},$$

where $\tau = 0.18$ is the progressivity parameter, $\lambda = 0.85$ is the scale parameter, and T_m is the guaranteed minimum income (\$8.606 thousand for singles, \$12.898 thousand for couples, yearly in 2016 dollars).

The stage-3 budget constraint is

$$c_t \cdot \text{den}(m_t, k_t) + i_t + a_{t+1} = (1 + r)a_t + y_t.$$

Child investment subproblem (stage 3, newborn only). When $k_t = 2$ (newborn in period t), child investment enters the continuation value through

$$V_k(i_t) = \omega_0 + \omega_1 i_t^{\omega_2}, \quad \omega_2 < 1.$$

The household's problem is

$$\max_{c_t, i_t} u(c_t) + V_k(i_t) + \beta V_{t+1}(a_{t+1}) \quad \text{s.t.} \quad c_t \cdot \text{den}(m_t, k_t) + i_t + a_{t+1} = (1 + r)a_t + y_t.$$

The first-order condition equates marginal utility per dollar:

$$\frac{u'(c_t)}{\text{den}(m_t, k_t)} = V'_k(i_t) \quad \Longleftrightarrow \quad c_t^{-\rho} = \text{den}(m_t, k_t) \omega_1 \omega_2 i_t^{\omega_2 - 1}.$$

Solution method. Given a_{t+1} , the budget constraint implies $c_t \cdot \text{den} + i_t = \text{available}$, where $\text{available} \equiv (1 + r)a_t + y_t - a_{t+1}$. Substituting into the FOC yields a single equation in i_t . The code solves this via bisection on $i_t \in [10^{-6}, 0.9999 \times \text{available}]$:

1. Compute $c_t(i_t) = (\text{available} - i_t)/\text{den}$.
2. Evaluate FOC residual: $r(i_t) = c_t(i_t)^{-\rho} - \text{den}(m_t, k_t) \omega_1 \omega_2 i_t^{\omega_2 - 1}$.
3. Update bracket: if $r(i_t) < 0$, increase i_t (consumption too high); else decrease.
4. Terminate when $|r(i_t)| < 10^{-10}$ or bracket width $< 10^{-10}$ (max 50 iterations).

The solution is unique because $r(i_t)$ is strictly increasing in i_t : $c_t(i_t)$ is decreasing in i_t , so $u'(c_t(i_t))$ rises, while $V'_k(i_t)$ falls when $\omega_2 < 1$.

Contraception choice (stage 2). Let $p_t(s_t)$ be the birth probability. Given stage-3 values with and without a birth, $(V_t^{\text{birth}}, V_t^{\text{nobirth}})$, the stage-2 objective is

$$V_t^{(2)} = p_t(s_t) V_t^{\text{birth}} + (1 - p_t(s_t)) V_t^{\text{nobirth}} - \phi_s s_t,$$

with an interior FOC $p'_t(s_t) (V_t^{\text{birth}} - V_t^{\text{nobirth}}) = \phi_s$. The code uses a closed-form solution for s_t under the implemented $p_t(s)$ specification.

Labor choice with taste shocks. In periods solved by DC-EGM, the labor decision has i.i.d. type-I extreme value taste shocks with scale $\sigma_l(e)$, implying an inclusive value (log-sum) aggregator and a logit work probability:

$$V_t = \sigma_l(e) \log \left(\exp(V_{t,l=0}/\sigma_l(e)) + \exp(V_{t,l=1}/\sigma_l(e)) \right),$$

$$P_t(l = 1) = \frac{\exp(V_{t,1}/\sigma_l(e))}{\exp(V_{t,0}/\sigma_l(e)) + \exp(V_{t,1}/\sigma_l(e))}.$$

OA.10.3 Solution algorithm (backward induction)

This section documents the solver `VFI_P_DCEGM` in `vfi_dcegm.jl`. The algorithm proceeds by backward induction, but uses different numerical routines depending on age.

Overview. Let T_R be the number of retired periods, and let T_{NF} denote the number of working periods after fertility ends. The code partitions the horizon into: (i) retirement ($t > T - T_R$), solved by EGM; (ii) non-fertile working ages ($T - T_R - T_{NF} < t \leq T - T_R$), solved by DC-EGM; (iii) fertile ages ($t \leq T - T_R - T_{NF}$), solved by VFI with grid search (plus analytical or one-dimensional inner problems for i_t and s_t).

The rationale for this partition is computational efficiency. In retirement, there is no labor-leisure choice and the problem reduces to a standard consumption-saving model, for which EGM is highly efficient. In non-fertile working ages, the discrete labor choice introduces non-convexities that can generate discontinuous policy functions; DC-EGM handles this by computing choice-specific consumption functions via EGM and then applying an upper-envelope algorithm to recover the global optimum. In fertile ages, the additional discrete margins (marriage, contraception, schooling) and the multi-stage within-period structure

make DC-EGM less practical, so the code reverts to grid search with analytical solutions for the continuous sub-problems.

Algorithm 1 (Retirement, EGM). In retirement, labor is absent and the problem is a standard consumption-saving model with a borrowing constraint. The EGM step for each discrete state (θ, e, m, mk, k) is:

1. Fix the exogenous grid for next-period assets $\mathcal{A} = \{a'\}$.
2. For each $a' \in \mathcal{A}$, compute expected marginal utility next period using the already-solved consumption policy $c_{t+1}(\cdot)$, and invert the Euler equation

$$u'(c_t(a')) = \beta(1 + r) \mathbb{E}[u'(c_{t+1}(a'))]$$

to obtain $c_t(a')$.

3. Use the budget constraint to map $(a', c_t(a'))$ into the endogenous current asset level $a_t(a')$.
4. Interpolate from the endogenous grid back to the exogenous grid, impose the borrowing constraint, and store $c_t(a)$, $a_{t+1}(a)$, and $V_t(a)$.

Algorithm 2 (Non-fertile working ages, DC-EGM). In working ages after fertility ends ($t \in \{T - T_R - T_{NF} + 1, \dots, T - T_R\}$), the household chooses labor $l_t \in \{0, 1\}$ and savings. Because labor is discrete and shocks are extreme value, the continuation value involves an inclusive value and choice probabilities. The code implements DC-EGM following [Iskhakov et al. \(2017\)](#).

For each period t (going backward) and each discrete state (θ, e, x, m, mk, k) :

1. *Choice-specific EGM step.* For each current labor choice $l_t \in \{0, 1\}$:
 - (a) Compute disposable income $y_t(l_t) = \tau(\text{gross}(l_t), m)$ where $\tau(\cdot)$ is the progressive tax-transfer function.
 - (b) For each $a' \in \mathcal{A}$ (exogenous next-period asset grid), compute expected marginal

utility at $t + 1$:

$$\mathbb{E}_t[u'(c_{t+1})] = \sum_{l'=0}^1 P_{t+1}(l' = 1 \mid a') \cdot u'(c_{t+1,l'}(a')),$$

where $P_{t+1}(l' = 1 \mid a')$ is the work probability from the previous iteration (logit).

(c) Invert the Euler equation to obtain consumption on the endogenous grid:

$$c_{t,l_t}(a') = [\beta(1+r) \mathbb{E}_t[u'(c_{t+1})]]^{-1/\rho}.$$

(d) Map to endogenous current assets using the budget constraint:

$$a_{t,l_t}(a') = \frac{c_{t,l_t}(a') \cdot \text{den}(m, k) + a' - y_t(l_t)}{1+r}.$$

(e) Construct the choice-specific value on the endogenous grid:

$$V_{t,l_t}(a_{t,l_t}(a')) = u(c_{t,l_t}(a')) + \mathbf{1}\{l_t = 1\} \psi_l(t, e) + \beta V_{t+1}(a').$$

2. *Upper envelope.* The endogenous grid $(a_{t,l}, c_{t,l}, V_{t,l})$ may be non-monotonic when labor decisions change discontinuously. Apply the upper-envelope method:

(a) Sort by endogenous assets $a_{t,l}$.

(b) Check monotonicity: if $a_{t,l,j+1} \geq a_{t,l,j} - 10^{-10}$ for all j , use direct interpolation.

(c) Otherwise, for each exogenous grid point $a \in \mathcal{A}$, compute $V_t(a) = \max_j V_{t,l}(\text{segment}_j(a))$ over all segments.

3. *Credit constraint region.* For $a < \min(\{a_{t,l}(a')\})$, set $c_t = (a(1+r) + y_t - \underline{a})/\text{den}$ and $a_{t+1} = \underline{a}$.

4. *Logit aggregation.* Aggregate choice-specific values with Type-I EV taste shocks (scale $\sigma_l(e)$):

$$V_t(a) = \sigma_l(e) \log(\exp(V_{t,0}(a)/\sigma_l) + \exp(V_{t,1}(a)/\sigma_l)),$$

$$P_t(l = 1 \mid a) = \frac{\exp(V_{t,1}(a)/\sigma_l)}{\exp(V_{t,0}(a)/\sigma_l) + \exp(V_{t,1}(a)/\sigma_l)}.$$

Algorithm 3 (Fertile ages and schooling, VFI with grid search). In fertile ages (and in early schooling periods), the code switches to grid-search VFI because the within-period structure (meeting/marriage, contraception and birth risk, newborn investment, schooling decisions, and experience dynamics) generates non-convexities and additional discrete margins that are not well suited for DC-EGM.

For each fertile period t (going backward) and each discrete state (θ, e, x, m, mk, k) :

1. *Stage 3 (given marital and fertility outcome).* For each labor choice $l_t \in \{0, 1\}$, the code searches over $a_{t+1} \in \mathcal{A}$ and computes implied consumption from the budget. If $k_t = 2$ (newborn), it solves (c_t, i_t) jointly using the FOC (bisection method described above) for each candidate a_{t+1} . It stores the maximizing a_{t+1} , c_t , i_t and the resulting choice-specific value.
2. *Labor aggregation.* For each state, it aggregates across l_t using the log-sum formula with scale $\sigma_l(e)$.
3. *Stage 2 (contraception and birth risk).* For states with no child ($k_t = 1$), it computes V_t^{birth} and V_t^{nobirth} from stage 3 and solves for optimal contraception analytically. It then forms the expected value integrating over the realized birth.
4. *Stage 1 (meeting and marriage).* For eligible singles, it applies the meeting probability $\mu_{t,e}$ and compares the stage-2 value under marriage versus remaining single, generating the marriage policy and the beginning-of-period value.
5. *Schooling decisions.* In the first periods, it solves high-school continuation and college attendance/continuation decisions using choice-specific value comparisons with extreme-value taste shocks.

Numerical details. (i) Grid search is accelerated by breaking when consumption turns negative and by exploiting local monotonicity in a' . (ii) The child-investment inner problem uses bisection with tolerance 10^{-10} and maximum 50 iterations. (iii) All consumption values are floored at 10^{-10} before utility evaluation to prevent numerical overflow.

Convergence and numerical tolerances. The solver employs the following numerical tolerances:

- *Consumption positivity:* $c_t \geq 10^{-10}$ (machine epsilon floor)
- *Child investment FOC:* Bisection terminates when $|u'(c_t) - V'_k(i_t)| < 10^{-10}$ or bracket width $< 10^{-10}$ (maximum 50 iterations)
- *Upper envelope:* Segments are considered monotonic if $a_{t,j+1} - a_{t,j} > -10^{-10}$
- *Interpolation:* Weights clamped to $[0, 1]$ using $w = \min(\max(w, 0), 1)$
- *Logit aggregation:* Uses log-sum-exp trick to prevent overflow: $\log(\sum_j \exp(x_j)) = x_{\max} + \log(\sum_j \exp(x_j - x_{\max}))$

OA.10.4 Forward simulation

The function `simulationF` takes the policy objects produced by `VFI_P_DCEGM` and simulates N life histories. It uses pre-drawn uniform random variables for fertility, meeting, and labor choices to ensure reproducibility across parameter vectors. In early periods, schooling continuation and college continuation/dropout are stage-3 policies that are indexed by the realized fertility outcome $j \in \{k, nk\}$; accordingly, these schooling rules are evaluated after the fertility draw and conditional on the realized j (see Section 5 and Appendix OA.10.3).

Algorithm 4 (Simulation). For each simulated woman $i = 1, \dots, N$:

1. Initialize $(a_1, \theta, e_1, x_1, m_1, mk_1, k_1)$ and store deterministic objects (age mapping, IDs).
2. For $t = 1, \dots, T$:
 - (a) Evaluate policy functions at the current asset level by linear interpolation on $\mathcal{A}$.
 - (b) If eligible and single, realize a meeting draw and apply the marriage decision rule (sub-stage 1).
 - (c) If in fertile ages and without a child, apply the contraception policy, compute $p_t(s_t)$, and realize conception with the fertility draw (sub-stage 2), obtaining $j \in \{k, nk\}$.

- (d) Apply schooling decisions in early periods using the stage-3 policy rules conditional on the realized j (high-school continuation, college attendance/continuation/dropout).
- (e) Realize labor supply using $P_t(l = 1)$ and the labor draw. Update experience deterministically when working.
- (f) Given realized discrete outcomes, update assets using the savings policy; store consumption, income, and other outcomes.

3. After simulating all individuals, compute model moments from simulated histories.

OA.10.5 Calibration (SMM) and optimization

I estimate the model using the simulated method of moments (SMM). SMM chooses parameters to minimize the distance between moments computed from simulated model output and their empirical counterparts.

Target moments and loss function. Let $m^{\text{data}} \in \mathbb{R}^{118}$ denote the vector of empirical moments and $m(\vartheta) \in \mathbb{R}^{118}$ the simulated moments under parameter vector ϑ . The SMM loss function is

$$\mathcal{L}(\vartheta) = \sum_{j=1}^{118} w_j \left(\frac{m_j(\vartheta) - m_j^{\text{data}}}{m_j(\vartheta) + 0.01} \right)^2,$$

where all weights $w_j = 1$ (equal weighting). The additive constant 0.01 in the denominator prevents division by zero for near-zero moments and scales the loss to be approximately unit-free. This formulation emphasizes percentage fit rather than absolute deviations, which is appropriate given the wide range of moment magnitudes (e.g., birth rates ~ 0.05 – 0.30 vs. college attendance ~ 0.10 – 0.70).

Choice of moments. The 118 target moments include: (i) fertility outcomes by age, education, and ability (birth rates, birth timing, births out of wedlock); (ii) labor supply by age, education, and maternal status; (iii) marriage rates and timing; (iv) educational attainment by ability; and (v) wage profiles by education and experience.

Algorithm 5 (SMM objective evaluation). Given a candidate parameter vector ϑ :

1. Map ϑ into model objects (e.g., the conception technology parameters, labor preference/taste-shock scales, meeting probabilities, and child-investment parameters).
2. Solve the model to obtain value and policy functions (Algorithm 1–3).
3. Simulate outcomes (Algorithm 4).
4. Compute $m(\vartheta)$ from simulated histories and return $\mathcal{L}(\vartheta)$.

Global optimization and parallelization. The file `calibration_hpc.jl` runs a global search using differential evolution through `BlackBoxOptim.jl` (variant: `de_rand_1.bin`). Differential evolution is a derivative-free global optimizer well suited to non-convex, high-dimensional problems where the objective function is noisy or discontinuous. It maintains a population of candidate solutions and iteratively improves them through mutation, crossover, and selection operations. The algorithm operates as follows:

1. Initialize 47 parallel workers, each with a perturbed starting parameter vector.
2. Each worker runs an independent differential evolution search with population size 10–15.
3. Terminate when all workers complete their allocated time budget (7 days per worker) or when the loss improvement falls below 10^{-6} for 1000 consecutive evaluations.

OA.10.6 Computational performance and implementation

Hardware and software. Estimation was performed on a high-performance computing cluster with Intel Xeon Gold 6248R processors (48 cores per node, 3.0 GHz base frequency). The code is implemented in Julia 1.9.3, leveraging multithreading for EGM/DC-EGM steps and distributed parallelism for calibration. Key packages: `Interpolations.jl` (v0.14), `BlackBoxOptim.jl` (v0.6), `Distributed.jl` (standard library).

Solution time. A single model solution at the estimated parameters requires:

- *VFI (backward induction):* ~15–20 seconds (30 asset grid points)
- *Simulation (10,000 agents):* ~8–12 seconds

- *Total (solve + simulate + moments):* $\sim 25\text{--}35$ seconds per parameter vector

Calibration runtime. The SMM estimation uses differential evolution (`de_rand_1_bin`) with 47 parallel workers, each running independent searches from perturbed starting values. Total calibration time: approximately 8,064 CPU-hours (168 hours wall-clock time with 48 cores). The algorithm evaluates approximately 420,000 parameter vectors before convergence.

Grid density and accuracy. The baseline specification uses $N_a = 30$ asset grid points with cubic spacing: $a_j \propto j^3$ to concentrate points near the borrowing constraint. Robustness checks with $N_a = 50$ yield moment differences $< 0.5\%$ for all targeted statistics, confirming numerical convergence. Child investment is solved analytically via the first-order condition (bisection with tolerance 10^{-10}), avoiding discretization error.

Numerical stability. To ensure stability: (i) All consumption values are floored at 10^{-10} before utility evaluation. (ii) Logit aggregation uses the log-sum-exp trick: $\log(\sum_j \exp(x_j)) = x_{\max} + \log(\sum_j \exp(x_j - x_{\max}))$ to prevent overflow. (iii) Interpolation weights are clamped to $[0, 1]$. (iv) The bisection algorithm for child investment uses robust bracketing with explicit checks for corner solutions.

Computational requirements.

- *Minimal replication:* Single model solution requires < 1 minute on a standard laptop (4 cores, 16GB RAM)
- *Full estimation:* Requires HPC access (48+ cores recommended); wall-clock time ~ 168 hours
- *Memory:* Peak usage $\sim 8\text{GB}$ per worker (solution), $\sim 2\text{GB}$ (simulation)

Random number generation. All stochastic elements (simulation draws for fertility, marriage, labor, education) use pre-generated uniform random variables with fixed seed (4546), ensuring exact replicability across parameter vectors. This design ensures that changes in moments reflect only parameter changes, not simulation noise. Calibration uses pseudo-random perturbations for initial parameter values (seed set per worker ID).

Software dependencies. Core packages with versions: `Parameters.jl` (0.12), `Interpolations.jl` (0.14), `BlackBoxOptim.jl` (0.6), `Distributed.jl` (standard library), `DataFrames.jl` (1.5), `Distributions.jl` (0.25), `CSV.jl` (0.10). Full environment specified in `Project.toml` in the replication package.